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Abstract:

The AI bubble is inflating, as so many have before it. The AI bubble will burst, as so many have before it. In its wake it will leave abandoned fantasies and real changes in social relations of production, as has each techno-institutional wave since the 18th century. To understand what these are likely to be requires a firm grasp on the dynamics of techno-institutional regimes in capitalism, the recurring class struggle over and consequences of extension or displacement of manual by mechanical manipulation of nature and of human-embodied by machine-embedded knowledge and codified knowledge in production, and the relevant class dynamics of the techno-institutional wave ending in the societies considered. The transition is occurring on the background of transition of the global manufacturing core in capitalism, requiring an understanding of how prior such reorientations of the global network of capitalism unfolded. The paper provides a discussion of these theoretical and historical elements to outline a way of understanding the institutional political economy of AI in the second quarter of the twenty-first century.

Introduction

- I. Basic concepts and analytic tools for an institutional political economy of capitalism
 - A. *Capitalism, the Red Queen Dynamic, Power and Productivity*
 - B. *Class in capitalism—adding the dimension of knowledge*
 - C. *Techno-institutional waves and class conflict*
 - D. *Class relations and macroeconomic stability*
- II. East from Eden—Deindustrialization and its Discontents
 - A. *Bifurcated development: Computation and information technologies vs. non-import-exposed low-productivity services*
 - B. *Class, global reorientation, and the differentiation between liberal oligarchic, corporatist oligarchic, and social democratic societies*
- III. Manual and low-skill cognitive labor: impact on the unskilled and skilled working classes and the clerical class
 - A. Robotics
 - B. Algorithmic Surveillance and Management Systems
 - C. Ghost work and the global division of labor
- IV. Cognitive Labor: LLMs and the change of professional and paraprofessional labor, class implications for the PMC, capitalists, smallholders, and clerical classes
 - A. Alternative trajectories for AI in the professions
 1. Law offices
 2. Physicians and healthcare providers
 3. Pharmacists and pharmacy technicians
 4. Investment advisors
 - B. Techno-institutional pathways for AI in Cognitive Labor
 1. AGI Capitalist Utopia

2. An incumbent-PMC led model and breakdown of the apprenticeship model
3. Capitalist-led trifurcation of professional cognitive labor
4. A social-democratic model

V. Conclusion

Introduction

A decade or at most two from now, AI will deliver human transcendence, practical immortality, or at least double the human lifespan to 150 years, and with the use of AI finance ministers and central bankers, eliminate global poverty.¹ A decade or at most two from now, AI will make Earth uninhabitable by humans because machines will rapidly and continuously self-improve, but their goals will be misaligned with ours.² At the limit, to prevent the emergence of superintelligence that will kill us all, datacenters should be globally prohibited and, if they continue to operate, bombed.³ If either utopians or dystopians are right, then we are merely rearranging the deck chairs on the Titanic. We might as well go out and enjoy what is left of the rest of our lives. The future will be so different from the past and present and its consequences will transform human life so vastly that nothing in our present experience or past can prepare us for what is to come.

Outlandish claims are a hardy perennial in capitalism over the past three hundred years, usually supporting silly stock valuations of anything that smells of being part of the transformation, and leaving behind lost fortunes, great wealth, and (sometimes) genuinely transformative technological change—without quite fulfilling either utopian or dystopian visions. I proceed on the assumption that neither option is realistic and that exponential technological change has repeatedly been predicted, and repeatedly failed to materialize.⁴ First, the scale of investment in data-centers and the sheer material requirements on GPU production and energy requirements required to continue to grow data-centers at the same rate to follow the same development path is unsustainable as a profitable exercise; second, all the reachable training data has already been used to train LLMs, and efforts to use machine-synthesized training data appear to be of doubtful use; and third, growth of algorithmic efficiency is contained without a genuine shift to new models, even if the material constraints on continued exponential growth of data-centers did not make the present path unlikely to be profitably continued. The present bubble-financed growth is producing debt overhang that is almost certainly speculative at least, if not at the Ponzi stage, in Minsky's terms.

¹ RAY KURZWEIL, *THE SINGULARITY IS NEARER: WHEN WE MERGE WITH AI* (2024); Marc Andreessen, *Why AI Will Save the World*, ANDREESSEN HOROWITZ (June 6, 2023), <https://a16z.com/ai-will-save-the-world/>; Dario Amodei — *Machines of Loving Grace*, <https://www.darioamodei.com/essay/machines-of-loving-grace> (last visited Aug. 19, 2025).

² NICK BOSTROM, *SUPERINTELLIGENCE: PATHS, DANGERS, STRATEGIES* (2017); STUART RUSSELL, *HUMAN COMPATIBLE: ARTIFICIAL INTELLIGENCE AND THE PROBLEM OF CONTROL* (2019).

³ Eliezer Yudkowsky, *The Open Letter on AI Doesn't Go Far Enough*, TIME, Mar. 2023, <https://time.com/6266923/ai-eliezer-yudkowsky-open-letter-not-enough/>.

⁴ Not only are the Jetsons flying cars not with us, but confident predictions in the early 2010s that continued exponential growth in computer vision will lead self-driving cars to become ubiquitous by the mid-2020s seem quite beside the point today, and the Uberization of everything, confidently predicted a decade ago, has become the Uberization of a few useful services, while cutting edge employers fight with their Gen Z employees to come back to the office rather than work remotely.

All these reasons suggest that, like prior major technological shifts, AI too will follow an S curve of growth, and indeed that it is already on the slower top part of the curve, past the period of fastest improvement. Nonetheless, AI in its present state combined with incremental improvement for years to come will be powerful enough to become, alongside energy transition, one of two carrier sectors in the coming techno-institutional wave. Its effects are therefore best predicted through the lens of the historical dynamics of techno-institutional change of the past three hundred years. These dynamics are characterized not primarily, much less exclusively, by technology, but by a technological-institutional-ideological ensemble, a regime that reflects power-seeking dynamics, as well as productivity-seeking investments, carried out on an institutional terrain inherited from prior rounds of transformation in capitalism. Finally, I proceed on the assumption that when the AI bubble collapses, there will be a window of political opportunity to pursue a meaningfully transformative agenda, and hence that developing an understanding of what would, and what would not, be transformative is a precondition to taking advantage of the opportunity to introduce transformative institutional change after the next crisis.

Technology does not develop exogenously and autonomously to a given society's institutional political economy, operating on society from the outside. Capitalism is the product of historically-specific institutional structures that produced the only form of social life in human history to trigger and sustain continuous waves of technological change. It is the form of social life that emerged in societies whose institutions made commodity exchange in markets the dominant form of subsistence, production, and protection. For the first time in human history, most people lived in households that depend on buying their basic survival needs in markets, most organizers of production depend on buying their basic factors of production—labor, knowledge, natural material resources, and the means to call on social resources for a given purpose and overcome time and uncertainty between conception and execution—in markets, and the political military formations depend on selling debt into financial markets and taxing market transactions and income from market exchanges. That universal imperative of markets triggered the Red Queen Dynamic: the imperative to run as fast as we can just to stay in place, and twice as fast if we want to get anywhere. It is that imperative that produces the deep instrumentalization of ourselves, our relations, knowledge, and culture, and the Earth we depend on for our very life in the pursuit of profit. It is that imperative that produced both the historically unprecedented material growth and innovation, and the historically novel but persistent patterns of instability, dislocation, and social and geopolitical conflict that shaped the world we inherited and continue to hold us in their grip today and propel us forward to an uncertain future.

At least since the 18th century, the content and nature of practical knowledge deployed in production has been a major terrain of struggle over power in social relations of production. By “production” I mean the coordinated application of labor and knowledge to natural resources over time, subject to contingency, to make things people need and want. This is true for a hunter-gathering band that must combine the labor of seeking with the knowledge of what to seek and how to consume it, and must survive the day, as it is for the startup entrepreneurs aiming to sell some new product and using VC funding to buy labor, resources, and knowledge to coordinate them toward producing their new product and to bridge time between conception and execution. In capitalism, the object of production is to produce a commodity—something other people will

buy for money—whether or not they need it, and whether or not they wanted it before it was produced or were induced to want it by advertising or manipulation.

In both liberal and Marxian analyses, property in the resources and physical tools or machines has traditionally been given pride of place in defining the class structure of production, the relative roles and contributions of capitalists to workers, and from these the power over production and the distribution of its fruits that these roles brought with them in capitalist social relations. Liberal versions interpret the ownership structure as natural and the relations as consensual, leaving the distribution of the fruits of production the result of the invisible hand of the market. Justice-focused liberal versions emphasize inherited advantage and opportunities that require redistribution to achieve the just result, while libertarian versions see nothing that can be fixed without doing more harm than good. In the Marxist version, property marked the foundation from which exploitation could be veiled as free exchange. But property in the means of production was the primary dimension of analysis.

Adding the understanding that control over knowledge in production is a distinct dimension of power in production, as necessary as resources, labor, and credit, opens up a second dimension along which to understand both class structure and conflict in social relations of production. Much of this has been implicit in various strands of critical literature, but making it explicit and a distinct object of analysis helps us to understand how the diverse dimensions of conflict that AI introduces relate to each other, and how they relate to the recurring patterns of struggle over social relations of production throughout the history of capitalism.

Production combines (a) embodied knowledge that workers possess in their own bodies and control, (b) materially-embedded knowledge, or technology, and (c) codified knowledge, that is, knowledge about the production process that can be rendered explicit and available for managerial control. The relative salience of embodied knowledge as opposed to embedded or codified knowledge is central to defining power in social relations of production and has been since early modernity. At least since the 17th century, capitalist organizers of production have deployed machines that shifted knowledge in production from workers to machines they could own, enabling them to hire more readily-replaceable workers with low skill and lower wage expectations—often children and women—so as to squeeze a greater share of the fruits of production, whether or not the new machines also increased productivity as a whole. At least since Wedgwood's ceramics factory in Etruria, capitalists producers have implemented processes that enabled them to shift some of the knowledge in production to a fine-grained, observable division of labor codified in discrete tasks. This too shifted knowledge in production from artisanal embodied knowledge to codified knowledge, and shifted control over knowledge in production from workers to managers—whether or not these were the capitalists themselves or the managerial employees.

Much of the literature on algorithms and algorithmic decision systems has focused on liberal concerns: of the vulnerability of individuals as consumers, borrowers, or citizens to big companies

and big government.⁵ Within computer science, there has been a heavy emphasis on fairness understood in terms of equal opportunity, or bias, consistent with prevailing progressive American concerns with racial and gender bias, and following a range of fairness-seeking measures consistent with those that characterize American antidiscrimination law.⁶ A major concern throughout this period, particularly focused on platforms and platform capitalism before the reorientation toward AI, was focused on the concentration of economic power in the hands of a handful of globally-dominant firms, quintessentially Google and Amazon.⁷ This concern had two main lines: one more consumer-welfare focused that assumed the primary concern was deviation from efficient markets,⁸ and a second more concerned with liberal polity and the translation of economic dominance into the accumulation of political power, sometimes called Neo-Brandeisian.⁹ This longstanding concern has now been extended and transposed to the large companies developing the leading edge LLMs.¹⁰

Most of the debates on data collection and governance, concentrated market power, and equal opportunity in employment, credit, or criminal enforcement exhibit quintessentially liberal political and social theory. Their primary object of concern are individuals—as consumers or citizens—in relation to market actors or the state. The primary focal points of critique are deviations from individual choice in markets or from equal treatment in individuals’ interactions with state or large market actors. The critical literature engaging in this debate took the form either of challenging the plausibility of characterizing the relation as one of choice, the individual as the correct level of analysis or concern, or the natural baseline of choosing individuals as opposed to

⁵ FRANK PASQUALE, *THE BLACK BOX SOCIETY: THE SECRET ALGORITHMS THAT CONTROL MONEY AND INFORMATION* (2015); CATHY O’NEIL, *WEAPONS OF MATH DESTRUCTION: HOW BIG DATA INCREASES INEQUALITY AND THREATENS DEMOCRACY* (2016); VIRGINIA EUBANKS, *AUTOMATING INEQUALITY : HOW HIGH-TECH TOOLS PROFILE, POLICE AND PUNISH THE POOR* (First Edition. ed. 2018); Solon Barocas & Andrew D Selbst, *Big Data’s Disparate Impact*, 104 CALIF. LAW REV. 671 (2016); Manish Raghavan et al., *Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices*, in PROCEEDINGS OF THE 2020 CONFERENCE ON FAIRNESS, ACCOUNTABILITY, AND TRANSPARENCY 469 (2020), <https://doi.org/10.1145/3351095.3372828>.

⁶ Cynthia Dwork et al., *Fairness Through Awareness*, in PROCEEDINGS OF THE 3RD INNOVATIONS IN THEORETICAL COMPUTER SCIENCE CONFERENCE (ITCS) 214 (2012); Michael Feldman et al., *Certifying and Removing Disparate Impact*, in PROCEEDINGS OF THE 21ST ACM SIGKDD INTERNATIONAL CONFERENCE ON KNOWLEDGE DISCOVERY AND DATA MINING (KDD) 259 (2015); Moritz Hardt et al., *Equality of Opportunity in Supervised Learning*, in ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 29 3315 (D. D. Lee et al. eds., 2016), <http://papers.nips.cc/paper/6374-equality-of-opportunity-in-supervised-learning.pdf>; Jon Kleinberg, Sendhil Mullainathan & Manish Raghavan, *Inherent Trade-Offs in the Fair Determination of Risk Scores*, in PROCEEDINGS OF THE 8TH INNOVATIONS IN THEORETICAL COMPUTER SCIENCE CONFERENCE (ITCS) 43:1 (2017); Raghavan et al., *supra* note 5.

⁷ Lina M Khan, *Amazon’s Antitrust Paradox*, 126 YALE LAW J. 710 (2016); NICK SRNICEK, *PLATFORM CAPITALISM* (2017).

⁸ THOMAS PHILIPPON, *THE GREAT REVERSAL: HOW AMERICA GAVE UP ON FREE MARKETS* (2019).

⁹ K. Sabeel Rahman & Kathleen Thelen, *The Rise of the Platform Business Model and the Transformation of Twenty-First-Century Capitalism*, 47 POLIT. SOC. 177 (2019); Lina Khan, *The End of Antitrust History Revisited*, 133 HARV REV 1655 (2020).

¹⁰ David Gray Widder, Sarah Myers West & Meredith Whittaker, *Open (For Business): Big Tech, Concentrated Power, and the Political Economy of Open AI*, SSRN ELECTRON. J. (2023), <https://doi.org/10.2139/ssrn.4543807>; Julie Cohen, *Oligarchy, State, and Cryptopia* (Aug. 25, 2025).

institutionally constructed power relations as the primary focus of normative concern.¹¹ Alongside these, the environmental costs of AI became the focus of critique since a pair of publications in 2021.¹²

Throughout the past decades there has also been a line of literature far closer to the concerns of class and the ways in which technology shapes class power. Most of this work, most prominently Karen Levy's, focused on data collection and surveillance, algorithmic decision making, and their integration into systems of control and decision we now associate with AI—shapes power in the labor relation, significantly reducing worker autonomy and worker power.¹³ This line follows in the footsteps of Harry Braverman's classic work, already focused specifically on how the assembly line was a surveillance system designed to extract maximum effort, or Sam Bowles' *Social Institutions and Technical Change*, which already identified keystroke logging and GPS in trucks in 1989 as early examples of digital surveillance deployed to shift power in the labor relations from workers to managers.¹⁴ A second line of work, following Mary Gray and Siddharth Suri's *Ghost Work*, underscored both that what counts as "AI" in fact embodies a large amount of human labor, and that much of this labor was performed in poorer countries under degraded conditions of work.¹⁵ A third line of work focused on the ways in which first platform work, then more broadly automated surveillance and management, undermine worker mobilization and unionization efforts.¹⁶ The most comprehensive recent effort to integrate this approach and analyze contemporary employment practices as moves in class power struggles over production is Brishen Rogers's *Data and Democracy at Work*.¹⁷

¹¹ Elettra Bietti, *Consent as a Free Pass: Platform Power and the Limits of the Informational Turn*, 40 PACE LAW REV. 310 (2019); Salomé Viljoen, *A Relational Theory of Data Governance*, 131 YALE LAW J. 573 (2021); JULIE E. COHEN, BETWEEN TRUTH AND POWER: THE LEGAL CONSTRUCTIONS OF INFORMATIONAL CAPITALISM (2019).

¹² Emily M. Bender et al., *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?* , in PROCEEDINGS OF THE 2021 ACM CONFERENCE ON FAIRNESS, ACCOUNTABILITY, AND TRANSPARENCY 610 (2021), <https://dl.acm.org/doi/10.1145/3442188.3445922>; KATE CRAWFORD, THE ATLAS OF AI: POWER, POLITICS, AND THE PLANETARY COSTS OF ARTIFICIAL INTELLIGENCE (2021), <https://www.jstor.org/stable/j.ctv1ghv45t>.

¹³ Karen E. C. Levy, *The Contexts of Control: Information, Power, and Truck-Driving Work*, 31 INF. SOC. 160 (2015); Karen E. C. Levy, *Digital Surveillance in the Hypermasculine Workplace*, 16 FEM. MEDIA STUD. 361 (2016); Solon Barocas & Karen Levy, *What Customer Data Collection Could Mean for Workers*, HARVARD BUSINESS REVIEW (Aug. 31, 2016), <https://hbr.org/2016/08/the-unintended-consequence-of-customer-data-collection>; KAREN LEVY, DATA DRIVEN: TRUCKERS, TECHNOLOGY, AND THE NEW WORKPLACE SURVEILLANCE (2023); Ifeoma Ajunwa, Kate Crawford & Jason Schulz, *Limitless Worker Surveillance*, CALIF. LAW REV. (2017), <https://scholarship.law.berkeley.edu/californialawreview/vol105/iss3/3>; CRAWFORD, *supra* note 12.

¹⁴ HARRY BRAVERMAN, LABOR AND MONOPOLY CAPITAL: THE DEGRADATION OF WORK IN THE TWENTIETH CENTURY (25th anniversary ed. 1998); Samuel Bowles, *Social Institutions and Technical Change*, 321 in TECHNOLOGICAL AND SOCIAL FACTORS IN LONG TERM FLUCTUATIONS 67 (Massimo Di Matteo, Richard M. Goodwin, & Alessandro Vercelli eds., 1989), http://link.springer.com/10.1007/978-3-642-48360-8_5.

¹⁵ MARY L. GRAY & SIDDHARTH SURI, GHOST WORK: HOW TO STOP SILICON VALLEY FROM BUILDING A NEW GLOBAL UNDERCLASS (2019); Adrienne Williams, Milagros Miceli & Timnit Gebru, *The Exploited Labor Behind Artificial Intelligence* (2022), <https://www.noemamag.com/the-exploited-labor-behind-artificial-intelligence>.

¹⁶ Lily Irani, *Justice for "Data Janitors,"* PUBLIC BOOKS, <http://www.publicbooks.org/nonfiction/justice-for-data-janitors> (last visited Apr. 7, 2015); Lilly Irani, *The Hidden Faces of Automation*, 23 XRDS CROSSROADS ACM MAG. STUD. 34 (2016).

¹⁷ BRISHEN ROGERS, DATA AND DEMOCRACY AT WORK: ADVANCED INFORMATION TECHNOLOGIES, LABOR LAW, AND THE NEW WORKING CLASS (2023), <https://direct.mit.edu/books/oa-monograph/5556/Data-and-Democracy-at-WorkAdvanced-Information>.

The fundamental point recurring throughout the history of capitalism and appearing in new form now is that there is class conflict over the degree and distribution of asymmetric power in social relations of production that plays out with every new techno-institutional wave. There is a high road institutional path to resolving this conflict, where workers and capitalists are yoked in a more-or-less cooperative dynamic of continuous improvement in both human embodied knowledge and materially embedded knowledge (technology). There is a low road path, where capitalists seek to disembody knowledge and locate it in technologically embedded form that capitalists can fully own as property or in codified form that shifts power to managers, again weakening labor. The low road path is the non-cooperative solution to this conflict, and dominates absent institutionalized counterpower able to raise the costs of low-road strategies and make them less profitable. Through its effects on the income level and security of the broad social base of the working and clerical classes, low-road dynamics to profitability introduce both significant social dislocation, anomie, and instability and systematic failures in macroeconomic growth.

Part I lays out basic concepts and analytic tools with which to pursue an institutional political economy analysis of technological change. It provides the context for understanding the present in light of long-term patterns of conflict over technology and knowledge in production in capitalism, and explains the theoretical framework of why social democratic institutions drive toward a high-road capitalism dynamic while liberal institutions drive toward low-road dynamic. Part II characterizes the core features of the institutional political economy of the techno-institutional wave now ending, and the global class dynamics it created on the eve of the Global Financial Crisis that signaled its likely end. Part III looks at the more mature areas of change—robotics and algorithmic surveillance and management—as domains of application where we have more evidence on the dynamics of technological change as a function of the institutional political economy of the receding wave. The primary observation is that in both domains patterns of deployment and use follow expected divergent pathways, with Western European countries, particularly the Nordic social democracies and Germany, maintaining high-road manufacturing excellence and higher robotics density than the United States while seeing increased wages and employment associated with robotics deployment, and see far less deployment of algorithmic surveillance and management systems focused on speeding up production and intensifying work effort through sanctions, as is becoming more common in the United States. Meanwhile, ghost work harnesses educated workers in developing countries to perform cognitive labor under exploitative conditions, providing a contemporary instantiation of a centuries-old pathway of jurisdictional arbitrage in the pursuit of profit. Part IV looks more specifically at LLMs and cognitive labor, analyzing how their potential capabilities interact with class structure and prevailing institutions to open up vastly divergent paths between the utopian capitalist model underlying the present AI bubble and what a social-democratic version could look like, and what achieving it would entail. Part V concludes.

- I. Basic concepts and analytic tools for an institutional political economy of capitalism
 - A. *Capitalism, the Red Queen Dynamic, Power and Productivity*

Capitalism, or market society, is a form of social life in which: (a) most people live in households that depend on commodity exchange in markets to satisfy their basic needs and wants;

(b) most organizers of production depend on commodity exchange in markets for access to labor, material and knowledge resources, and credit or finance to deal with time and uncertainty; and (c) the political military formations depend primarily on selling public debt or taxing market activity as commodity exchange in markets to satisfy their fiscal needs. What defines a society as “market society” (as opposed to a society with markets, such as have existed since antiquity) is the conjunction of market dependence for subsistence, production, and protection as the dominant form of each. Market *reach* (how pervasive and all-encompassing commodity exchange is as the dominant social form throughout a society) rather than market power, is what makes a society a market society.

The combined effect of market dependence for subsistence and for production is that more-or-less everyone must acquire more-or-less everything through commodity exchange. The imperatives this produces are new compared to those in pre-capitalist and non-capitalist societies, where access to, and risk of loss of, the means of subsistence and production are governed by institutionalized relations of kinship, reciprocity, and direct coercion. This historically unique combination of market *freedom* (liberated from institutionalized relations of kinship, reciprocity, and direct coercion) combined with generalized market *dependence* as the new source of imperatives marks a society as a market society, or a capitalist society. It forces all workers to instrumentalize their own work, education, relations, and daily efforts, and all organizers of production—smallholders like family farmers, plumbers, or shopkeepers; professional and managerial services firms; or the largest corporate entities—to increase their productivity continuously to, at least, stay abreast of competitors, or to increase their power to become price makers rather than takers. It is this incessant instrumentalization of labor, knowledge, and nature that produces the historically unprecedented dynamics of continuous improvement in productivity through innovation, coupled with recurring struggle over the structure of social relations of production. I refer to this incessant pursuit of advantage, the imperative to run as fast as we can, continuously increasing our ability to turn labor, nature, knowledge, and relations into money, the Red Queen Dynamic.¹⁸

That power-seeking is the driving force of innovation in capitalism, and that innovation (rather than efficient allocation) is the whole ball of economic growth, is the core Schumpeter’s theory of innovation.¹⁹ All firms and smallholders depend on markets to obtain labor, knowledge, natural resources, credit and finance, and must run as fast as they can just to stay in place. That is, unless firms can develop some level of market power, they must operate at peak efficiency just to meet payroll and debt obligations on pain of losing their ability to continue to be organizers of production. Markets are never perfect because competition endogenously drives everyone subject to these market imperatives to identify or create, and exploit, non-competitive niches in time, space, and context. Firms that fail to do so are eventually outcompeted in access to resources, labor, and finance by those that do, and driven out of the market. The imperative to pursue power is reinforced by

¹⁸ “Here, you see,” the Red Queen tells Alice, “[I]t takes all the running you can do, to keep in the same place. If you want to go somewhere else, you must run at least twice as fast as that!” LEWIS CARROLL, *ALICE THROUGH THE LOOKING-GLASS* (Illustrated ed. 2009); Yochai Benkler, *Power and Productivity: Institutions, Ideology, and Technology in Political Economy*, in *A POLITICAL ECONOMY AND JUSTICE* (Danielle Allen et al. eds., 2021).

¹⁹ JOSEPH A. SCHUMPETER, *CAPITALISM, SOCIALISM, AND DEMOCRACY* (1st ed ed. 2008).

investors and lenders who cannot price uncertainty, and require the prospect of quasi-rents as a precondition to financing capital.²⁰ This is because true uncertainty, not only about the outcomes but also about the probabilities of different outcomes, cannot be priced efficiently *ex ante*. To compensate for the inevitable component that cannot be priced efficiently *ex ante*, financing projects under true uncertainty requires that the outcome of the investment includes meaningful pricing power to avoid being a price-taker at efficient prices: that is, at prices that, by definition, reflect only knowable, not unknowable, costs.²¹

Both innovation and finance under uncertainty therefore depend on the capacity of organizations to create and reap rents, rather than to receive efficient prices in perfectly competitive markets. While some firms may secure a durable monopoly, most firms cannot. Instead, the imperatives to find sources of power forces most firms to create quasi-rents, and then partially spend these down to increase their magnitude and half-life while developing new sources of quasi-rents as the first set diminishes. For firms, technical and organizational innovation is a major pathway to escape competition. This is the heart of Schumpeterian innovation.²² Firms invest in the expensive and uncertain process of innovation in expectation of gaining quasi-rents during the period when their innovation enables them to do things that their competitors cannot, or the same things at lower cost than their competitors can match. This enables them to reap supra-competitive returns for a while. Schumpeter insisted that the core assumption of neoclassical economics about markets clearing in equilibrium at marginal cost necessarily leads to stagnation.²³

If the distinctive characteristic of successful firms is their ability to create and sustain *pricing power*, rather than a productivity edge *per se*, there is no systematic reason to believe that firms will prefer productivity-enhancing strategies (most importantly technological and organizational innovation) to extractive power-seeking strategies that simply increase their share of a non-growing pie. Profit-seeking firms are agnostic between pricing power born of what Thorstein Veblen called “business sabotage”—destroying competitors and competitive conditions in their sector—and pricing power built on new abilities to make more of what people need and want for less.²⁴ What matters is maximizing the value of anticipated quasi-rents. There is every reason to think that firms will pursue both strategies—increasing both productivity and power—in various measures that depend on how useful they are in creating, increasing the magnitude, and extending the duration of the quasi-rents obtainable from their activity.²⁵

²⁰ HYMAN P. MINSKY, *STABILIZING AN UNSTABLE ECONOMY* (1986). The full specification of Minskyian endogenous instability as it emerges out of, and endogenously destabilizes, capitalist economies with sufficiently broad financial markets is outside the scope of this article, as is integration of the macroeconomic implications of the drive to extract profits by grabbing rather than growing into a Neo-Kaleckian framework. I lay these out in chapters 13 and 14, respectively, of BENKLER, *supra* note 16.

²¹ MINSKY, *supra* note 22, at 179–96, 213–18.

²² JOSEPH A. SCHUMPETER, *CAPITALISM, SOCIALISM, AND DEMOCRACY* 81–106 (3d ed. 2008).

²³ *Id.* at 83.

²⁴ THORSTEIN VEBLEN, *ABSENTEE OWNERSHIP: BUSINESS ENTERPRISE IN RECENT TIMES: THE CASE OF AMERICA* (Routledge 1996) (1923); Janet T. Knoedler, *Veblen and Technical Efficiency*, 31 J. ECON. ISSUES 1011 (1997); Jesse F. Dillard & Linda V. Ruchala, *Veblen's Placebo: Another Historical Perspective on Administrative Evil*, 38 ACCT. HISTORIANS J., no. 1, 2011, at 1.

²⁵ Benkler, *supra* note 18.

Growing the pie and grabbing a larger slice contribute equally to the bottom line. A profit-maximizing firm is indifferent whether it gets profits from introducing a new assembly line that is more efficient but requires highly-skilled unionized workers, or building a less efficient new assembly line in a right-to-work state that enables it to reduce labor costs because the workers cannot unionize. Ideally, from management's perspective, the firm would build the most efficient assembly line it can, in a labor-unfriendly jurisdiction, and incorporate in a jurisdiction with corporate laws that weaken shareholders and weak antitrust enforcement, gaining productivity-based and distributional-power-based profits. Horizontal power vis-à-vis competitors and entrants, and vertical power vis-à-vis workers, suppliers, and consumers, and for managers, shareholders, are complementary in this regard. Two additional implications follow from this. First, without active social steering, that is, industrial policy, innovation will not be systematically socially beneficial, but would more likely be so if forced by institutions away from power-seeking pathways toward productivity-seeking pathways. Second, firms will oppose or circumvent institutional changes that are socially beneficial at the macro level to the extent these reduce their bargaining power in relation to workers, consumers, suppliers, or vendors more than they increase the firm's private capacity to capture rents.

Capitalists have, historically, deployed seven primary pathways as power-seeking strategies. First, technological innovation that shifts power over knowledge in production from embodied knowledge controlled by skilled workers into knowledge embedded in machines that capital can own as property. Second is manipulating consumer demand toward what is most profitable for them to produce. Here, machine learning in particular suggests the potential for ever-more refined and personalized manipulation to drive consumer demand toward goods and services most profitable to advertisers, a major concern of the existing literature on consumer surveillance and price discrimination. Third is transforming institutions, most directly law, so as to weaken labor, consumers, suppliers, or vendors. Fourth is jurisdictional arbitrage, locating operations in jurisdictions where a given factor of production—primarily labor and nature—is more systematically accessible at lower prices, whether due of efficiency or of greater asymmetric power. This has been a major source of social destabilization in the United States and some Western European societies, as offshoring and outsourcing circumvent pockets of powerful labor primarily by moving manufacturing and, more recently, services to Asia, and partly, for Western European societies, to post-Soviet Middle-European societies. A similar effect is obtained by, fifth, harnessing the political military formations of dominant capitalist societies to project force across the globe to obtain better terms of trade in various foreign jurisdictions that can be forced to accept asymmetric terms—for their natural resources, workers, or consumers. Sixth, is harnessing status-subordinated workers as an underclass, usually racialized or gendered, often non-citizen immigrants, to provide a workforce with low wage expectations and weaker bargaining power than populations of workers with full juridical equality. Seventh, and finally, is an escape to finance, where redistributive, rather than productive gains, repeatedly become available through endogenous cycles of speculative and Ponzi finance that have recurred roughly on a roughly decadal basis since the middle of the 18th century.²⁶ Since the Global Financial Crisis, financial rentiers have been pursuing a series of small

²⁶ I review each of these seven power-seeking pathways, and provide the historical evidence for their emergence as early as fourteenth and fifteenth century Northern Italy, and seventeenth century Dutch Republic and England, in

bubbles—primarily various cryptocurrencies and briefly SPACs, and in the current decade this pursuit of bubbly profits in deeply uncertain contexts has found its way into dizzying stock valuations and deep real economy investment in data-centers, catapulted by the promise of exponential improvements in AI and the near achievement of Artificial General Intelligence (AGI). A full analysis of the institutional political economy of AI would require pursuing each of these distinct power-seeking strategies in the new techno-institutional wave. Here I focus more narrowly on the interaction between technological change to increase productivity and shift power in production, as these reshape the prevalence of class composition and the distribution of class power.

B. Class

Class is asymmetric position in social relations of production, reflecting asymmetric power over factors of production—natural resources (land and energy most foundational among them), labor, knowledge, tools or technology, and credit or finance (the capacity to harness existing social resources to future production and bridge the time and uncertainty from conception to execution and payment of the planned activities). This asymmetric power is a function of institutions—patterns of habituated behavior into which individuals must fit under penalty of coordinated social resistance, and the social mechanisms of their production or enforcement. Institutions emerge in historical time and are inherited by any given society from its own history and, through habituation and internalization, structure both social relations and the prevailing common sense, *mentalité*, or ideology in a given society and time as to how the world works, what causes what, what is valued and what relations and roles are normal, conceivable, or desirable. These institutionalized relations, socially inherited, operative in historical time, and never chosen, are the microfoundations of social analysis, rather than any imaginary natural, universal, or otherwise pre-social and ahistorical individuals with beliefs, preferences, or constraints expressed as choices.

Unlike the standard liberal “upper, middle, and working class” structure based on income distribution, and unlike Marxist “capitalists/proletariat” binary division, with “petite bourgeoisie” occupying a purgatory in between, Talha Syed and I explain why understanding class on the background of institutionally-structured control over elements of production produces eight classes that systematically reappear throughout the history of capitalism, albeit with different prevalence.²⁷ Specifically, we add to property as a crucial marker of class differentiation the dimension of whether the work is manual or cognitive, underscoring the centrality of control over and application of knowledge in production. Knowledge in production comes in three forms: embodied in human beings, embedded in material culture, and codified in a form susceptible to explicit communication in language. Embodied knowledge is skill or tacit knowledge that human beings possess in their bodies and is sufficiently resistant to codification that it must be internalized by human beings in order to be useful. Embodied knowledge is the kind of knowledge in production that workers have the most withholding power over. Because it takes time to train any worker to have it, skilled or experienced workers whose embodied knowledge is important for the

chapters 5, 6, 8, and 10-13 of YOCHAI BENKLER, *THE GLOBAL ORIGINS OF CAPITALISM: POWER, PRODUCTIVITY, AND THE EVOLUTION OF MODERN MARKET SOCIETIES* (2026).

²⁷ Yochai Benkler & Talha Syed, *Reconstructing Class Analysis*, 4 J. LAW POLIT. ECON. 731 (2024).

profitability of an enterprise have more bargaining power vis-à-vis capitalists than do workers whose contribution is easy to replicate.

Technology is knowledge useful in production embedded in material culture. Some definitions treat any practical knowledge that allows us to do new things or old things more efficiently as “technology” or “technological progress.” That definition is less useful for distinguishing between institutions and technology. Take Taylorism. Taylor’s *Principles of Scientific Management* is a quintessential *institutional* innovation: It consisted in a new set of rules and practices for how managers should obtain information about their production process and translate it into a precise set of behavioral rules imposed on workers. It is a change in social relations—in the roles workers play vis-à-vis each other in the coordinated activity of producing, or vis-à-vis foremen, process engineers, and higher-level managers. It is not “wrong” in some metaphysical sense to call Taylorism—the application of new information (e.g., Taylor’s time motion studies) to make production vastly more efficient—a “technology.” It just makes it less useful in understanding the difference between, say, the clothes dryer or the electronic spreadsheet, on the one hand, and Title VII of the Civil Rights Act or options theory, on the other hand, as contributing factors to the increase of women’s labor force participation in the former case or financialization in the latter. In the case of industrial organization, the assembly line is a technological embodiment of a managerial philosophy, but Taylorism itself would fall in my definition under the rubric of institutional, rather than technological, change, and the knowledge it brings to bear in production as codified, rather than materially-embedded, knowledge. I am not claiming a metaphysical truth to this definition. Only its practical utility for understanding the distinctive features of the contribution of technology to political economy, whose other major elements are institutions and ideology.

What Taylorism produces is codified knowledge: knowledge that can be written down explicitly and become directly useful in production by anyone reading and following the procedure or practice set out in codified form. Taylor’s time motion studies were an effort to convert embodied knowledge into codified knowledge, shifting power to managers so they could shrink the variability among workers and reduce workers’ ability to exercise power over the speed and intensity of work on the shop floor. Checklists that govern surgeons are a form of codified knowledge, as are workplace manuals. That codified knowledge is rarely perfect is well-understood by workers who have long used Italian strikes, or “work-to-rule” labor actions. In these, workers follow the codified process to the letter. They withhold their active cooperation, which under normal circumstances fills in the blind spots of the codified process with their own embodied knowledge, usually learned on the job in a process of day-to-day flexible adaptation to the noisy environment of real life production and the lack of fit between the formalized context required to produce a codified manual and the reality of the physical world.

Knowledge in production has been an important site of contestation between workers and capitalists throughout the history of capitalism, as it has been between workers and the professional and managerial class (PMC), the latter and the capitalists, and the skilled workers who possess significant degrees of embodied knowledge and the unskilled or semi-skilled workers who possess smaller quanta of embodied knowledge, generally easier to replace by equally low-skilled workers.

In Syed and my framework for understanding class, there are three classes that make up the profit-reaping classes. These include those who own property in calls on existing resources and tools, but do no work in production (rentiers), those who both own property and organize production and do managerial cognitive work in it (capitalists), and those who do managerial and symbolic expert cognitive work, but do not have property in the resources used in production (professional and managerial). The rentiers historically were landowners but, throughout the history of capitalism, have generally also included the owners of (often inherited) claims on past resources that could be applied to current work in anticipation of future income—money and secure obligations, that is, financial rentiers. The capitalists own resources and organize production, buying labor, materials and resources, and credit as commodities to produce profit. What distinguishes them is that they are market-dependent organizers of production, who may own or acquire proprietary control over the core inputs as commodity, including land and other natural resources and the labor of workers who are themselves market-dependent for subsistence, and, unlike the rentiers, manage production as well as owning resources, unlike the rentiers. Alongside these there has been a professional class and a managerial class—in the persons ships’ captains and clerks, notaries and lawyers, plantation and plant managers since early modernity.²⁸

Smallholders own the property in their core inputs—whether land, for rural smallholders, or their tools and workshop or shop, for urban smallholders, both craft manufacturers and shopkeepers. They mostly produce using family labor, although they may supplement it with a small number of paid workers, historically mostly in long-term relationships of apprenticeship. The knowledge they use is mostly embodied, that is to say, tacit skill they possess in their own bodies, rather than formal codified knowledge that would allow anyone to perform the task by following explicit instructions. Marxists mostly ignore these as either petite bourgeoisie if urban, or peasants if rural, but in fact they have always been an important economic and political force in the emergence and shaping of class relations in capitalism. Craft or skilled workers, by contrast, do not own the workshop or buy the material resources generally, but they apply embodied knowledge in production, over which they exercise control simply from the fact that their skills are in their body and cannot readily be codified and turned into a process capitalists can oversee for deploying unskilled labor, or turned into a tool, that is, turned into technology that the capitalists can own outright. As the capital requirements of smallholder craft manufacturing increased in late 17th century in Holland and early 18th century in Britain, a growing number of perpetual journeymen with no hope of becoming urban smallholders began to form a distinct class of skilled workers, replacing the model of lifecycle apprenticeship as a transitional path to craft smallholding.

At least since the early 17th century there have been struggles between skilled workers and capitalists over technology, usually centered on efforts of the latter to embed the practical knowledge that made skill so powerful into machines, and so to increase their power to control the rents from production by circumventing the power of the skilled workers over the production process and replacing them with weaker, unskilled or semi-skilled workers. These unskilled

²⁸ Claims I make throughout this article about the historical prevalence of some classes and shifts are all based in the detailed historical analysis in BENKLER, *supra* note 26.

workers control only their labor, and are “unskilled” in the sense that the embodied knowledge required to perform their functions in production is easily and quickly acquired by anyone, leaving them far less bargaining power than workers with embodied knowledge who are much harder to replace.

Orthogonal to either property or knowledge, capitalist producers have, throughout the history of capitalism, also sought to increase profits by harnessing workers who, under the prevailing institutions and ideology of their society and time, were juridically not recognized as full persons. These historically included, most prominently, racialized enslaved workers and later racialized “free” workers after emancipation, as well as women, children, and unskilled or semi-skilled immigrants. Figure 1 maps these classes, including the clerical class, who perform cognitive work using symbolic and codified knowledge at a lower level of complexity than the professional class, a class that became an important part of the dynamic in the twentieth century with the expansion of white-collar workers to deal with the increasing complexity of capitalist production.

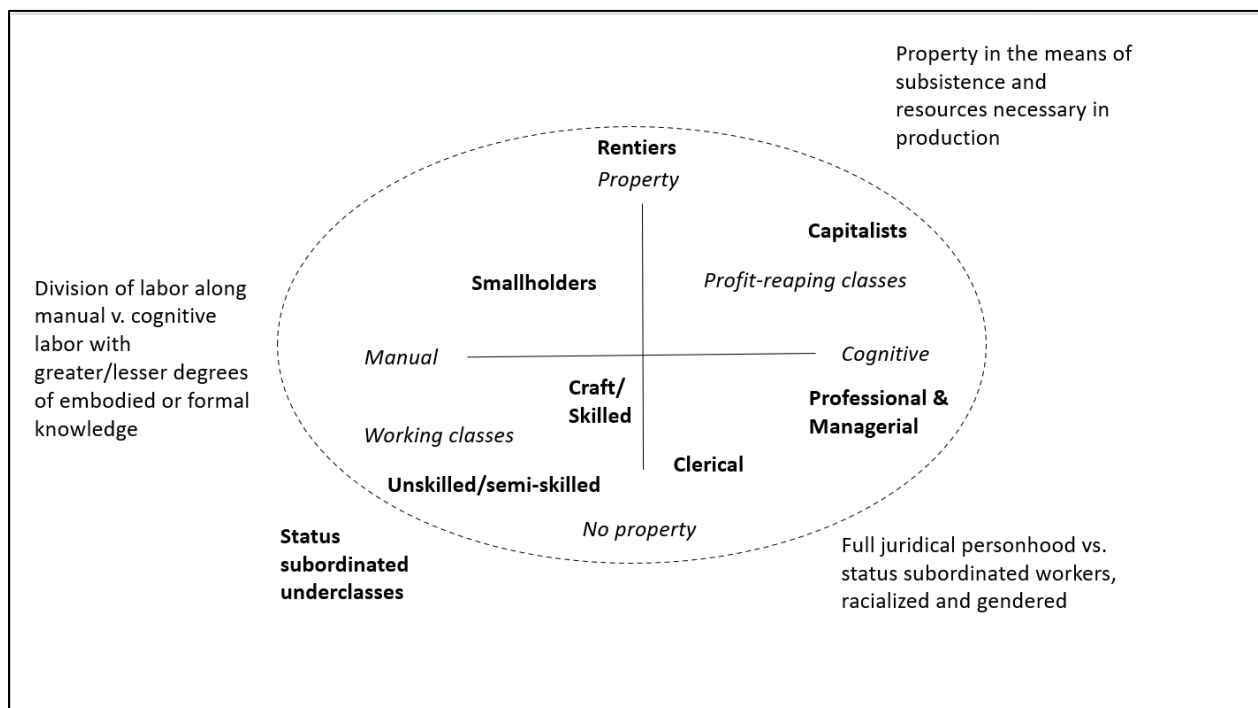


Figure 1 Class structure based on property in factors of production and role in division of labor. From Benkler & Syed 2024.

C. *Techno-institutional waves in capitalism and class conflict*

The Red Queen Dynamic drives a wave pattern that Schumpeter, and in his footsteps significant strands in the Schumpeterian innovation literature, have captured in the term Kondratiev waves.²⁹ This line of work observed that major transformations of the leading

²⁹ CHRISTOPHER FREEMAN & LUC SOETE, THE ECONOMICS OF INDUSTRIAL INNOVATION (3rd ed ed. 1997); CHRIS FREEMAN & FRANCISCO LOUÇÃ, AS TIME GOES BY: FROM THE INDUSTRIAL REVOLUTIONS TO THE INFORMATION REVOLUTION (2002); CARLOTA PEREZ, TECHNOLOGICAL REVOLUTIONS AND FINANCIAL CAPITAL: THE DYNAMICS

technological innovations and sectors in capitalism recur in roughly fifty to seventy years-long waves. The basic dynamic is that innovation is driven by the effort to create quasi-rents. A business at the cutting-edge can produce the same things at lower cost, or new things others cannot easily replicate, and so for a while has pricing power to extract rents—to price above marginal cost. Such profits draw financial investors seeking the assurance of above-marginal-cost pricing power to deal with the uncertainty—the incalculable aspects of contingency associated with all production. Over time, the incremental improvements after initial large gains slow down as the technology matures, lagging enterprises learn how to incorporate the technology into their own operations, and the profits obtained by the early movers shrink. As the profits shrink, innovators whose efforts suggest the possibility of escaping from the pack are able to attract finance more readily than they could when the margins from investing in the pack leaders were still high and reliable. Eventually some new technology among those novel ideas indeed produces enough rents to focus investments on it, and it becomes the new leading edge of a new wave of innovation-driven growth.

The process is not efficient. A lot of silly money is lost on harebrained schemes and outright fraud. But when the roughly decadal financial bubbles that have recurred over the past two hundred and sixty years coincide with the end of a wave, some of the silly money fuels the successful innovations as well as the outlandish ideas and Ponzi schemes.³⁰ A new wave begins. Chris Freeman and his coauthors in particular emphasized that the Kondratiev waves also coincide with labor conflict, as at their start the dislocation to lagging sectors causes structural un- and underemployment, and at their end the profit squeeze leads to conflicts again, while at their sustained growth stage organized labor is sometimes able to capture a share of the above-marginal-profit returns in the leading sectors, particularly in societies where law and violent coercion have not made it impossible to unionize.

The observation about labor conflict and unions underscores a more fundamental point about this half-century-long wave pattern in capitalism. Technology does not operate separately from the institutional structure out of which it emerges and into which it is introduced. Nor is it systematically productivity-increasing, but rather is profit-increasing, whether that profit is grounded in higher productivity or more effective grabbing of a larger share of a slow-growing pie. Institutional structure, just like technology, is itself the product of struggle and investments that can increase productivity and shift the balance of power between classes and within classes. Each major techno-institutional wave comes out of technological and institutional adaptations to the prior wave and conflicts over the changes it brought, and produces new pressures and conflicts as the techno-institutional regime shifts from the older model whose profitability was squeezed into a new one offering new sources of quasi-rents to sustain profits for a new wave. Table 1 offers a condensed overview of these waves. It borrows from the structure introduced for Kondratiev waves starting with the British industrial revolution, but extends it back to the 16th century, and emphasizes both the class dynamics distinct to each period, with an emphasis on the major groups of status-subordinated workers subject during to hyper-exploitation during this period and the

OF BUBBLES AND GOLDEN AGES (Repr ed. 2003); Carlota Perez, *Microelectronics, Long Waves and World Structural Change: New Perspectives for Developing Countries*, 13 *WORLD DEV.* 441 (1985).

³⁰ MINSKY, *supra* note 20; ROBERT Z. ALIBER & CHARLES POOR KINDLEBERGER, *MANIAS, PANICS, AND CRASHES: A HISTORY OF FINANCIAL CRISES* (Seventh edition ed. 2017).

institutional dimensions of the new techno-institutional regime characteristic of the wave, particularly as it relates to class power.

Technology has been a terrain of struggle between capitalists and skilled workers since at least the 17th century, when English weavers protested the importation of the Dutch Loom, arguing the dangers of technological unemployment to no avail. Even the quintessential technology of the British Industrial Revolution, Arkwright's water frame, was no more quantitatively productive than its primary substitute, Crompton's mule, while the mule produced substantially finer thread. The difference was that the mule required skilled male workers, while the water frame employed women and children with socialized lower wage expectation and no alternatives. Only when the powered mules were spinning thread five times the fineness of the water frame were Lancashire's leading industrialists forced to abandon the frame with its profitable low-wage workforce for the more productive mule with its more skilled and better organized workforce. The original Luddites, the Nottinghamshire frame knitters, were trying to get a bill through Parliament that would have forced clothiers who were replacing existing stocking frames with a new kind that, instead of producing finished seamless stockings produced flat pieces of textile that were then sown together to make a stocking and allowed them to displace skilled, organized, male frame knitters with unorganized women. The proposed bill would have required manufacturers to mark their stockings so as to distinguish the poor quality of inferior copies, regulate and regularize the measurement of lace for piece rates, prohibit payment in truck (goods, rather than cash), and give magistrates the power to regulate the rents on frames that weavers rented to produce at home.³¹ It was not fear of technology, but an effort to regulate the use of technology specifically designed to undercut the power of skilled labor, that drove the original Luddites. The historical evidence of power-seeking technological change is substantial, and reaches from loom designs in the 17th century, through a Derbyshire silk-throwing factory that employed hundreds of children and women decades before Arkwright's water frame (and was its direct inspiration), to iron molding in the Gilded Age, automobile production automation in the 1950s, and machine tool design in the 1960s.³²

³¹ I lay out the evidence to support this and describe other work documenting this power-seeking technological change dynamic in chapters 11 and 12 of BENKLER, *supra* note 26.

³² MAXINE BERG, *THE AGE OF MANUFACTURES, 1700-1820: INDUSTRY, INNOVATION, AND WORK IN BRITAIN* (2nd ed ed. 1994); Tine Bruland, *Industrial Conflict as a Source of Technical Innovation: Three Cases*, 11 *ECON. SOC.* 91 (1982); ROBERT W OZANNE, *A CENTURY OF LABOR-MANAGEMENT RELATIONS AT MCCORMICK AND INTERNATIONAL HARVESTER* (1967); THOMAS J. SUGRUE, *THE ORIGINS OF THE URBAN CRISIS: RACE AND INEQUALITY IN POSTWAR DETROIT* (First Princeton classics edition ed. 2014); DAVID F. NOBLE, *FORCES OF PRODUCTION: A SOCIAL HISTORY OF INDUSTRIAL AUTOMATION* (1st ed ed. 1984); LEVY, *supra* note 13; Barocas and Levy, *supra* note 13.

Wave	Technologies	Carrier industries	New class dynamics/Status-subordinated workers	Institutional dimensions (power-seeking technologies)
~1580s–1650s Dutch Republic England	Water-powered pumping and mechanical power; Fluyt; Defensible ship; East Indiaman Dutch loom	Land reclamation; Powered sawyering, manufacturing; Industrial-input-focused commodity agriculture Woolen textiles Commodity-exporting colonial plantations	Rural migrants into growing urban centers; Rising capitalist landowners; shrinking rural smallholding employment, converted to rural proletariat or urban new migrant workers	Short-term leasing of reclaimed lands shrinks smallholder farming; increased rural dependence on manufacturing Public utility style bread market regulation (Holland); Elizabethan labor and poor laws Capitalist revolutions vs. Absolutist consolidation efforts; liberal ideology and legality lead to first liberal oligarchic states
1650s–1710s Dutch Republic shifting to England	Agronomic innovation: Seed improvement; Norfolk four-crop rotation; seed drill; “factory in the field”	English grain; Dutch finance speculation Sugar and associated products	Enslaved Africans in the Caribbean; emergence of a rural population with partially-decommodified access to commons and wastes and opportunities for market-oriented cottage production, rising real wages and rising women’s wages in England.	Seed markets; profit-accounting in agriculture Conflicts over legal bases of enslavement; monopoly vs. open trade in the Atlantic and Indian Oceans;
1710s–1770s Britain; Midlands, Birmingham	Coke as primary furnace fuel; makes possible new chemical reactions; Rolling, stamping, and printing in metals, furnishings, and fabrics. Crucible steel and new machine tools Factory silk throwing	Metalware; home furnishings; textiles, as import substitution; guns and naval artillery	Women, children; unskilled demobilized soldiers; emergence of the permanent journeyman with no prospects of becoming master due to capital costs. Seasonal migrant workers in Holland (mostly German); Coerced labor in Java; enslaved labor in the Caribbean and Suriname—mostly squeezing profits not increased productivity;	Early sweated trades; combinations laws to suppress organizing by the new permanent journeyman class; Parliamentary Enclosures from the 1750s to suppress wages of the relatively (to other European societies) high real wage rural cottage producers. (multiple spinning and loom designs in silk, and wool aimed to harness children to work)
~1780s–1850s Great Britain (US South for raw materials)	Arkwright’s mill; Crompton’s mule; Whitney’s cotton gin	Mechanized cotton manufacture; Wrought iron	Skilled mule spinners organize; most skilled men stay out of industrial manufacturing Women & children in UK primary vector of proletarianization Enslaved Black workers in the US South Young women + immigrants in New England	Britain: Factory System Intensive violent suppression of labor organizing and universal suffrage in Britain US: Domestic slave trade; intensification of legal degradation of free Black people The Lowell system (Arkwright water-frame; cylindrical copper roller; small mules in rows; ring frame)

Wave	Technologies	Carrier industries	New class dynamics/status-subordinated workers	Institutional dimensions (power-seeking technologies)
1830s-1890s Great Britain, spreading to US and Europe	Steam engines Steam ships and railroads; Iron and Coal; Early American system of replaceable parts	Railway construction; Rolling stock manufacture; World shipping; Machinery	Irish Catholic immigrants in East and Midwest Black enslaved workers into sharecroppers and share tenants in South Chinese workers in West Women in the sweated trades	Integrated national markets through transportation and shipping networks South after Reconstruction: sharecropping (self-acting mule in Britain; shift to ring frames in US)
1880s-1940s US and Germany overtake Great Britain	Steel; Electricity; Chemistry; Finance	Heavy engineering; Chemistry and civil engineering; Electrical supply and distribution; Canned and bottled food	Until 1916 in the US: Southern European (Italians); Eastern & Central European (Jews) Women in the sweated trades After 1916-1924 (immigration exclusion): Black workers: the Great Migration & still sharecroppers in the South Women flooding low-end clerical work	Modern corporation; Consolidation and scale; Global markets Dense urbanized manufacture; sweatshops & tenements; displacing craft unionized workers with unskilled and semiskilled immigrants (capping machines; McCormick reaper iron-molding machines) Authoritarian corporatism emerges in Germany & Japan; social democratic institutions emerge in Denmark and Sweden
1920s-1980s US, spreading to Europe	Assembly line; Internal combustion engine; Petroleum and petrochemicals	Automobiles; Energy (oil); Electrical appliances	Treaty of Detroit constructed around the ideal male head of family earning family wage Black workers excluded Rosie the Riveter mobilized/demobilized Expansion of pink collar work	Fordism Institutionalized labor unions; Beveridgian social institutions; social democracy generalized; capital controls and international financial markets constrained; in US, early retrenchment; (1950s early automation; computerized numerical control machine tool designs)
1970s-2020s US, spreading to Europe and Asia Japan, spreading to Korea and China	Telecommunications; Computers; Information technology Robotics in manufacturing; advanced manufacturing techniques	Information and communications technologies; in rich countries—shift to services;	Retrenchment of social democratic institutions across the board; deepest in liberal oligarchic societies; dualization in corporatist societies; liberalized social democratic structures in the Nordics Women moving into services “flexibility”, leading edge of part-time & precarious work; immigrants; undocumented in US; guest workers in mainland Europe; MOSTLY: offshoring to harness workers in the Global South	Globalized supply networks; (harnessing the Global South) Financialization (disinvestment from labor and production capital) Fissured workplace

Table 1: Techno-institutional waves in capitalism from early modernity to the present

The fundamental tension is between developing and using technology that increases productivity by deploying more skilled labor, combining embodied and materially embedded knowledge and therefore shifting some power to the skilled workers over the joint returns, as compared to using a technology that may increase productivity no more, or even less, but provides distributive gains by employing a more vulnerable and replaceable workforce—unskilled workers, or workers who are status subordinated, historically women, children, and immigrants—whose low wages contribute to the bottom line more than the incremental increase in productivity offered by the higher-skill-dependent technology.

Moreover, the capital costs of some technologies can and have historically put smallholder artisanal producers in direct competition with capitalist enterprises, usually with the latter squeezing out the former. Such conflicts go back to the use of water-powered mechanical fulling by Genoese capitalist clothiers who offshored cloth finishing to the English West Country in the late 14th century, circumventing Flemish woolen craft manufacturers constraints, through Dutch wind-powered sawyering crucial to the construction and ship-building industries of the Dutch Republic in the late 16th and early 17th century, Wedgwood's ceramics on English craft potters in the 18th century, Singer's sewing machines on tailors in the 19th century, or the dramatic rise of large scale retailers at the expense of shopkeepers associated with the impact of barcodes, optical scanners, and container shipping in the 1970s on the manageable size of inventories and customer bases.

Nonetheless, the impact of technologies on class structure and class power is not deterministic. Usually, technologies can be instantiated in different ways, based on the inherited institutional structure and class power into which they are introduced. Capitalists may seek to leverage a given technological change to obtain greater power over workers or smallholders, but they are not always successful. While the introduction of the self-acting mule in Lancashire did weaken the mule-spinners union and reduce the skills content of the work, the new self-acting mule-minders were still relatively skilled and well-organized, and were able to maintain greater control over wages and work conditions in Lancashire than were the women spinners in Glasgow, where better-organized capitalists succeeded in deploying smaller mules yoked in rows and operated by unmarried women supervised by a handful of men at lower wages; or in Lowell, where the cotton mills deployed the Glasgow configuration from the start, before a powerful male spinners union could form, and later replaced the Yankee farm girls using those smaller mules with immigrant women working ring frames when the former began to organize themselves. Similarly, Danish land reforms in the early nineteenth century meant that by the third quarter of that century, when creamery centrifuges were introduced, transforming dairying, on which the Danish economy's integration into the global capitalist network depended, smallholder farmers had already established stable smallholder property and a thriving dairy export sector to Britain's growing population, backed by cooperative credit institutions. As a result, the high capital costs associated with the new centrifuges resulted in cooperative ownership structures, despite early mover efforts from investor-owned dairy operators. By contrast, the same exact technology, introduced in Ireland where capitalist relations had long prevailed in the dairy export industry, the new creameries merely reinforced the incumbent model

of capitalist dairy processing with asymmetric power over dependent smallholder producers.³³ While similar comparative studies are rare, the phenomenon recurred in the different instantiations of barcodes, optical readers, and grocery chains,³⁴ and we return to it in the discussion of robots in part III.

The bottom line is a simple one. Because knowledge is a crucial component of production, control over knowledge in production is a crucial dimension of class definition and class power in social relations of production. The specific combination of human embodied, machine embedded, and symbolically codified knowledge in any given regime will have significant effects on the distribution of class power in that regime, and patterns of technological adoption reflect the strategic investments of capitalists, workers, both skilled and unskilled, smallholders, the professional and managerial class, and the clerical classes in pushing the production system in a direction that leaves them with more power in the new regime. The technological affordances, however, neither develop, nor are instantiated once introduced, independently of the institutional structure out of which they come or into which they are introduced. Instead, both the institutions and the technology are arenas of struggle over the definition of class lines in the new wave of production and the relative power each class has in social relations of production under the new techno-institutional regime.

D. Class relations and macroeconomic stability

Class relations in a given society during a given techno-institutional wave directly affect macroeconomic stability. To make money individual capitalists must sell to someone—and for a capitalist society to grow, they must do so not only as individual enterprises, but as a class. That is the problem of aggregate demand. In standard macroeconomics models there are four possible sources of aggregate demand: consumption by households, investment by capitalists (in real economy capital, not in financial claims which are necessarily about distribution, not production) and households (investments in real estate), government real expenses—salaries or battleships, not transfer payments (because these show up in household consumption)—and net exports—that is, exports minus imports (because the imports show up as consumption). Household consumption in market society is overwhelmingly paid for by labor income, primarily wages. There are two opposing patterns to consumption—broad-based and elite-based—with the former reflecting demand by the middle seven or eight deciles of income distribution and the latter reflecting spending by the professional and managerial classes (which comes from wages, which also account for some of the spending of capitalists) as well as profits taken by capitalists and rentiers. In other words, there can in principle be a (a) broad-based consumption regime that builds on high wages across the economy and (b) a pyramid consumption pattern that depends on a high-profit regime built at least partly on suppressing the wage bill.

There are two potential constraints on household consumption as a driver of growth. The first, relatively trivial constraint, is the micro-economic one: What would make households buy more

³³ I review the literature on power-seeking technological change primarily in chapter 11 of *The Global Origins of Capitalism*, and the Danish/Irish case in particular, including the specific drivers of early 19th century land reform in Denmark that led to the specific institutional political economy behind this change, in chapter 15 of the book.

³⁴ Bartholomew C. Watson, *Barcode Empires: Politics, Digital Technology, and Comparative Retail Firm Strategies*, 11 J. IND. COMPET. TRADE 309 (2011).

new stuff if they can satisfy their desires and needs with the old? Neoclassical models ignore this, treat demand as exogenous, and assume that “more is better” is the usual state of affairs. The actual solution is that manufacturers create and drive demand toward what they can profitably sell. As Lyons silk producers discovered in the 1670s and Wedgewood perfected in the 1760s, as Nicolas Barbon theorized in 1690 and Thorstein Veblen reinvented in 1899, status anxiety, keeping up with the Joneses—that is, consuming as a means of competing for status and social standing—appears to be a source of insatiable demand, and one that capitalist producers have been leveraging consciously since the 17th century.³⁵ This solves the microeconomic problem of insatiable demand endogenously, rather than simply assuming it exogenously. Doing so rejects the exogeneity of consumer choice and the intellectual coherence of welfare economics as a normatively neutral conceptual apparatus. While normatively problematic if one wishes to insist that welfare economics is an apolitical approach, the microeconomic problem of demand increasing to meet supply is a “solved problem” descriptively. Satiation of consumer demand will not limit the expansion of capitalism. There is no repose in the time of “our grandchildren.” Marketing can, and has since the late seventeenth century, reshaped demand enough to avert crises of overproduction in the long run, and will continue to do so until we run up against the carrying capacity of Earth. That is why the leading demand manipulation companies like Meta and Google have the valuations they do, and is why the use of AI for increasingly refined demand manipulation is a certainty.

Less easily resolved is the macroeconomic question: who will buy the expanding output of increasingly productive processes so as to sustain their profitability? If the answer is household consumption, then the question is where the households will get the money to spend on the things that producers produce? If it is investment, then where will the profits come from to fund it? And so too for government expenditures and net exports. As far as domestic household income from which to make these, the answer in capitalist societies is wages from labor for the working, clerical, and professional and managerial classes, market sales income for smallholders, and income from profits for the rentiers and capitalists. Alternatively, in an interconnected world, it can come from similar sources in other societies through net exports. How each of these is leveraged to answer the question of “where will aggregate demand come from” defines the growth pattern and trajectory of a society, and endogenously introduces sources of stability and instability to that social order and its capacity to sustain growth, and hence its social, economic, and political trajectory.

In the recently prominent Growth Models Approach in the field of comparative political economy, a “wage-led economy” is one where, as a matter of pure descriptive statistics, household consumption is the largest component of growth and reflects spending out of wages.³⁶ The standard mid-twentieth century social democratic theoretical assumption was that working class households spend more of their money than do rich households, because there is only so much money a person can spend on \$5000 bottles of wine or private jets. As a result, high working-class wages lead to high aggregate demand, which then provide capitalists a demand sink into which to sell their

³⁵ *Global Origins* chapter 11.

³⁶ Lucio Baccaro & Jonas Pontusson, *Rethinking Comparative Political Economy: The Growth Model Perspective*, 44 *POLIT. SOC.* 175 (2016).

products.³⁷ By contrast, in this Growth Models literature a “profit-led economy” is one where wages are too low to sustain this kind of broad-based consumption-driven economy, and instead profits lead the economy—primarily as business investment, but also (though this is harder to measure) as luxury spending by wealthy households whose income comes from profits (rentiers and capitalists), with the latter providing the trickle-down demand for businesses ranging from domestic servants and dog-walkers to luxury yachts and thousand-dollar menu items at fancy restaurants. Increasing capitalist profits so that investment and rich household consumption drive economic growth is the foundation of “trickle-down” economics.³⁸ Although the Growth Models literature does not focus on government spending, the basic components of aggregate demand tell us that there can also be government-expenditure-led growth, but that requires some source of government income—in a profit-led economy this can come from selling public debt into financial markets to the profit-reaping classes (increasing the salience of Minskyian endogenous instability); in a wage-led economy it can come from broad-based income taxes; in a mineral rich country it can come from charging the extractive sectors, as is common in petroleum producing states that then engage in substantial domestic expenditures; or in a state with an aggressive military strategy it can come from military spending, financed by borrowing from the rentier class against the promise of acquiring riches in terms of tributes (in money or natural resources) or in asymmetric trade relations (enabling domestic producers to gain from improved terms of trade). Britain’s growth in the long 18th century reflected a large component of this latter strategy as it emerged to the role of global hegemon, as has American military Keynesianism since World War II during the era of *Pax Americana*.

A high-road, broad-based wage-led growth model represents a cooperative solution to the problem of how to maintain a dynamically growing and stable society. Achieving and sustaining that cooperative solution requires an institutional structure that forces most firms in a society to adopt some version of what business school literature calls a “high-commitment, high-performance” strategy.³⁹ Profitability in such a strategy is driven by productivity growth enabled by intensive, high-quality contributions from labor in exchange for high wages consistent with productivity growth and a cooperative, dignity- and creativity-enabling environment. Living in such a society does not mean every single organization adopts such a strategy, any more than it means that in a profit-led model all firms necessarily adopt exploitative low-road strategies. But it does mean that most employers have no choice but to adopt such a strategy, and as a result in fact invest in technological growth oriented toward skills-enhancing technologies, organizational strategies that depend on knowledge and skills intensification to increase their productivity and profitability, and a high-road strategy, because it is hard to attract labor to any other strategy, and

³⁷ Amit Bhaduri & Stephen Marglin, *Unemployment and the Real Wage: The Economic Basis for Contesting Political Ideologies*, 14 CAMB. J. ECON. 375 (1990); WAGE-LED GROWTH: AN EQUITABLE STRATEGY FOR ECONOMIC RECOVERY (Marc Lavoie & Engelbert Stockhammer eds., 1st ed. 2013 ed. 2013).

³⁸ WAGE-LED GROWTH, *supra* note 37.

³⁹ MICHAEL BEER, RUSSELL A. EISENSTAT & NATHANIEL FOOTE, HIGH COMMITMENT, HIGH PERFORMANCE: HOW TO BUILD A RESILIENT ORGANIZATION FOR SUSTAINED ADVANTAGE (1st ed ed. 2009); REBECCA HENDERSON, REIMAGINING CAPITALISM : HOW BUSINESS CAN SAVE THE WORLD (2020); ZEYNEP TON, THE GOOD JOBS STRATEGY : HOW THE SMARTEST COMPANIES INVEST IN EMPLOYEES TO LOWER COSTS AND BOOST PROFITS (2014); THE FIRM AS A COLLABORATIVE COMMUNITY: RECONSTRUCTING TRUST IN THE KNOWLEDGE ECONOMY (Charles C. Heckscher & Paul S. Adler eds., 2006).

because labor law, social insurance, unemployment and wage policies, and banking laws create a mutually-reinforcing institutional framework best suited to this strategy.

That the strategy can be stable and self-stabilizing does not mean that it is not susceptible to defection and destabilization. For one, some firms in such an economy may try to increase their own profitability by producing to sell into the broad population with high-wages, while adopting as their own production strategy a low-wage, lower productivity but nonetheless high-profit strategy. Such a firm could employ workers who fall outside the institutional matrix that supports the model and who cannot get employment in the high-wage enterprises, often immigrant or other status-subordinated workers. The narrower the base of social insurance, and more tied it is to already-existing employment in the core high-road firms, the easier it is for such “defectors” to establish their low-road strategy businesses. If enough firms follow this strategy, the high-wage model unravels. In dualized economies where there is a core sector, usually export-focused, these firms diverge into two distinct market segments—the core, usually manufacturing (often export-oriented) sector that is high-wage and provides corporatist (firm or sector-specific) social insurance and services, and a periphery that provides low-productivity, low-cost services and basic goods that reduce the wage bill of the core sector. This was the institutional evolution of corporatist authoritarianism in pre-World War II, and corporatist oligarchic in postwar, Germany and Japan.

Symmetrically, in the high-wage, high-employment economy workers can develop enough bargaining power to demand wage increases that outpace their productivity growth. Faced with this state of affairs, the “profit squeeze,”⁴⁰ investors begin to pull out of the investments, because their profitability cannot keep up with wage demands that increase faster than productivity growth. They begin to refocus on low-road strategies that use low-wages in a low-growing productivity model as their source of profitability: on technology that allows them to replace skilled and juridically high status workers with unskilled and status subordinated workers; on institutional transformation to weaken labor, whether through increased unemployment, reduction in social insurance (to make the alternative to wage work increasingly unpleasant), or changes in labor law that make it harder for workers to cooperate with each other; or they might refocus their efforts on exports or offshore production to countries where labor is weaker or cost of living lower, so that they can pay “efficiency wages” at a lower level that does not cut into their profits as much. Alternatively, they stop investing productively altogether, retain their profits from existing operations, and use them to engage in speculative financial operations. This is what happened in the Dutch Republic in the 18th century, in Britain after the Long Depression of the late 19th century, and in the United States since the 1970s.

Preventing defection by either capitalists or workers from this cooperative high-road strategy requires institutions that constrain these paths of defection. It requires that labor be organized and strong, partly to discipline employers to keep the wages growing with productivity as well as

⁴⁰ Stephen A. Marglin & Amit Bhaduri, *Profit Squeeze and Keynesian Theory*, in THE GOLDEN AGE OF CAPITALISM: REINTERPRETING THE POSTWAR EXPERIENCE 153 (Stephen A. Marglin & Juliet B. Schor eds., 1992), <https://doi.org/10.1093/acprof:oso/9780198287414.003.0004>; Samuel Bowles & Robert Boyer, *A Wage-Led Employment Regime: Income Distribution, Labour Discipline, and Aggregate Demand in Welfare Capitalism*, in THE GOLDEN AGE OF CAPITALISM: REINTERPRETING THE POSTWAR EXPERIENCE 187 (Stephen A. Marglin & Juliet B. Schor eds., 1992), <https://doi.org/10.1093/acprof:oso/9780198287414.003.0005>.

denying workers to defecting firms, but equally important, to discipline the labor formations in the various sectors of the economy to enforce wage solidarity across sectors, so there will not be beggar-thy-neighbor conflicts among different sectors and subsections within the working classes (e.g., manufacturing vs. service, export-oriented vs. domestic, etc.). It also requires that capitalists be well-coordinated and able to enforce broad wage-setting standards that prevent firms following low-road strategies from undermining the cooperative solution.

While in theory these institutional arrangements may come from union or employer association rules alone, in actual historical practice the cooperative solution only prevailed for meaningful periods where there was effective state coordination and, where necessary, enforcement and mediation to anchor high-road institutions. Moreover, the high-road cooperative solution is strengthened in a society in which the market imperatives on labor are weaker, and hence the pressure on marginal employees to accept jobs defecting employers offer on poorer terms are low. A high level of social services, unemployment insurance, retraining to new jobs, and social insurance, all of which require a large and effective public fisc, operate to restrict the opportunities for strategic defection by low-road employers, simply because the alternatives to accepting employment on bad terms are not as dire as they are in societies without these means to partially decommodify access to fulfillment of basic household needs and wants. This is why Sweden and Denmark are able to continue to sustain a balanced growth model even after liberalization, combining export with wage led growth, while neither the corporatist nor the liberal oligarchic societies have been able to do so since the 1980s.

Wherever the cooperative high road solution falters because of either the internal tensions and modes of defection by capital or overreach by labor, or because of changes in the global network of production and consumption, or in those societies where it never took off because of the history of that society's institutional political economy, as was the case in many post-colonial countries, growth can come from any combination of drivers: profit, exports, debt, or government expenditures. Each of these comes with its own endogenous instability drivers, and none has succeeded in sustaining growth over time except for export-led strategies or institutional arbitrage services (tax shelters and shady banking) in some small economies. Profit-driven growth increases as capitalists increase profits by depressing the wages that would otherwise fund household consumption. The direct effect on productivity on the supply side is that this strategy reduces the labor-effort-skill component of productivity growth in order to keep wages down. The macroeconomic effect is that it reduces aggregate domestic demand by the reduction in household consumption associated with lower wages. The lower productivity growth component is hard to overcome with a purely profit-driven solution, because any level of technological productivity growth must be compared to the counterfactual of technological change designed to harness higher quality labor ability and effort at higher wages. This problem of business "sabotage" is integral to the Red Queen Dynamic, and can only be overcome in a high-wage, high-productivity growth state that sufficiently restrains the low-road path to profitability. The macroeconomic challenge of lower aggregate demand can, however, be postponed through one of several approaches—none of which is itself stable for more than a decade or two because of its endogenous destabilizing dynamics.

A historically important solution to aggregate demand shortfalls in the absence of wage-led growth was export-driven growth. As long as early capitalist societies were islands in an ocean of non-capitalist societies, their profit-reaping classes could profit by selling to the political military and ecclesiastic elites of these other jurisdictions, to the urban craft manufacturers and merchants of those societies, and to their own profit-reaping classes and political military elites. As more countries shifted to capitalism over the course of the nineteenth and twentieth centuries, however, a solution that depended overwhelmingly on exports became harder to sustain, particularly for the wealthy countries who were among the earlier movers. Fundamentally, it is impossible for all countries in the world to adopt export-led growth because it is impossible for all countries to be *net* exporters. For any country to be a net exporter, there have to be other countries that are net importers. One important dimension of capitalist military history is precisely the imposition of asymmetric terms of trade requiring weaker countries to absorb the exports of the militarily and economically more powerful country.

Another temporary and destabilizing short-term solution to failures of aggregate demand is expansion of credit that enables households to use debt to continue to consume at higher levels than their wages can support. Here, firms depress wages to maintain or increase profitability, but investment and rich-household expenditures alone are not enough to provide an adequate sink for the production output. Instead, the increasing profits that are not invested become the source of credit through the intermediation of the financial sector, funding household consumption based on the promise of future earnings that the wage stagnation that led to the collapse of wage-led growth assures will never come. In other words, by definition an economy will only really be debt-led, rather than wage-led, if finance is already in at least the second Minskyian stage—the “speculative finance” stage—at least with respect to household credit. From there, the need to attract and retain investors against competitors who show higher returns, however unsubstantiated by real economic value, accelerates the process into Ponzi finance.⁴¹ The most obvious example of this model is the United States since the 1990s, where mortgage and credit card debt-fueled consumption (and household investment) sustained aggregate demand well beyond the possible wage growth of a working population faced with intensified cost-cutting, welfare benefit cuts, and increasingly precarious work lives could hope to pay off. The combination of a low-wage, high-profit domestic economy with credit-driven consumption, and profit-driven offshoring to reduce cost of imports led to sustained, arguably increasing standards of living even in the low-wage, high-profit equilibrium, because the wage suppression in the offshore production sites overcame the stagnant and uncertain wages. This is the US economy with ballooning credit card debt plus Walmart. The expanded use of collateralized debt obligations and mortgage-backed securities in the 1990s contributed to obscuring the speculative and eventually Ponzi nature of these credit facilities, and provided an avenue for financial investments of the profits captured from the strategy until it collapsed in the Global Financial Crisis.

The destabilizing effect of this solution emerges as a side effect of the endogenous instability of financial markets. But this is not the only source of instability in the political economy of societies that depend on such profit- and speculative credit-led model. Rather, it is the inevitable

⁴¹ MINSKY, *supra* note 20.

anomie or alienation of large swaths of the working population that arises from the daily experience of having to meet credit payments in the teeth of poor wage prospects, and the day-to-day experience of degrading wage/labor conditions by employers whose profit-led strategy makes them see their employees as cost centers to be squeezed to the lowest possible cost. Even if here or there a high-road employer can provide examples of inspiring labor relations, the question is not what is possible in discrete examples, but what model pervades the lives of most working families in a society at a given historical period. Just as the high-road wage-led institutional political economy sets the baseline institutional constraints on defection and makes cooperation the default strategy for most firms, so too does the low-road institutional structure. As a society, on an economy-wide basis, these high-road strategies do not become generalized when the institutional political economy of a society makes it easier and cheaper for firms to aim for a profit-led model, with or without the boost to GDP offered by an export-based or credit-sustained consumption element.

The anomie driven by a precarious, debt-dependent daily grind and a lack of meaningful dignified work is far harder to deal with by pure manipulation of sources of income. It feeds the deep disaffection of broad swaths of the populations of many of the wealthiest countries in the world with their elites and the democratic process that brought them to government. It undermines the legitimating force of the political formations, particularly as the structural dynamics of the system appear resilient to changing party-political processes, and “voting the bums out” stops being a sane strategy unless the challenger promises a fundamental rejection of the entire prevailing system—whether it is Brexit, Trump, or the Euroskeptic parties. This is the most fundamental challenge of the profit-led growth strategies, one that none of the spending of elites, stagnant investment of companies, or skyrocketing stock market returns can solve.

The upshot of this understanding of the relation between wages, employment, and productivity growth is that the interaction between power and productivity is not a micro-economic dynamic in the decisions of any given firm, aggregated to the level of society at large. Rather, it is a basic tradeoff in the institutional structure of capitalist societies. There is one relatively narrow path that drives towards both dynamic productivity growth and sustained aggregate demand growth. It is the model driven by the consumption made possible with broad-based high wages growing in line with productivity growth, coupled with broad social investment in knowledge intensification, and in social insurance to make low-road competition harder to sustain. From this understanding, it is hardly surprising that Sweden and Denmark are not only among the most egalitarian societies in the world, but also among those ranked as more economically free than the United States.

II. East from Eden—Deindustrialization and its Discontents

A. From the Golden Age of Capitalism to Neoliberal retrenchment

The last thirty years of the wave of Fordist mass production—where mass production, automobiles, and home appliances led the wave—coincided with the so-called “Golden Age of Capitalism.” The moniker reflected the fact that this was the only time in the long history of capitalism that strong growth and productivity growth were coupled with rising standards of living for the working classes across most of already-industrialized capitalist societies—in Western Europe, North America and Oceania, and Japan.

The vast devastation left by World War II required expansive reconstruction in Europe and Asia. Wartime production systems and collective mobilization left both institutions and ideology demanding and making possible solidaristic recovery. Several institutional planks characterized all the wealthier capitalist societies that shared in the prosperity of this period, although with variations that reflected their inherited institutional political economy from their 19th century (or in the case of Britain and the United States, 17th and 18th century) adoption of capitalism. These included, above all, (a) a tightly regulated banking and financial system, combined with well-coupled international monetary coordination backed by capital export controls in all wealthy countries, limiting private international financial flows; (b) strong labor unions; (c) a large role for government in wage-setting institutions and managing unemployment; (d) expanding social insurance programs encompassing large portions of the population; (e) public ownership or close regulation of monopoly (formal or de facto) public utilities in energy, transportation, and communications. The tight regulation of finance prevented escape from productivity-enhancing investments in that direction, while also constraining the path of jurisdictional arbitrage to seek out sources of purely distributive profits. Strong unions contained the ability of capitalists to take low-road strategies based on squeezing labor, as did relatively generous social insurance and clear wage determination institutions that led to at least sectoral, or even more broadly national, solidaristic wage setting. How broadly these applied in the population depended on the institutional political economy these countries had developed in their own transitions to capitalism, resulting in Esping Andersen's differentiation of the "Three Worlds of Welfare Capitalism."⁴²

Least generous to begin with, and first to unravel, were these institutions in the liberal oligarchic republics that were first to adopt capitalism—the United States and the UK. These were followed by the corporatist oligarchic republics, most prominently Germany and Japan, where corporatist social insurance, knowledge, and wage determination institutions resulted in dualized economies, as they had been since the two countries intentionally implemented their own model of authoritarian corporatist capitalism in the 19th century. The smallest group were the Nordic social democracies, again reflecting their own institutional struggles in their transition to capitalism in the late-nineteenth, early-twentieth centuries.⁴³

For our purposes here, what matters most was the dimension of knowledge in production. Fordist mass production was based on prioritizing technologically embedded and codified

⁴² GÖSTA ESPING-ANDERSEN, *THE THREE WORLDS OF WELFARE CAPITALISM* (1990).

⁴³ The comparative political economy literature on this variation is vast. At the height of neoliberalism, the most prominent school was "varieties of capitalism," primarily anchored by American political scientists, and the "power resources" theory, primarily anchored by sociologists working in Nordic social democracies. After the Global Financial Crisis a "growth models" approach, mostly integrating Neo-Kaleckian models with French Regulation school insights to provide a distinctive, though at present largely descriptive, approach. *VARIETIES OF CAPITALISM: THE INSTITUTIONAL FOUNDATIONS OF COMPARATIVE ADVANTAGE* (Peter A. Hall & David W. Soskice eds., 2001); Torben Iversen & David Soskice, *Distribution and Redistribution: The Shadow of the Nineteenth Century*, 61 *WORLD POLIT.* 438 (2009); Baccaro and Pontusson, *supra* note 36; M. (MARC) LAVOIE, *POST-KEYNESIAN ECONOMICS: NEW FOUNDATIONS* (2014); *WAGE-LED GROWTH*, *supra* note 37. The best synthesis of the varieties of capitalism and three worlds of welfare capitalism, as these applied to the fifth Kondratiev retrenchment, is KATHLEEN ANN THELEN, *VARIETIES OF LIBERALIZATION AND THE NEW POLITICS OF SOCIAL SOLIDARITY* (2014). I provide a macro-theoretical explanation for this variation and anchor it in the observed variation in the 19th century roots of the adaptations of the relevant societies in chapters 15 and 16 of BENKLER, *supra* note 26.

knowledge in the production process. It combined the Taylorist model of scientific management, the ambition to codify to the greatest extent possible all the necessary motions of workers, while Ford's assembly line embedded this optimized codified process in a mechanical process that could set and monitor the time and sequence of human operations. The broad high school movement in the United States produced a flow of young men with sufficient literacy and numeracy skills to be incorporated into this system that sought to use embedded and codified knowledge, rather than embodied knowledge. The different structures of initial and continuing vocational training in various European countries and Japan, in turn, played a large role in explaining the differentiation described by the varieties of capitalism literature. Fordist production required large numbers of shopfloor foremen rising from the semi-skilled working class, and process engineers from the professional class, as well as large numbers of clerical-class workers, bifurcated into supervisory clerical jobs as "white collar," which remained male and paraprofessional or professional, and line-level "pink collar" jobs in a gendered division of labor filled mostly by unmarried women. The result was a rising new "middle class" of workers, whose career trajectory from young adulthood was often tied to a single employer, particularly outside the US, with union representation producing wage gains and seniority-based raises, with clerical wage bill suppressed by using gender norms to suppress wages. This broad-based compression between the semi-skilled working classes and the clerical classes, with the paraprofessional and professional classes, backed by significant labor power to demand a significant share of the income sustained broad, wage-led growth translated through household consumption in the United States as the global demand sink while the European and Japanese economies recovered from the war. Countries with more social democratic structures were able to combine net-export-led with wage-led household consumption as bases for balanced growth, while those countries that depended on net export led growth, Germany and Japan in particular, developed along the dualized structure they had already developed before World War II in their own early process of industrialization.

For all the glorification of the "Glorious Thirty," the peak of this regime lasted primarily during the 1950s. In the United States, the rebellion of capital began as early as the Taft Hartley Act of 1947, and with it the effort to escape to newly-legalized "right-to-work" states with antiunion institutional structures. It continued in the 1950s and 1960s with the strategic automation of production in the most heavily unionized segments.⁴⁴ In Europe, the model lasted longer because the reconstruction from the war sustained growth while the expansion into export markets dominated in the late 1940s by American manufacturing provided more than ample expansion room. In Japan, the system of lifetime employment, internal training, and dualized structure of a core of industrial production supplied by a periphery of fragmented small-scale suppliers was established by the *zaibatsu* as anti-unionization institutions in the 1920s. These structures were re-established in slightly modified form in the postwar era after the newly-reknit *keiretsu* and the American occupation administration's anti-Communist legislation quashed a brief and unsuccessful effort to create independent unions.

The recovery of postwar European and Japanese manufacturing, coupled with the gains labor made, put a significant squeeze on profits of American capitalist enterprises. They began to look

⁴⁴ SUGRUE, *supra* note 32.

for ways to escape the shrinking profits regime. Most directly, American manufacturing share of global exports shrank from 25.6% in 1948 to 16.7% in 1966 and only 13.4% by 1980. Meanwhile, Mainland Western Europe's share rose from 13.4% in 1948 to 30.6% in 1966 and 31.6% in 1980, and Japan and the East Asian "tigers" (Korea, Taiwan, Hong Kong, and Singapore) rose from negligible in 1948, to 7.4% in 1966 and 12.3%, almost as much as the United States, in 1980. The American balance of payments consequences, compounded with increasing government debt to cover Viet Nam war expenditures and the Great Society social spending, led to increased inflation and pushed Richard Nixon to suspend gold convertibility of the dollar, bringing the Bretton Woods system of postwar international monetary system to collapse. This was soon followed by financial deregulation both domestically and in terms of international financial flows. Jack Welch's legendary position as a superstar CEO of GE marks the primary pathways of capitalist escape during this period—over the course of the 1980s and 1990s GE Capital came to account for 60% of GE's profits as a mark of the role of financialization, before its collapse in the Global Financial Crisis, while his ruthless annual workforce downsizing and outsourcing significantly reducing labor costs and relied increasingly on precarious, unskilled labor with no training pathway within the company as a mark of the role of workplace fissuring.

In the United States, financialization and fissuring increased the role of the professional and managerial class doing cognitive work, both in finance and control over intellectual property and in managing the orchestration of fissured production networks, while reducing the need for, and prevalence of, the skilled working class foremen, the process engineers, and the workers who possess embodied knowledge about the specific production system and context of the particular employer. The precarious nature of employment, the profitability of placing contractors within the fissured workplace in cutthroat competition against each other, squeezed the share of embodied knowledge in production and reduced the contexts and value of acquiring firm-specific knowledge in production. The result was high profitability with low productivity growth, and a shrinking capacity to compete in world markets for manufactured goods as the centers of excellence in manufacturing shifted more decisively to Europe and East Asia.

As Thelen described in her comparative political economy of liberalization, the same was not true in Europe.⁴⁵ There, institutional constraints on financialization combined with longstanding institutionalized vocational training systems resulted in continued skills upgrading in industry. While European manufacturing exporters, like the United States, lost ground to Asian manufacturers, the increased skills component enabled them to shift to capital goods exports to those countries as well as focusing on high-end consumer products able to absorb the higher wage bills. Nonetheless, these European societies also differentiated internally. Germany and similar Christian Democratic societies remained competitive at the high end of industrial manufacturing by leaning into their dualized class structures—investing in vocational training and social insurance for the relatively stable, high-embodied-knowledge core, while maintaining a more precarious services sector left outside the training and social insurance institutions that supported the core. Denmark and Sweden, by contrast, saw expanded investment in continuing vocational training and educational upgrading, expanded the scope of their universal programs to cover white

⁴⁵ THELEN, *supra* note 43.

collar and professional workers as well, resulting in a distinctive form of dynamic stability different from either the US or UK, on the one hand, or Germany, Austria, and similar systems, on the other hand. Thelen's careful analysis offers more fine-grained differentiation and overlaps than this basic tripartite typology covers, but for our purposes what is important is that she identifies differentiation along the axis of knowledge in production, particularly the development and leveraging of embodied knowledge in workers who as a result retain relatively high power in relations to capitalists and the professional and managerial class, as a crucial vector of institutional differentiation among the various societies at the cutting edge of the global technological frontier.

B. Bifurcated development: Computation and information technologies vs. non-import-exposed low-productivity services

The most recent techno-institutional wave is conventionally understood as anchored by information and communications technologies.⁴⁶ We have been talking about “the computer revolution” or “the knowledge economy” at least since the early 1960s, adding terms like “postindustrial society” or “information economy” since the 1970s.⁴⁷ Yet as Robert Solow famously quipped in 1987, “we see the computer revolution everywhere except in the productivity statistics.”⁴⁸ Solow was writing specifically about the United States, focusing on the paradoxical mismatch between the leading role American companies and startups took in developing computers and information technologies and the lack of meaningful productivity growth in the American economy as a whole. A quarter of a century later, Acemoglu, Autor, and their co-authors reinforced Solow's observation, disputing intervening economists' efforts to dispel “Solow's Paradox” at least for the United States.⁴⁹ They showed that there was a brief period of productivity growth in industries that used more computers and information technology-enabled manufacturing tools and processes in the 1980s and 1990s relative to industries that did not, but that, beginning in the early 1990s, IT-intensive firms reduced output and employment faster than they increased their productivity. They were not expanding output while shedding workers, the standard story behind “the robots will eat all the jobs.” They were also shrinking output, while dropping jobs at an even faster rate during the 1990s, a process that slowed down in the 2000s but did not bounce back toward growth.

Solow's paradox becomes easier to understand if one shifts from a framework that sees technology and technological growth as an exogenous event that should operate in free market economies as a uniform boost to productivity to a framework that sees productivity and power as potential substitutes, as well as potential complements, depending on the institutional framework. Table 1 lists the top twenty companies in the United States by number of employees globally and market capitalization, in each case with the company's primary lines of business.⁵⁰ What becomes

⁴⁶ FREEMAN AND SOETE, *supra* note 29; FREEMAN AND LOUÇÃ, *supra* note 29; PEREZ, *supra* note 29.

⁴⁷ JAMES RALPH BENIGER, *THE CONTROL REVOLUTION: TECHNOLOGICAL AND ECONOMIC ORIGINS OF THE INFORMATION SOCIETY* 4–5 (5. print ed. 1997).

⁴⁸ R. Solow, *We'd Better Watch Out*, 36 N. Y. TIMES BOOK REV. 36 (1987).

⁴⁹ Daron Acemoglu et al., *Return of the Solow Paradox? IT, Productivity, and Employment in US Manufacturing*, 104 AM. ECON. REV. 394 (2014).

⁵⁰ Number of employees derived from CompaniesMarketCap's “Top publicly traded American companies by number of employees,” which aggregates the 10-K figures. The counts are for global employees, and typically include full time and part time workers, as these are usually reported in company 10-K disclosures. Market

clear is that during the present wave the United States, still the leading global economy at its start, featured a bifurcation—a shift to information technology *design and software*, rather than manufacturing, and finance, at the higher end of what stock markets sought and valued, while most workers were forced to accept jobs in non-import-exposed retail chain stores combining logistics and place-based delivery of finished goods (consumer products, grocery) and services (food & hospitality, healthcare services), through outsourcing firms in fissured workplaces. The consumer goods, in turn, were mostly manufactured in China and elsewhere in Asia, with the share of imported consumer products (excluding grocery and health products) that Amazon and Walmart sold in the years before the pandemic disrupted global networks likely around 65%-85%.⁵¹

The most profitable sectors were staffed by professional class workers—finance professionals, software developers, biopharmaceutical researchers, and managers who developed expertise at orchestrating logistics networks and squeezing the most out of non-professional workers. It was a wave during which the PMC in the United States used its embodied knowledge, focused on high-end cognitive work in symbolic manipulation, to capture most of the market value of growth. Although growth was lower than in the “Golden Age,” and productivity-growth more persistently so yet, the PMC was able to increase its own wealth dramatically by grabbing a larger share of that slow-growing economy. The working classes, most starkly unionized men who had been the primary working class direct beneficiaries of Fordist broad-based growth, lost their path to economic security and respectability, while married women in general, and women of color in particular, were increasingly recruited to fill in the new, degraded terms of labor in the rising services sector. Apologists for the declining wages of the American working classes spent a quarter of a century developing and rearranging data to assert that the bifurcation between the PMC and the working classes was a natural result of “skills-biased technical change,” assuming markets actually reflected marginal contribution to value.⁵² By the mid-2010s even the labor economists whose most ingenious work kept the model going in the teeth of decadal changes in data shifted

capitalization is price × total shares outstanding at the observation time. The list above is a snapshot from CompaniesMarketCap’s U.S. ranking page in September 2025.

⁵¹ Walmart is the leading importer of containers into the US. While the large share of Walmart’s grocery business leads its overall share of imported goods somewhere between 30%-45%, its General Merchandise category—electronics, toys, apparel (which is 98% imported)—plausibly range from 65%-85%. Amazon, whose increased share of focus on third party sellers includes heavily Chinese sellers, likely ranges between 65%-85%. All these are author’s estimates based on publicly available sources. There is no single authoritative source for making this estimate more formal.

⁵² L. F. Katz & K. M. Murphy, *Changes in Relative Wages, 1963-1987: Supply and Demand Factors*, 107 Q. J. ECON. 35 (1992); Autor, H. DAVID, LEVY FRANK & J. MURNANE RICHARD, *The Skill Content of Recent Technological Change: An Empirical Exploration*, 118 Q. J. ECON. 4 (2003); Daron Acemoglu & David Autor, *What Does Human Capital Do? A Review of Goldin and Katz’s The Race between Education and Technology*, 50 J. ECON. LIT. 426 (2012). The challenge was that in the 1980s, wage gaps increased monotonically with educational attainment measured by number of years, in the 1990s, wages increased in the bottom and top of the income distribution, but not in the middle, while after 2000, the college premium pretty much disappeared, leaving some growth at the bottom, while throughout this period the very top of the income distribution showed high returns to higher education, but differentiated heavily in favor of finance over engineering. Lawrence Mishel, Heidi Shierholz & John Schmitt, *Dont’ Blame the Robots* (2013); T. Philippon & A. Reshef, *Wages and Human Capital in the U.S. Finance Industry: 1909-2006*, 127 Q. J. ECON. 1551 (2012). All this was far more consistent with an analysis of institutional differentiation along class lines than with a simple, efficient labor market story about

Top 20 by Market Cap	Business	Top 20 by Employees	Business
NVIDIA	Tech: Chip design and software	Walmart	Retail chain store
Microsoft	Tech: software	Amazon	Tech, retail, logistics; cloud infrastructure
Apple	Tech: hardware design and software	UPS	Shipping; logistics
Alphabet	Tech: software; advertising	Home Depot	Retail chain store
Amazon	Tech, retail, logistics; cloud services	Concentrix	Outsourcing
Meta Platforms	Tech: software, advertising	Target	Retail chain store
Broadcom	Tech: chip and infrastructure software	Marriott International	Hospitality
Tesla	Tech, manufacturing: electric vehicles	Kroger	Grocery chain store
Berkshire Hathaway	Finance, conglomerate with operating subsidiaries	UnitedHealth Group	Insurance; healthcare delivery
Oracle	Tech: software	Berkshire Hathaway	Finance, conglomerate with operating subsidiaries
JP Morgan Chase	Finance & retail banking	TJX Companies	Retail chain store
Walmart	Retail chain store	Starbucks	Food service chain store
Eli Lilly	Pharmaceuticals	Cognizant Technology Solutions	IT services (most employees in India)
Visa	Finance: consumer credit	TriNet	Professional employer organization (outsourced HR operations)
Mastercard	Finance: consumer credit	Costco	Retail chain store
Netflix	Tech: video platform	PepsiCo	Beverage & food
ExxonMobil	Energy: fossil fuels	JPMorgan Chase	Finance; retail banking
Johnson & Johnson	Pharmaceuticals; medical devices; consumer health products	Inspireity	Professional employer organization (outsourced HR operations)
Costco	Retail chain store	Lowe's	Retail chain store
Home Depot	Retail chain store	FedEx	Shipping; logistics

Table 2: Top 20 firms by market capitalization and employees. Companies that appear in both lists appear in **bold italics**.

their explanation to import-exposure as the primary driver of both degraded terms of labor and political rebellion toward ethnonationalist populism.⁵³

relative marginal contribution to production under new technological conditions. L. Mishel, J. Schmitt & H. Shierholz, *Wage Inequality: A Story of Policy Choices*, 23 NEW LABOR FORUM 26 (2014).

⁵³ David H Autor, David Dorn & Gordon H Hanson, *The China Syndrome: Local Labor Market Effects of Import Competition in the United States*, 103 AM. ECON. REV. 2121 (2013); David Autor et al., *Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure*, 110 AM. ECON. REV. 3139 (2020).

C. Class, global reorientation, and the differentiation between liberal oligarchic, corporatist oligarchic, and social democratic societies

To imagine the past fifty years in terms of “postindustrial” society is to take an exclusively western, not to say North American or even US-UK-centric viewpoint. Rather, the shift is to an “information economy” in important ways, but also a shift in the global location of manufacturing toward Asia, and its integration into global manufacturing chains that integrate production across diverse jurisdictions seeking the most profitable combinations of productive labor and low wages to maximize profitability.⁵⁴ The global relocation of capital to manufacturing in Asia, and in the narrower European case, to the post-Soviet economies of east-central Europe, becomes clear when we look at the average annual change in labor productivity growth across most OECD countries as well as several major rising Asian manufacturing centers (Figure 2). In Europe this mini-version

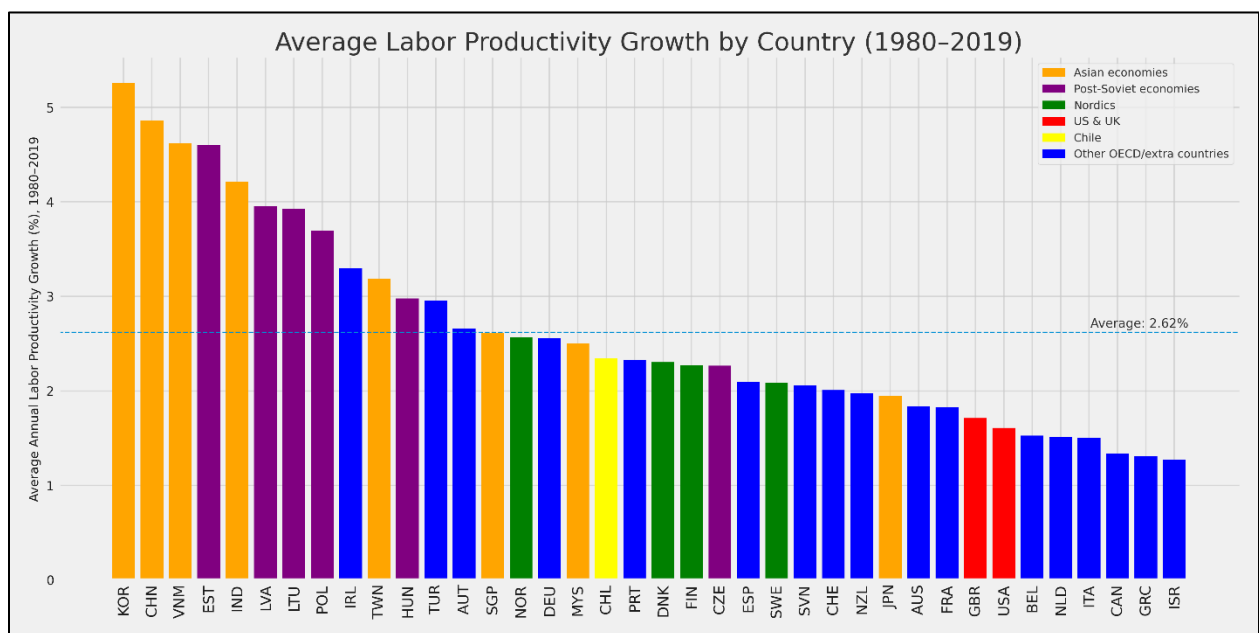


Figure 2. Average annual productivity growth 1980-2019 OECD countries, additional Asian economies; and post-Soviet economies. Source: Penn World Table.⁵⁵

⁵⁴ JOHN HAGEL & JOHN SEELY BROWN, *THE ONLY SUSTAINABLE EDGE: WHY BUSINESS STRATEGY DEPENDS ON PRODUCTIVE FRICTION AND DYNAMIC SPECIALIZATION* (2005); Ariell Reshef & Gianluca Santoni, *Are Your Labor Shares Set in Beijing? The View through the Lens of Global Value Chains*, 102; Judy Fudge & Hila Shamir, *Strike Law and Workers' Power Resources in Global Supply Chains and Platform Giants*, 164 INT. LABOR REV. (2025).

⁵⁵ Ireland uses the Irish government's estimate, which reduces Irish labor productivity growth otherwise inflated by high-tech firm's tax evasion policies of booking IP profits in Ireland. NATIONAL COMPETITIVENESS COUNCIL, *OVERVIEW OF IRELAND'S PRODUCTIVITY PERFORMANCE, 1980-2005* (2006), https://competitiveness.ie/media/kz4p4toj/ncc061019_productivity_performance_overview.pdf; NATIONAL COMPETITIVENESS DEPARTMENT FORFAS, *BACKGROUND REPORT DETAILED ASSESSMENT OF IRELAND'S PRODUCTIVITY PERFORMANCE 1980-2005* (2006), https://www.competitiveness.ie/media/v43juctv/ncc_productivity_background_report.pdf.

of offshoring took the form of companies from wealthier European societies, particularly Germany, locating manufacturing operations in Poland, the Czech Republic, and to an extent, Hungary, to take advantage of lower wage expectations and in some cases weaker labor protection.⁵⁶

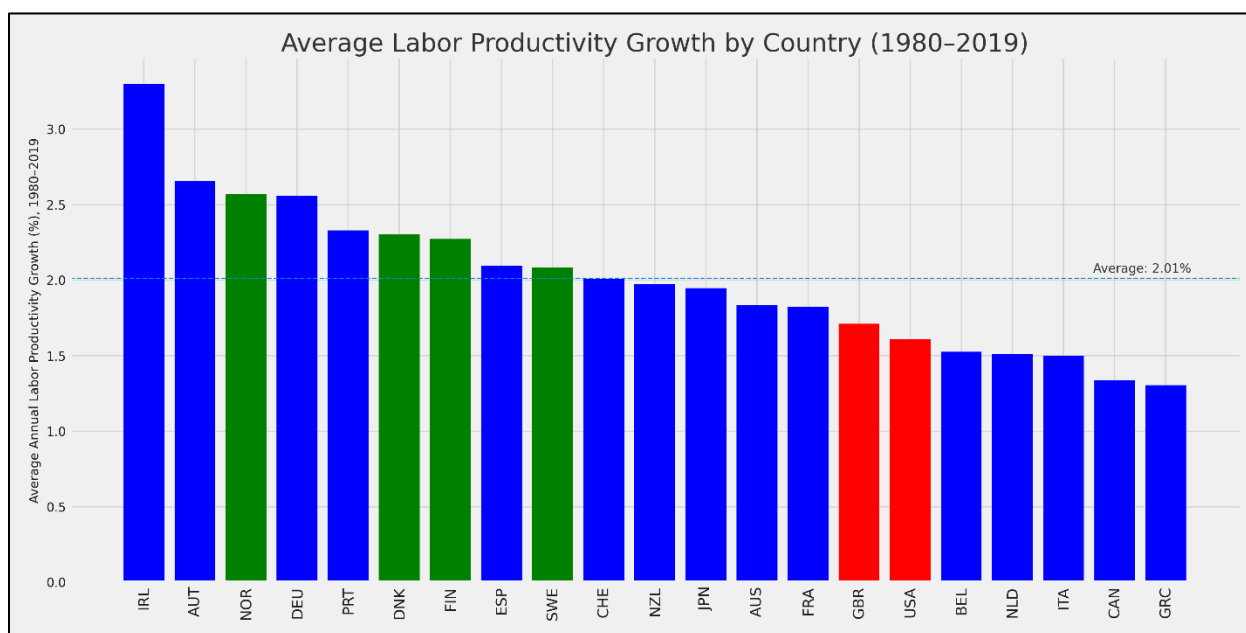


Figure 3. Average annual productivity growth 1980-2019 OECD countries subset already well-developed by 1980. Source: Penn World Table. Ireland reduced to reflect Irish government estimate.

Equally revealing, if not more so, is the difference in labor productivity growth among those countries that were already advanced market societies by the end of the Fordist wave. Figure 3 represents the subset of labor productivity growth statistics using only those countries. The two flagships of the Reagan-Thatcher revolution perform below most mainland-European economies, and behind all the Nordic social democracies. The difference reflects the institutional political economy of these European countries, where liberalization was adapted and incorporated into a different institutional political economy, which enabled them to focus on continuing improvement in manufacturing in a narrower sector, losing some ground in global manufacturing, but not nearly as much as the United States.

The crucial lesson is this. The information economy and its neoliberal institutional structure were driven by the imperatives of capital to escape the profit squeeze of the late Fordist “Golden Age” institutional structures. In part, capital escaped to finance. Undermining Bretton Woods with novel financial instruments, like the Eurodollar market, and leveraging the collapse of financial regulation and containment of cross-border flows, finance became its own focus of profit, restarting the bubble-crash dynamics that had characterized global capitalism since 1763 and were tamed briefly between the 1930s and 1970s. In part, capital flowed into information technology with inefficient, but nonetheless transformative levels of investment in the cutting edge of information technology development. How these investments manifested reflected the institutional

⁵⁶ A clear example is Amazon’s opening of fulfillment centers right across the border from Germany in Poland to head off the “Make Amazon Pay” worker mobilization. Fudge and Shamir, *supra* note 54.

political economy diverse societies inherited from their own histories. The Reagan Thatcher revolutions came in the liberal oligarchic republics that had been the hegemonic forces of global capitalism since the early 17th century.⁵⁷ The corporatist oligarchic societies that grew out of nineteenth century corporatist authoritarianism—most prominently Germany and Japan—developed dualized economies, in Germany along the lines of migration and age, in Japan along the lines of gender. The social democracies invested heavily in continuous educational upgrading, retreated in important ways from their wage solidarity policies during the Fordist wave and introduced more flexibility into their workforce, but, crucially, socialized the risk of dislocation associated with rapid change in both technology and global trade and manufacturing dynamics. These latter have seen the most stable dynamic growth among the wealthy countries, although the global forces did force a partial retreat from their high egalitarian standards without giving up on the universalism of their programs to sustain sufficiency level public provisioning of access to basic goods. The result is that these societies systematically outperform both the United States and the UK, and Germany, France, and similar mainland-European societies not only in terms of the human development index, but also in terms of the democracy index published by Freedom House *and* the economic freedom indexes published by the conservative Heritage Foundation and World Bank.⁵⁸

As we turn to consider the likely development and effects of AI, it is crucial to bear in mind that these will primarily be shaped by the institutional political economy of the societies in which it is incorporated, and not by some internal, universal characteristics of the technology. AI as a technology in production will undoubtedly reshape social relations of production across the globe. But it will do so as a function of, and in interaction with, the inherited institutions, class structures, and class power of the societies it touches. There will be no single, all-overpowering technological effect that will sweep away history and the historically-specific structure of the world's societies.

III. Manual and low-skill cognitive labor: impact on the unskilled and skilled working classes and the clerical class

A. Robotics

In 2014 Erik Brynjolfsson and Andrew McAfee, then both at MIT's business school, published a highly influential book called *The Second Machine Age*.⁵⁹ The book reflected the apotheosis of the “skills-biased-technical-change” era and triggered a wealth of reactions on “the future of work” and will robots take all the jobs. Like today's more aggressive predictions about AI, the book took discrete successes in machine intelligence and robotics—the successes of machines against humans in chess, Go, and eventually *Jeopardy!*; self-driving cars, humanoid robots, and the vast improvements in speech recognition software—and combined them with teaching their readers

⁵⁷ The Dutch Republic was such a society at the time, preceding England but following a similar long-term trajectory from rapid industrial and trade growth with high productivity gains, to colonial exploitation, milking of existing assets, and eventually financialization by the time it collapsed in 1795.

⁵⁸ UNDP, HUMAN DEVELOPMENT REPORT, <https://hdr.undp.org/> (last visited Sept. 23, 2025); *Freedom in the World Report*, FREEDOM HOUSE, <https://freedomhouse.org/country/scores> (last visited Sept. 23, 2025); HERITAGE FOUNDATION, INDEX OF ECONOMIC FREEDOM: PROMOTING ECONOMIC OPPORTUNITY AND PROSPERITY BY COUNTRY (2024), <https://www.heritage.org/index/>.

⁵⁹ ERIK BRYNJOLFSSON & ANDREW MCAFEE, *THE SECOND MACHINE AGE: WORK, PROGRESS, AND PROSPERITY IN A TIME OF BRILLIANT TECHNOLOGIES* (1 edition ed. 2014).

how to think in terms of exponential growth. Together, this led them to predict vast improvements in machine capacities to replace human labor and increase productivity dramatically, while leaving large questions about what humans would do, with a heavy emphasis on the future exponential improvement of artificial intelligence. Crucially, robots are distinguished from other computerized machines by their capacity to manipulate objects in the world. Their direct impact is therefore primarily on workers who do manual labor—whether skilled or unskilled—rather than on the professional and managerial class or even on the clerical class. Speaking in the past tense in 2014, Brynjolfsson and McAfee declared that “self-driving cars went from being the stuff of science fiction to on-the-road reality in a few short years.”⁶⁰ As they summarized precisely the class conflict (in their terms, the promise of intelligent humanoid robots) typified by iRobot’s “Baxter”:

Baxter has a few obvious advantages over human workers. It can work all day every day without needing sleep, lunch, or coffee breaks. It also won’t demand healthcare from its employer or add to the payroll tax burden. And it can do two completely unrelated things at once; its two arms are capable of operating independently.⁶¹

The point was not only the mechanical advantages of the industrial robot, but also the elimination of labor conflict and wages. The following half decade saw a veritable cottage industry, among both non-profit foundations and global corporate leadership, of conferences and workshops to consider the social and worklife consequences of “the future of work.” Universal Basic Income (UBI) became a widely considered option—because, after all, if people can’t do most of the useful work, they still need to get money to live. While most industrial robots are far from the image of powerful AI, the studies of the relation between labor, particularly jobs and wages, and the distribution of robots in manufacturing does provide us with some purchase on the slightly more mature areas of economic activity where technology has been able to provide a significant substitute or complement for human labor—both in its mechanical side, replacing manual labor, whether or not it combined embodied knowledge with manual manipulation or not, and to a lesser degree, its cognitive side.

The most influential work has been a series of articles by Daron Acemoglu and Pascual Restrepo, combining theoretical economic modeling with empirical studies of the United States. The theoretical model assumed exogenous technological change, predicting initial loss of jobs but, over the long term, as new technologies reduced the cost of using labor in newly-invented tasks relative to further intensification of automation, the two (labor and automation) will settle on a new equilibrium growth trajectory. Nonetheless, assuming that automation is skills-biased, this will result in increased inequality between skilled and unskilled workers, until the cost of skilled labor makes it more profitable to standardize some of the skilled work to leverage unskilled labor at lower labor cost.⁶² In this model, the more labor is able to organize and demand high wages, the more capital will invest in automation, and the more jobs will be lost. The order of operations is always that automation runs ahead of the creation of new tasks, eliminating jobs, and then new

⁶⁰ *Id.* at loc 273.

⁶¹ *Id.* at loc 444.

⁶² Daron Acemoglu & Pascual Restrepo, *The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment*, 108 AM. ECON. REV. 1488 (2018); Daron Acemoglu & Pascual Restrepo, *Automation and New Tasks: How Technology Displaces and Reinstates Labor*, 33 J. ECON. PERSPECT. 3 (2019).

tasks catch up and create more new tasks that leverage skilled workers, and later still, unskilled workers. In later empirical work, the two authors conducted an empirical study of the impact of the introduction of robots into manufacturing in the United States, using local labor markets, and found that introduction of robots reduced employment and wages. As they put it succinctly, “One more robot per thousand workers reduces the employment-to-population ratio by 0.2 percentage points and wages by 0.42%.”⁶³

Studies of robotics and labor markets in Europe, however, showed the opposite results. Early and influential in this was the work of Georg Graetz and Guy Michaels, looking at the impact of robotics adoption in manufacturing across seventeen countries from 1993 to 2007.⁶⁴ Graetz and Michaels found that robotic density increased labor productivity and total factor productivity, did not reduce the number of hours worked, but did increase wages, without changing labor’s share of income relative to capital. While there was no wage polarization, the productivity benefits worked to the benefit of mid-skilled workers against low-skilled workers, although the study did not observe whether this reflected upskilling of the low-skilled workers or their displacement. The countries that increased robotics density the most in their sample were Germany, Denmark, and Italy, all three countries with strong labor protections and significant (though very different) vocational training programs, either in public settings or within firms. In a counterfactual analysis of loss of productivity absent intensified robotics used, the countries that benefitted most from roboticization were Germany, Denmark, and Finland, while the United States benefitted among the least—slightly ahead Hungary, Australia, Ireland, and Greece, but behind the twelve other countries in the study.⁶⁵

This work was further deepened and extended in later work by Lene Kromann and collaborators, studying the effects of robotics in both developed and developing economies, as well as the differentiated effects based on industry sector and import exposure.⁶⁶ The paper strongly suggests that the institutional political economy of a society, as well as its position in the global trade and manufacturing network, influence the effect of robots on employment and wages, rather than the generic features of the technology as compared to human workers. The core of their finding was that robot intensity is associated with increased labor productivity and total factor productivity, higher average wages, and either unchanged or increased employment levels in manufacturing, with no decrease in the share of income going to labor relative to capital. Robotics density increased faster in sectors of the economies of European societies that were more exposed to increased competition from China. Robot adoption in European manufacturing sectors thus appears to be a strategy to increase labor productivity in the teeth of global competition as manufacturing has shifted to Asia. Additional papers on Europe and a broader range of countries, using somewhat different specifications in each case, suggest either less significant positive impact

⁶³ Daron Acemoglu & Pascual Restrepo, *Robots and Jobs: Evidence from US Labor Markets* (2017), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2941263.

⁶⁴ Georg Graetz & Guy Michaels, *Robots at Work*, 100 REV. ECON. STAT. 753 (2018).

⁶⁵ Georg Graetz & Guy Michaels, *Online Appendix for “Robots at Work,”* Table A10, https://personal.lse.ac.uk/michaels/Graetz_Michaels_Robots_appendix.pdf.

⁶⁶ Lene Kromann et al., *Automation and Productivity—a Cross-Country, Cross-Industry Comparison*, 29 IND. CORP. CHANGE 265 (2020).

on wages, no negative impact on low-skilled employment, or significant variation by sector.⁶⁷ By contrast, recent studies of the adoption of robots in emerging economies suggest that the increase in labor productivity and positive effects on employment appear only in wealthier economies, whereas adoption in emerging economies tends to come at the expense of reducing labor share of income.⁶⁸ What is quite consistent is that Acemoglu and Restrepo's observations in the United States do not reflect the impact of manufacturing robotics adoption in other societies, but reflect the specific conditions of American capitalism since the Reagan revolution. In another widely cited study comparing German to American firms, the authors found that adoption by the former did not result in job losses, but rather in reassignments of unionized workers to new tasks, but did result in reduced hiring of younger workers⁶⁹—reflecting different institutions, and consistent with the dualized structure of the German corporatist institutional model—protecting insiders at the expense of outsiders to the existing alignment of employer associations and unionized workers.

Looking at all these cross-country studies, the conclusion that emerges is that there is no single outcome from the introduction of robots in manufacturing. Rather, the impact of a change in technology in a capitalist society depends on the inherited social relations of production as they evolved in that society, on the position these inherited social relations puts that society in the global network of manufacturing, trade, and knowledge, and on how these instantiate the technology as part of a techno-institutional regime through successive rounds of high growth, slowdown, crisis, struggle, and institutional adaptation: including institutional innovation, evasion, decay, and escape. In the United States, where capitalist enterprises early adopted automation partly as a union-avoidance strategy even in the 1950s, and shifted since the 1970s toward systematically treating labor as a cost center and escaping to finance or to PMC-heavy, worker-thin sectors, robotics entered as labor-cost-savings devices.

In several European countries, while overall share of manufacturing declined, robotics investments combined with worker training and educational upskilling led to increased productivity that enabled the firms remaining in the manufacturing sector to hold their own vis-à-vis Asian competition, while strong unions were able to cooperate with capital in making these investments, secure in the knowledge that their power and the institutional structures that made their relative power possible would protect their capacity to secure their share of the productivity increases. The empirical evidence about robotics from the past twenty years suggest that this was in fact the trajectory these countries followed—both those that had a strong export sector supported by a dualized economy, and those that had a strong export sector embedded in a social democratic system that socialized the costs of continuous change and transition.

⁶⁷ FRANCESCO CHIACCHIO, GEORGIOS PETROPOULOS & DAVID PICHLER, THE IMPACT OF INDUSTRIAL ROBOTS ON EU EMPLOYMENT AND WAGES: A LOCAL LABOUR MARKET APPROACH (2018); Anton Jose-Ignacio et al., *The Labour Market Impact of Robotisation in Europe*, IDEAS WORK. PAP. SER. REPEC (2020); David Klenert, Enrique Fernández-Macías & José-Ignacio Antón, *Do Robots Really Destroy Jobs? Evidence from Europe*, 44 ECON. IND. DEMOCR. 280 (2023).

⁶⁸ Xiaoqing (Maggie) Fu et al., *Diffusion of Industrial Robotics and Inclusive Growth: Labour Market Evidence from Cross Country Data*, 122 J. BUS. RES. 670 (2021); Luis R. Díaz Pavez & Inmaculada Martínez-Zarzoso, *The Impact of Automation on Labour Market Outcomes in Emerging Countries*, 47 WORLD ECON. 298 (2024).

⁶⁹ WOLFGANG DAUTH ET AL., GERMAN ROBOTS – THE IMPACT OF INDUSTRIAL ROBOTS ON WORKERS (2017).

The International Federation of Robots releases annual figures for robotics density, in terms of number of robots per employee in manufacturing.⁷⁰ The top ten countries in 2024 were South Korea, Singapore, China, Germany, Japan, Hong Kong, Taiwan, Sweden, Denmark, and the United States. The United Nations Comtrade dataset reports exports.⁷¹ Based on that data, the top ten robotics exporters are Japan, Germany, China, Italy, Denmark, Sweden, the United States, France, South Korea, and the Netherlands. The largest individual companies by robot sales are FANUC Robotics division (Japan), ABB Robotics (Swiss-Swedish), Yasakawa robotics division (Japan), KUKA (robotics plus digital divisions) (Germany), Denso Wave (Japan), and Universal Robots (Denmark).⁷² What these rankings make clear is that learning about the impact of robotics on labor from the United States overemphasizes the experience of an unimportant player occupying a weak position in global manufacturing that, barring a turnaround, is shrinking further. Acemoglu and Restrepo's observations describe the impact of robotics under the liberal oligarchic variant of capitalism, not of robotics *per se*. The findings of Graetz and Michaels, Kromann and colleagues, and other Europe and emerging-market-focused studies provide insights into the impact of robotics under corporatist and social democratic institutional political economies, showing that for the already-wealthy countries at least, robotics present a path toward sustaining broader based growth than is feasible under the liberal-oligarchic model, albeit at the expense of the potential development of emerging economies. Roboticization in these emerging economies, for its part, puts them on a path to highly unequal development as capital investment and low-employment output shut off for the kind of broad employment in export-oriented manufacturing that has raised hundreds of millions of people in East Asia out of poverty in the past thirty years.

A turnaround for the United States would require significant importation of technologies and know-how, which the United States currently lacks—as evidenced by NVIDIA's announcement that it would construct three manufacturing plants for its most advanced AI chip in Arizona, Houston, and Dallas, each dependent on allying with one of the major Taiwanese chip manufacturers.⁷³ The difficulty is that, to counter the hegemonic ideology of free markets that capitalists and the PMC developed so effectively to legitimate their extraction of almost all gains from productivity growth in the past forty years, both the Republican Party (the alliance of capitalists and the white working class) and the Democratic Party (the alliance of the PMC and the status-subordinated underclass) have turned to ethnonationalist jingoism, primarily anti-China. The tension between the manifest need to import knowledge—embedded, embodied, and codified—to catch up to the global technological frontier, and the political rhetoric required to overcome the dominant ideology that served both party elites so well for over a generation, makes reliable, stable institutional structures to support such knowledge importation difficult to sustain. The most obvious signal of the problem are the recent immigration raids that resulted in repatriation of high-skilled Korean workers whose work was crucial to setting up Korean-owned

⁷⁰ *Statistics - IFR International Federation of Robotics*, <http://www.ifr.org/industrial-robots/statistics/> (last visited Apr. 29, 2015).

⁷¹ *UN Comtrade*, <https://comtradeplus.un.org/TradeFlow> (last visited Sept. 13, 2025). (using HS 847950)

⁷² Based on business news, industry journals, and company annual reports.

⁷³ NVIDIA Newsroom, *NVIDIA to Manufacture American-Made AI Supercomputers in US for First Time*, NVIDIA BLOG (Apr. 14, 2025), <https://blogs.nvidia.com/blog/nvidia-manufacture-american-made-ai-supercomputers-us/>.

manufacturing facilities required to import the most advanced battery manufacturing technologies to the United States, resulting in halting operations.⁷⁴

By contrast to the laggard position of the United States, the remarkably high position of two small countries—Denmark and Sweden—in all these rankings suggests that social democracy can be more, rather than less, dynamic, as well as more egalitarian. In the presence of a potential tension between power and productivity, where technological investment can go either toward power-increasing paths designed to weaken labor and increase the share of income going to capital, or to productivity-increasing paths, the degree to which the power-increasing investments have a plausible chance of success can shape the investment path of capital. Strong labor, on the one hand, coupled with universalized socialization of both the risks of continuous dislocation and of continuous updating and upgrading of embodied knowledge necessary to adapt to the rapid technological change the expected returns from each strategy. Strong labor in an institutional political economy that provides labor with both economic and political power makes power-seeking investments highly risky. As Fudge and Shamir underscore, Tesla lost its battle against Swedish workers because the legal framework enabled dock workers, postal workers, and cleaners across Sweden, Denmark, and Norway to act in a sympathy strike to force Tesla to the table.⁷⁵ The same happened to the far more powerful Swedish umbrella manufacturers association, the lockout world champion until the 1930s, when in 1980 it tried to flex its muscles to intensify liberalization in the largest lockout in Swedish industrial relations since the early twentieth century.⁷⁶

The example of robotics, and the difference between the United States and European countries, particularly the Nordic social democracies, offers a clear example of the “high-road” vs. “low-road” institutional structure for technological adoption. When grabbing a larger share of a slow-growing pie is cheap, low-road businesses gravitate toward that non-cooperative solution. When doing so expensive and risky, and particularly where strong labor can be a partner in continuous upgrading of technology and skills, options focused on growing the pie become more robust. Moreover, Swedish and even more so Danish social democracy socialize both the risks of dislocation—through high unemployment and retraining-period pay—and the costs of retraining workers repeatedly as the requirements of embodied knowledge change with technological change. The result is that it lowers the cost and time lag of matching the embodied knowledge workers possess with the continuously changing technology. In combination with the data concerning average annual productivity growth since (Figure 2) showing that all Nordics have exhibited higher average productivity growth than did the United States or Britain since the Reagan-Thatcher revolution, it appears that a well-structured social democratic institutional political economy can sustain more dynamic stability in a sector that involves significant potential conflict between labor and capital better than the liberal oligarchic model typified by American capitalism.

B. Algorithmic Surveillance and Management Systems in rich countries

⁷⁴ Christian Davies & Song Jung-a, *Korean Companies Admit Cutting Corners on US Visas but Say They Have Little Choice*, FINANCIAL TIMES, Sept. 8, 2025, <https://www.ft.com/content/c677b9aa-2e89-4feb-a56f-f3c8452b3674>.

⁷⁵ Fudge and Shamir, *supra* note 54.

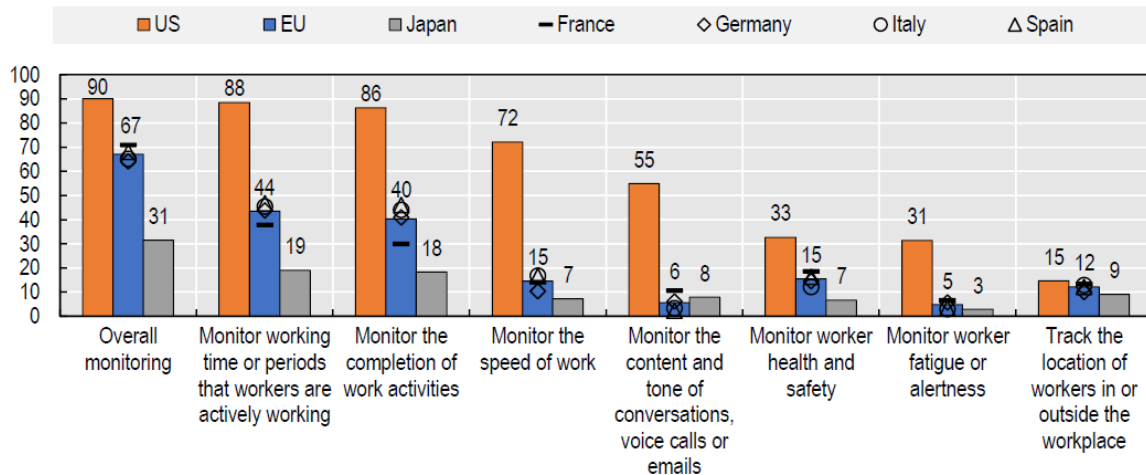
⁷⁶ Anders Kjellberg, *Storkonflikten 1980 och andra stora arbetskonflikter i Sverige*, 35 ARBETARHISTORIA MEDD. FRÅN ARBETARRÖRELSENS ARK. OCH BIBL. 33 (2011).

Review of the expansive work already done on automated surveillance. Clear summary of Levy et al; Dubal; completed with:

This is the money shot from the OECD Automated management report:

Figure 4. Monitoring tools are prevalent, with use decreasing as data collection may involve more personal data

Percentage of managers reporting their firm uses algorithmic management tools for the specific task

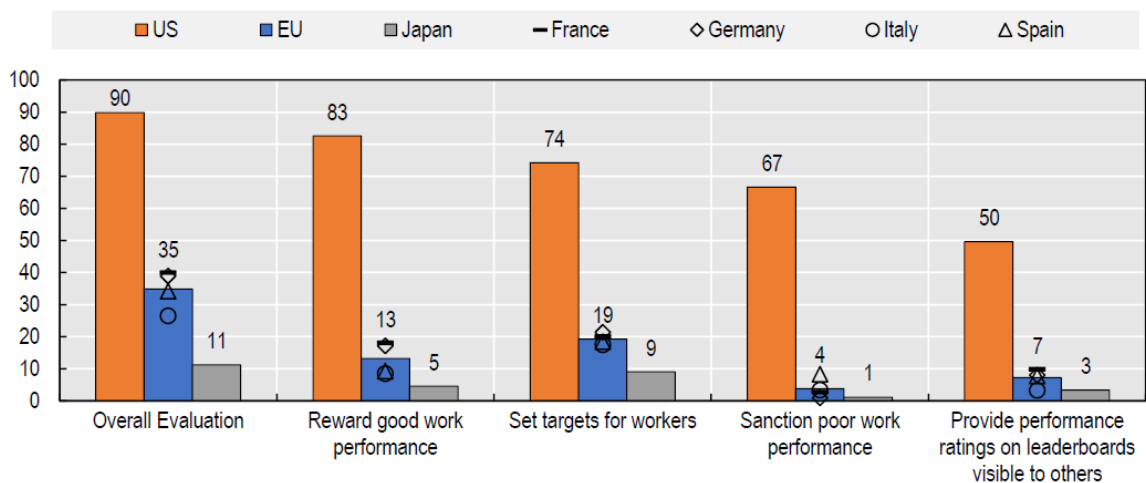


Note: Managers were asked to indicate whether their workplace provides software to partially or fully automate the monitoring of workers according to the answer options presented above.

Source: OECD employer survey on algorithmic management (2025).

Figure 5. Outside of the United States, evaluation tools are used far less

Percentage of managers reporting their firm uses algorithmic management tools for the specific task



Note: Managers were asked to indicate whether their workplace provides software to partially or fully automate the evaluation of workers according to the answer options presented above.

Source: OECD employer survey on algorithmic management (2025).

C. Ghost work and the global division of labor

Summary of the literature.

Adding the newer survey literature on the educational level, gender, high levels of use in India Philippines, trends in Brazil, Venezuela, Kenya, South Africa.

IV. AI and Cognitive Labor: LLMs and the Crisis of the Professional and Managerial Class

The PMC as we saw refers to the class of people in capitalism who do cognitive work in capitalist enterprises but do not have property rights in the enterprise or resources it uses in production. While the term originates in the work of Barbara and John Ehrenrich in the 1970s to characterize the growing role of an educated, salaried worker with distinct, often antagonistic, interests to those of the working class or the capitalists,⁷⁷ the class as characterized by its function in production (high cognitive work, using embodied and codified knowledge, without property in the enterprise or resources it depends on) has existed since late medieval proto-capitalist societies, and played a large role in the early capitalist takeoff of both the Dutch Republic and England as early as the sixteenth and seventeenth centuries.⁷⁸ As soon as one understands that production includes not only physical manipulation of natural resources, but also cognitive work, and adds the dimension of knowledge as a crucial input in production, control over knowledge as well as over natural resources through possessing calls on resources in production (property and obligations—credit and finance) becomes a crucial dimension of asymmetric position in social relations of production—that is, in class.

Managers—like non-owner ship captains, payment clerks, plantation managers—operated as paid cognitive workers of capitalist-owned enterprises throughout this period. Members of the professional social formations—lawyers (& notaries) and doctors—interacted with capitalist enterprises and the political military formations primarily as craft-based smallholders or in small partnerships of equals, usually employing a small number of clerical class workers, or in the case of doctors, care workers, not on track to join the institutionalized profession. The gendered structure of care work in both pre-capitalist and non-capitalist societies as well as capitalist societies, and the large role of organizations rooted in social relations of meaning-making, primarily religion, have always placed health care work in a distinct relationship to capitalist enterprises, different to that of lawyers or finance professionals, much less managers. All these social formations were self-reproducing through training and habituation into the institutions of the formation by following a craft structure: combining senior practitioners with junior apprentices on a training track to become future senior members of the social formation, similar to the craft model of production. The increase in scale of operations of capitalist enterprises, and the intensification of knowledge in production—science and scientific management—since the 1890s increased the salience and importance of the PMC over the course of the twentieth century. By the

⁷⁷ Barbara Ehrenreich & John Ehrenreich, *The Professional-Managerial Class*, 11 RADIC. AM. 7 (1977); Barbara Ehrenreich & John Ehrenreich, *The New Left: A Case Study in Professional-Managerial Class Radicalism*, 11 RADIC. AM. 7 (1977).

⁷⁸ Chapters 2, 3, 6, 8, 12 of BENKLER, *supra* note 26.

1970s, as the shift to an “information economy” made the binary workers/capitalists simply unable to capture how social life in modern market societies actually unfolded, the conditions for theorizing this class had ripened.

The tensions between the PMC and both capitalists and the working classes, particularly the skilled working class, were already present since the early twentieth century. Easier to see and more widely acknowledged is the tension between managers and process engineers, on the one hand, and skilled workers, on the other hand, implicit in scientific management and the assembly line. Less commonly noted is that the Berle and Means corporation—the corporation in which ownership and control were separated and where managers ran the operations according to their increasingly-professionalized judgment—reduced the shareholders in many cases primarily to financial rentiers, rather than capitalists actively managing the enterprises they owned. This tension became clearest in the assault on managerial professionalism in the name of shareholder value, popularly spearheaded by Milton Friedman and academically by Michael Jensen. Both focused on the same tension: the managers who were not owners managed the firm to serve diverse values, including community, workers, and their own perks, rather than doing the only thing managers of capitalist enterprises should do—maximize shareholder value.⁷⁹ The whole point of payment in stock, according to this work, was to “align the interests” of the managers with those of the shareholders—in effect, to turn salaried professional managers at the top echelons into capitalists actively managing a business they co-owned. This line of tension then matured in the conflicts over executive compensation since the late 1980s, as more scholarship made it clear that managers, now rewarded in stock options, managed the firms in ways that maximized their own returns rather than actually increasing the overall productivity and growth of the firm.⁸⁰

The escape of capital to finance also depended heavily on an intensified role for financial professionals and lawyers. The whiz kids with the new electronic spreadsheets could calculate the payoffs of leveraged buyouts for the first time, while others were now able to calculate collateralized debt obligations in different tranches reflecting different risk profiles for resale into financial markets. The result was that even though the alleged core of the “information economy” was research in information technology and in biotechnology, the pay of PhDs who went into finance dramatically outpaced the pay of PhDs who went into engineering after 1980.⁸¹ The intensified focus on stock market returns, particularly in the short term, further drove capitalists into the position of financial rentiers, leaving the finance professionals in the driver’s seat, and their interests were not necessarily aligned with those of the rentiers whose money they invested or, for that matter, with the professional managers whose job it was to run actual businesses rather

⁷⁹ Milton Friedman, *The Social Responsibility of Business Is to Increase Its Profits*, NEW YORK TIMES, Sept. 13, 1970, at 12; Michael C. Jensen & William H. Meckling, *A Theory of the Firm: Governance*, 3 J. FINANC. ECON. 3 (1976); Michael C. Jensen & Kevin J. Murphy, *Performance Pay and Top Management Incentives*, 98 J. POLIT. ECON. 225 (1990).

⁸⁰ Olivier Jean Blanchard, Florencio Lopez-de-Silanes & Andrei Shleifer, *What Do Firms Do with Cash Windfalls?*, 36 J. FINANC. ECON. 337 (1994); LUCIAN BEBCHUK & JESSE FRIED, *PAY WITHOUT PERFORMANCE: THE UNFULFILLED PROMISE OF EXECUTIVE COMPENSATION* (1. Harvard Univ. Press paperback ed ed. 2006); LUCIAN BEBCHUK & SCOTT HIRST, *INDEX FUNDS AND THE FUTURE OF CORPORATE GOVERNANCE: THEORY, EVIDENCE, AND POLICY* (2019), <http://www.nber.org/papers/w26543.pdf>.

⁸¹ Philippon and Reshef, *supra* note 52.

than increase stock market valuations.⁸² Mergers and acquisitions required not only financial professionals, but lawyers as well. So too did the mounting campaign to quash unionization efforts and lobby for deregulation. Meanwhile downsizing, outsourcing, offshoring, required a new set of managerial skills, as well as creating significant room for management consultants, providing ample room for new HR professionals, union avoidance professionals, and brand managers.

The result of the dynamic that pits the PMC against both the working class and all but the largest capitalists is that, as a class, the PMC has extracted most of the market value of productivity growth—limited as it has been—over the past half century. At the individual billionaire level this is not true. If one looks only at the billionaires, one sees primarily capitalists: the tech billionaires at the very top joined by Buffett, and the Waltons (Wal-Mart), Kochs, as well as some financial investors and hedge fund managers rounding out the top twenty or so.⁸³ But in a nation of 340 million people, the top one percent comprises 3.4 million, and even the top 0.1% is 340,000 strong. Once one begins to expand one's vision from the handful of individuals at the very top to the class that has benefited the most, the picture changes. There are few studies that try to break down the composition of the top 1% and 0.1%, but those few make clear that professionals and managers comprise the majority of that segment of the population. One study based on individual income tax returns and focused on the period from 1979 to 2005, just before the global financial crisis, showed that professionals in finance, whether managerial or not, and executives and managers in non-finance businesses, made up about 43-44%% of the top 1% (excluding/including capital gains), and 59-60% of the top 0.1%. In combination, that share held constant from 1979 to 2005, but saw a shift from non-financial executive and managerial professionals to financial professionals, and from salaried executives in public companies to executives and managers in closely held businesses.⁸⁴ Equally revealing in that study was how small a proportion of the top 1% and 0.1% was filled by computer, math, engineering, and technical professionals (outside of finance), growing from a mere 3.8% to 4.6% of the top 1% (a little less when including capital gains), except for the brief period that overlapped with the dot.com bubble, and the same was true for real estate professionals during the real estate bubble. By comparison, doctors grew from 16.8% of the top 1% to 20.4%, declining after the changes in medical payment systems to 15.7% by 2005. Lawyers' share remained stable, growing from 6.7% in 1979 to 7.7% in 2005. Other "superstars" whose success long served to legitimate the rising wealth of the top 1%—in arts, media, and sports—did not change their share of the top 1% and 0.1%, remaining about 1.6% of the top 1% throughout the period. The category of "entrepreneur, not elsewhere classified" shrank, both including and excluding capital gains, as a share of the top 0.1% (from 3.9% to 3.0% or 4.3% to 3.8% excluding/including capital gains). It is obvious why the individuals who capture the news headlines remain memorable—the athletes and performers with their multimillion dollar salaries,

⁸² JOHN COATES, *THE PROBLEM OF TWELVE: WHEN A FEW FINANCIAL INSTITUTIONS CONTROL EVERYTHING* (2023); BEBCHUK AND HIRST, *supra* note 80; GERALD F. DAVIS, *MANAGED BY THE MARKETS: HOW FINANCE RESHAPED AMERICA* (2009); RANA FOROOGHAR, *MAKERS AND TAKERS: THE RISE OF FINANCE AND THE FALL OF AMERICAN BUSINESS* (First edition ed. 2016); William Lazonick, *Profits without Prosperity*, 92 HARV. BUS. REV. 46 (2014).

⁸³ *The Forbes 400 List 2025 - The Richest People in America Ranked*, FORBES, <https://www.forbes.com/forbes-400/> (last visited Sept. 24, 2025).

⁸⁴ Jon Bakija et al., *Jobs and Income Growth of Top Earners and the Causes of Changing Income Inequality: Evidence from US Tax Return Data*, UNPUBL. MANUSCR. WILLIAMS COLL. (2012), <http://piketty.pse.ens.fr/files/Bakijaetal2010.pdf>.

the tech billionaires—and that is not to say that these are unimportant. But it is crucial to remember that at the population level, the past fifty years have been far more importantly an era of the rising share of the professional and managerial class at the very top of the income distribution.

More recent work with data running until 2014 reinforces this conclusion in part, but also renders its analysis in class terms more complex. In this 2019 paper, Smith and collaborators studied 11 million pass-through enterprises, that is, S-corporations and partnerships.⁸⁵ S-Corporations are owned by fewer than 100 shareholders, who are natural persons and U.S. residents, holding only a single class of stock. These need not be small—indeed, the authors underscore that firms such as Fidelity Investments, the U.S. subsidiary of PriceWaterhouseCoopers, as well as Hobby Lobby and home-improvement big box chain Menards are all held in this form. Such corporations, like partnerships, do not pay taxes at the entity level, but rather their shareholders or partners pay taxes on their profit distributions from the corporations directly. Since the Tax Reform Act of 1986 owners that fit these constraints have generally found it advantageous, in terms of tax bills for the owners, to incorporate as S corporations rather than as C corporations, or the usual publicly owned company. While the legal cap on number of owners is 100, Smith et al found that the average number of owners in pass through firms ranges between 2 and 3.4, particularly when setting apart legal services partnerships, which may have hundreds of partners. Three aspects of this paper are crucial to our analysis here. First, the typical C corporation, with its conflict between management and distributed ownership, covers many fewer people in the top income bracket. Executives who have been the subject of the executive compensation literature account for less than one-tenth the number of owners taking pass-through income from firms they own and manage, and take home an aggregate income one-eighth what their enterprise-owning peers take home in pass through profits and owner wages. Second, C-corporations tend to dominate in manufacturing and high tech, where large capital investments from diverse investors in different classes of shares are necessary, while these pass-through entities are the dominant form in the services industries, particularly professional services. This leaves us with the third point—that since 2001, more of the professional class entrepreneurs have begun to function like capitalists—as owners of the enterprise they work through, who deploy their own cognitive skills and work but also hire workers to provide various support functions, including as professionals, paraprofessionals, and clerical class workers. If in the 1980s and 1990s they still functioned as professional class workers, the sectors they dominate, like legal services, doctors and dentists, financial advisors, business services and consultants, have begun to shift to becoming capitalists. I will describe the findings first, and then offer some concrete examples and theoretical framing for how this should shape our thoughts about AI in general, and LLMs in particular.

The two largest categories of pass-through entities whose owners earn one million dollars or more are legal services and financial investment services, each taking profits of a little over \$28 billion in 2014, as compared to physicians' offices (\$8B), other professional and technical services (\$8B), and management consulting and real estate (~\$6B each). What we normally think of as professional services—legal services, physicians, dentists, professional services, insurance agents and the like tend to dominate the category of pass through entities whose owners occupy the top

⁸⁵ Matthew Smith et al., *Capitalists in the Twenty-First Century*, 134 Q. J. ECON. 1675 (2019).

1%, but this is not true of the top 0.1%. By contrast, financial investment activity, including private equity, venture finance, and hedge funds, as well as automobile dealers and oil and gas extraction fill the top spots among pass-through entities whose owners are in the top 0.1% of income (although there are entities providing the traditional professional services lower down the list of entities owned by 0.1% owners).

Incorporating these pass-through entities does three things to the common narrative. First, it suggests that the professional class workers functioning as capitalists are alive and kicking and taking a large share of the top incomes—only taking it as pass-through profits—at least in the United States. Second, it means that actively managing capitalists taking both profits and wages play a far larger role than financial rentiers taking returns on accumulated wealth than they play in the Piketty, Saez, Zucman framework. Third, it suggests that focusing on the largest corporations provides a very partial and incomplete picture of the class structure of the late information-economy techno-institutional wave and the emerging new wave. Unfortunately, the paper treats all work by the owners as “human capital,” and so it does not allow us to distinguish between the contribution of capitalist lawyers or dentists in their practice as professionals, and the contribution of the automobile dealer in his dealership, a more quintessentially capitalist application of cognitive managerial labor to an owned enterprise using hired labor. Nonetheless, the paper ends with one observation that is crucial for our purposes here.

Pass-through enterprises, and particularly those that have been most profitable, owe their high profitability in about equal measure to increased productivity (increase in total profits plus wages divided by the number of workers) and to an increase in the share of value added going to the owners, rather than the workers, whether as profit or as owner wages. That is, owners succeeded both in growing the profitability-per-worker their workers produced, and in grabbing a larger slice of this growing profitability relative to what they pay workers, to produce increased wealth. Smith and collaborators acknowledge that what they call “value added” in their assessment of “productivity,” however, is in fact profitability per worker, and could partly be growing because of rents, not actual productivity. This suggests that a roughly equal contribution of power-based, rather than productivity-based, increased returns is a lower bound of how much this increased profitability comes from power-increasing investments. More plausible is that some of the profit side of what Smith et al characterize as “productivity” is itself due to markups—that is, to power, over and above the portion of profits owing to power asymmetry between the owners and the workers in these enterprises.

A. Alternative trajectories for AI in the professions

Over the past thirty years, law and medicine have experienced similar patterns of increased utilization of paraprofessionals (paralegals and nurse practitioners/physician assistants (PPAs), respectively) and decline in the proportion of solo practitioners, the quintessential smallholder model. But the emerging alternatives are different, and different yet across jurisdictions, in ways that will impact the future development and instantiation of AI as a new knowledge infrastructure within these professions, and in the relations between capitalist owners of enterprises in these fields, the professional class workers, and the clerical and paraprofessional workers. In finance, in the United States, the opposite trend has developed since the Global Financial Crisis, with a larger

portion of financial advisors adopting a smallholder partnership model with very thin employment of clerical or paraprofessional workers. All these suggest quite distinct alternative pathways for AI-dependent professions, with very distinct implications for class relations within the professions, and for the capacity of these professions to deliver broad improvements for the population as a whole or intensified dislocation and decline in crucial access to legal, medical, and financial services in the population as a whole, as well as deepening macroeconomic instability. As was true of industrial production, automation, and robotics, there are possible high-road and low-road paths that could either produce a large and growing class of service sector clerical class and care workers with increased levels of embodied knowledge utilizing increasingly effective embedded knowledge tailored to enable them to deliver higher quality services than they could in the past, or an increasingly concentrated class of technologically-enabled capitalist-professionals leaving large swaths of the population unemployed or underemployed in the least creative parts of production and suffering alienation and anomie.

1. Law offices

Legal practice has seen a long-term decline in solo practitioners, an increase in the prevalence of corporation-like mega firms, and within practices, an increasing use of paralegals and permanent journeymen lawyers not on the path to full partnership, or full master craftsmen status as the partners, in turn, take on roles of capitalist owner-managers rather than fellow professionals. From 1980 to 2005 the American Bar Foundation published consistent statistics about the share of lawyers in different practice sizes.⁸⁶ Over the course of that period, solo practitioners declined from representing 49% of the profession to 32%, and all small firms below 10 practitioners dropping from 80% of lawyers to 58% of lawyers. Meanwhile, firms with more than one hundred lawyers skyrocketed from less than 3.5% of all lawyers to over 21%, with firms below one hundred but above fifty lawyers more than doubled from about 3% to about 7%. While no comparable statistics have been published since, it is possible to piece together from reports in professional media that large mega-firms continued to grow into multinational legal services firms, but Census Bureau data suggests that overall employment (including clerical staff and paralegals) by legal offices is still overwhelmingly provided in smaller law firms: with firms employing 1-4 employees making up 73% of law firms in the United States and employing nearly 20% of all employees in legal services, firms with 5-9 employees making up another 15% of legal offices, and employing nearly 16% of employees, and firms with 10-19 employees another 7% of offices and 15% of employment.⁸⁷

The strong emphasis of the professional press on elite lawyers in megafirms is understandable given their power and prominence in the profession, but for actually understanding the overall impact of AI on employment in legal services the dynamics in smaller firms employing fewer than

⁸⁶ BARBARA A. CURRAN & CLARA N. CARSON, *LAWYER STATISTICAL REPORT: A STATISTICAL PROFILE OF THE U.S. LEGAL PROFESSION IN THE 1980s* (1985). (providing the baseline data on which later reports relied, based on Martindale Hubbel listings); JOHN P. HEINZ ET AL., *LAWYER STATISTICAL REPORT: A STATISTICAL PROFILE OF THE U.S. LEGAL PROFESSION IN THE 1990s* (1994); JOHN P. HEINZ ET AL., *LAWYER STATISTICAL REPORT: THE U.S. LEGAL PROFESSION IN 2005* (2005). (providing additional years).

⁸⁷ This from the 2021 CBP report. US Census Bureau, *County Business Patterns*, CENSUS.GOV, <https://www.census.gov/programs-surveys/cbp.html> (last visited Sept. 29, 2025).

100 employees in all roles, not only lawyers, account for about 80% of the people employed by legal practices in the United States. A negligible proportion of practitioners in their first five years of practice and a very small share of lawyers with 5-9 years operate in these small firms, suggesting this model plays an insignificant role in lawyer training, while the distribution of lawyers with more experience suggest that solo practice is largely staffed by older lawyers, while younger experienced lawyers tend to practice in firms with between two and thirty lawyers.⁸⁸ This matters because as megafirms becoming more conventionally capitalist organizations, with a small number of actual partners playing more of a marketing or business development and managerial role, and a larger number of apprentice and journeymen associates maturing into various categories of permanent journeymen non-partners bearing various names such as of counsel, senior counsel, or non-equity partners. It appears that while thousands of lawyers may remain in these positions, many more leave for smaller firms, whose overall impact on employment and the forms it takes in the legal services sector may be far larger than most contemporary discussions of the profession consider.

A second trend that appears from looking at the economic census data is that there has been a significant upskilling of the non-lawyer staff in legal services. Figures 5a and 5b show the breakdown in occupations in legal practices between 2005 and 2023. In combination, they show that across law offices, the ratio of lawyers to non-lawyers has not change dramatically, but the mix of non-lawyers has shifted dramatically from clerical staff (legal secretaries and other support) to paraprofessional staff—paralegals. This suggests a far different story to the deskilling or technological unemployment story over the past two decades. Rather, we see that the quality of work and training of the non-professional staff has increased, providing less ministerial and more professional work, and with it their relative wages. This is not to idealize the life or income of a paralegal, in either a big firm or a small law office. It is to underscore, however, that with increased use of information technology and automation of simple ministerial tasks, there remains an important pathway for skills upgrading and more meaningful work for workers who had previously been in the clerical class. As small firms report in surveys that they adopt various AI-enabled tools to fill more of these ministerial tasks and manage billing, it is entirely possible that these tools will support the expansion of the smallholder or small partnership model with alliance with a more highly-skilled paraprofessional class, rather than it will result in a further immiseration of a stagnating clerical class.

⁸⁸ THOMPSON REUTERS & AMERICAN BAR ASSOCIATION, 2023 STATE OF U.S. SMALL LAW FIRMS A YEAR OF CHANGE (2023).

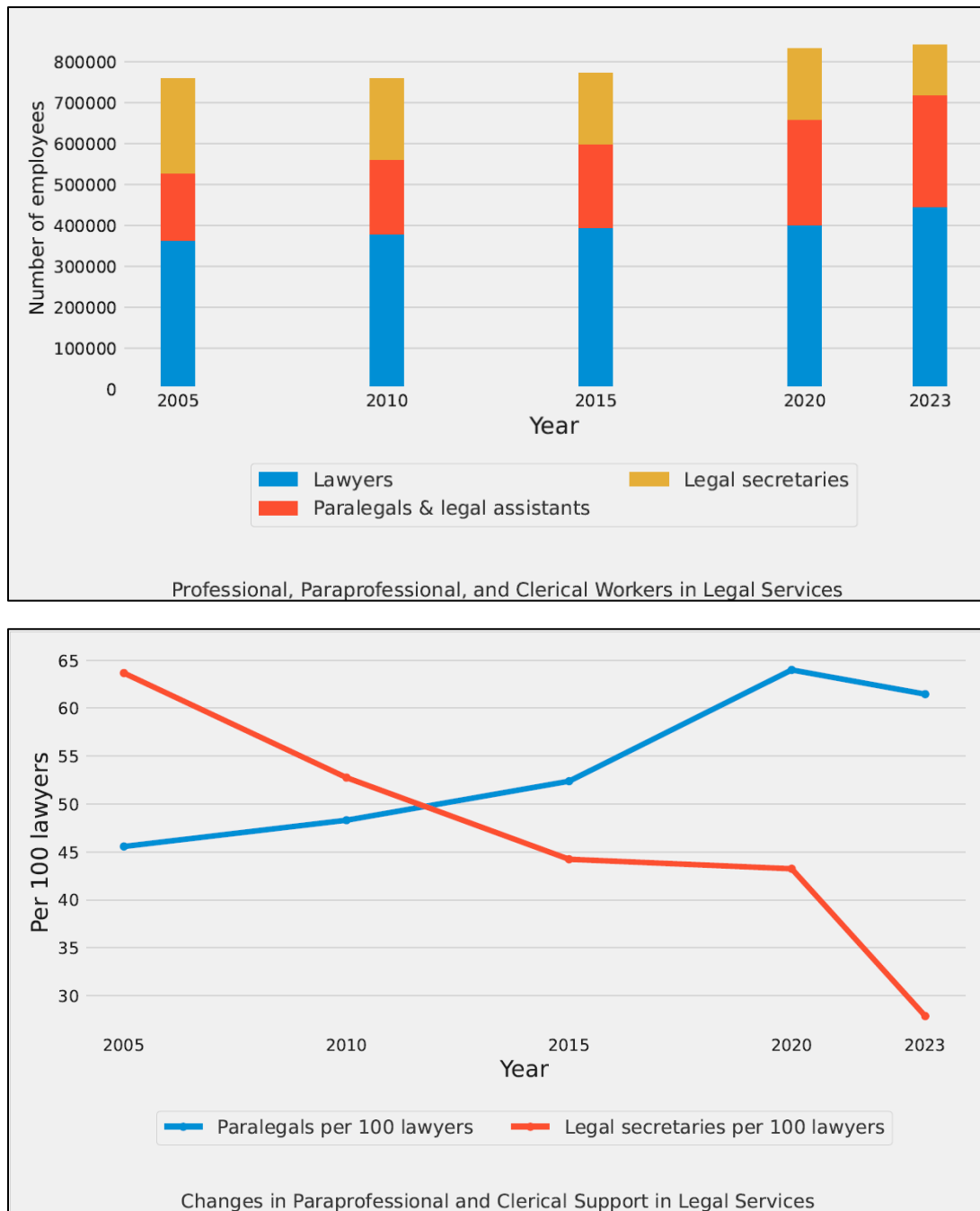


Figure 5a, 5b. Number and proportion relative to lawyers of legal secretaries and other clerical support vs paralegals. Source: U.S. Bureau of Labor Statistics, Occupational Employment and Wage Statistics.

2. Physicians and health care providers

In the 1980s, about eighty percent of physicians worked in practices with ten or fewer physicians, and nearly all these were physician-owned, with about seventy five percent of all

physicians being whole or part owners of their practice.⁸⁹ By 2012, 61.4% of physicians worked in such practices, and only 53% of physicians had an ownership stake in their practice. In 2024, only 47% of physicians worked in such small private practices, and only 35.4% had an ownership stake in a practice. The remaining slightly over half of physicians worked in practices owned by hospitals, healthcare systems, or private equity, or were directly employed or contracted (as independent contractors) by a hospital. The first wave of shifts from owners of professional practices to salaried employees of non-physician owners in the 2000s and 2010s took the form of hospital purchases of physician practices and recruiting of physicians to work in hospital-owned practices. The more recent wave is acquisition of physician practices by private equity firms.⁹⁰ Studies focused on physician practices tend to underestimate the role of private equity in reshaping the American healthcare system toward a for-profit, capitalist-owned framework. Private equity initially focused on purchasing hospitals and long-term care facilities, since the 1990s. These highly leveraged acquisitions were often followed by exits through public stock issues or sale to for-profit hospitals.⁹¹ By 2022, about 36% of hospitals in the United States were owned by for-profit entities, over half of hospital *beds* in the United States were owned by hospital chains with at least three hospitals, and of these, nine of the ten largest chains were for-profit.⁹² Just under half of U.S. hospitals remain non-profits, and nearly another 15% are government owned.

The primary drivers that drove physicians to sell their practices were institutional, not technological. Physician surveys emphasized that it was not new technology or capital costs of running a practice that led to the decline in physician-owned practices, but dealing with the powerful insurance industry that controlled payments and with medical supplies distributors who set prices in negotiation with buyers at scale. Surveyed physicians placed “the need to ‘better negotiate higher payment rates with payers, (70.8 percent),” “‘improve access to costly resources’ (64.9 percent) and ‘better manage payers’ regulatory and administrative requirements’ (63.6 percent)” as the leading reasons they sold their practice, followed by making it easier to participate “‘in risk-based payment models’ (55.1 percent).”⁹³ Three of these reflected bargaining power: either with payers—that is private healthcare insurance companies, as Medicare sets flat reimbursement rates—both on payment rates and on administrative requirements that would delay payment, and with medical supplies distributors (the costly resources over which they negotiated). The fourth is a secondary effect of the bargaining power of insurers—their willingness to increase physician payments if these take the form of reimbursement per patient, or capitation, rather than per procedure. This is a model by which the insurance company shifts the risk of patients with difficult and expensive needs to physicians in exchange for a fixed per-patient payment. In short,

⁸⁹ CAROL K. KANE, PHYSICIAN PRACTICE CHARACTERISTICS IN 2024: PRIVATE PRACTICES ACCOUNT FOR LESS THAN HALF OF PHYSICIANS IN MOST SPECIALTIES (2025).

⁹⁰ CAROL K. KANE & DAVID W. EMMONS, NEW DATA ON PHYSICIAN PRACTICE ARRANGEMENTS: PRIVATE PRACTICE REMAINS STRONG DESPITE SHIFTS TOWARD HOSPITAL EMPLOYMENT (2013); KANE, *supra* note 89.

⁹¹ Eileen Appelbaum & Rosemary Batt, Private Equity Buyouts in Healthcare: Who Wins, Who Loses? (Mar. 15, 2020), <https://papers.ssrn.com/abstract=3593887>.

⁹² W. PETER WELCH ET AL., OWNERSHIP OF HOSPITALS: AN ANALYSIS OF NEWLY-RELEASED FEDERAL DATA & A METHOD FOR ASSESSING COMMON OWNERS (2023), <https://aspe.hhs.gov/sites/default/files/documents/582de65f285646af741e14f82b6df1f6/hospital-ownership-data-brief.pdf>.

⁹³ KANE, *supra* note 89.

the long-term trend to consolidate physicians' practices into enterprises, first as hospitals and more recently their acquisition by private equity, reflects the institutional political economy of American healthcare institutions. It is driven by the need to compensate for the highly asymmetric of power health insurance companies (including managed care companies) under the uniquely privatized and profit-oriented health insurance system that the United States retains despite decades of efforts to shift to a universal public health system along one of the models used in all other wealthy countries.

Private equity, for its part, moved to increase the market power of hospitals and practices investors owned through a strategy of incremental acquisitions surrounding an initial purchase.⁹⁴ Precisely the highly fragmented, smallholder model of physician practices made it easy to pursue such a consolidation strategy—acquiring one major hospital or practice in a metropolitan area or region, and then expanding sideways by buying up small practices in transactions too small to trigger reporting requirements under merger regulations. The result has been both consolidation into large hospital networks, and concentration of market power in specific specialties and metropolitan statistical areas, with single private equity firms holding a 30% market share in at least one subspecialty in 108 Metropolitan Statistical Areas (MSAs) and over 50% in 50 MSAs.⁹⁵ After acquisition, PE-acquired physician practices saw a 20% increase in the charges per claim and an 11% increase in the allowed amounts per claim, they drew and served more new patients, and their patients overall saw physicians and other healthcare professionals more, and the practices increased the number of procedures they performed, at least in some subspecialties. Whether these additional services meant greater access to healthcare or unnecessary additional procedures and better negotiating position against the payers is unclear. Studies of hospital and nursing homes acquired by private equity suggest that there are real changes in care and staffing, but that the patient outcomes are far from obviously better.

Private-equity owned hospitals are more likely to add high-profit services to their hospital lineup after acquisition (like interventional cardiac catheterization, hemodialysis, or labor and delivery), and profitable technologies like robotic surgery and digital mammography, as well as freestanding or satellite emergency departments.⁹⁶ They charge more per inpatient day, and have a higher ratio of the charges to the costs overall, and in particular in their emergency departments, take fewer Medicare patients, and show slightly improved outcomes in specific procedures, although these appear to be accounted for entirely by the largest PE-owned hospital chain, Hospital Corporation of America (HCA).⁹⁷ Another study focused specifically on Medicare patients found overall no difference in outcomes between PE-acquired acute care hospitals and non-PE acquired hospitals except for acute myocardial infarctions, where the standard of care is well established

⁹⁴ Appelbaum and Batt, *supra* note 91.

⁹⁵ *Id.*; Ola Abdelhadi et al., *Private Equity–Acquired Physician Practices And Market Penetration Increased Substantially, 2012–21*, 43 HEALTH AFF. (MILLWOOD) 354 (2024).

⁹⁶ Marcelo Cerullo et al., *Private Equity Acquisition And Responsiveness To Service-Line Profitability At Short-Term Acute Care Hospitals*, 40 HEALTH AFF. (MILLWOOD) 1697 (2021).

⁹⁷ Joseph D. Bruch, Suhas Gondi & Zirui Song, *Changes in Hospital Income, Use, and Quality Associated With Private Equity Acquisition*, 180 JAMA INTERN. MED. 1428 (2020).

and its elements involve resource-intensive use of highly billable diagnostic and intervention cardiology.⁹⁸

The results appear quite different for nursing homes. A study of private equity owned nursing homes compared to matched non-PE owned nursing homes resulted in reduced number of nursing assistants, the most intensively connected care, but increase in registered nurses who deliver medical procedures that can be billed in addition to basic fees; decreased mobility, increased ulcers, and increased pain, and associated with these an overall increase of 11% in mortality, driven primarily by deaths of low risk patients who are kept longer in PE-owned nursing home (generating daily reimbursement income for longer stays) than in non-PE owned nursing homes.⁹⁹ The PE firms convert the low and steady Medicare payments that support this model primarily through “monitoring fees” portfolio companies owe the PE management firm, real estate sales cash return followed by rental payments by the nursing home for the same facilities they sold to pay off the debt used to purchase the facility, and interest payments on that debt—all of which squeeze the nursing homes’ income available to pay for caregivers and care.

For our purposes here, what matters about the rise of private equity in healthcare, converting and consolidating both for-profit hospitals and smallholder physicians’ offices into capitalist enterprises lies in the impact this has on the ways in which profit-driven imperatives are likely to drive technology adoption and deployment. All these studies reflect the fact that profit-driven investment and staffing changes patterns of care and staffing toward two goals—increasing services that can be billed profitably—possibly beyond what is medically necessary but useful when the most profitable care happens to align with the highly-imperfect compromises that set reimbursement rates per procedure; selecting patients with high-remuneration insurance over patients whose costs are covered by low-paying insurers, most importantly the Federal government through Medicare; and decreasing costs that cannot be billed separately, like the daily care of nursing assistants in nursing homes. In acute care contexts this appears to not meaningfully harm quality of care, and for specific disciplines that benefit from procedure-intensive practices, may improve quality of care. In context of low-tech long-term care, however, the results seem quite negative, but the evidence is sparse.

The most important aspect for our purposes is the change in staffing structure associated with the shift to private equity ownership and a for-profit model. Since the emergence of managed care, expanded Medicaid coverage, and increasing for-profit health insurance that followed the failure of the Clinton healthcare reform initiative in 1994, the proportion of nurse practitioner (NP) and physician assistants (PA) (collectively, Advanced Practice Providers (APPs)) providing healthcare care instead of, under the supervision of, or in concert with physicians has increased.¹⁰⁰ These

⁹⁸ Marcelo Cerullo et al., *Association Between Hospital Private Equity Acquisition and Outcomes of Acute Medical Conditions Among Medicare Beneficiaries*, 5 JAMA NETW. OPEN e229581 (2022).

⁹⁹ Atul Gupta et al., *Owner Incentives and Performance in Healthcare: Private Equity Investment in Nursing Homes* (Feb. 2021), <https://www.nber.org/papers/w28474>.

¹⁰⁰ Jonathan Oberlander, *The US Health Care System: On a Road to Nowhere?*, 167 CMAJ CAN. MED. ASSOC. J. 163 (2002); Kristin N. Ray et al., *Trends in Visits to Specialist Physicians Involving Nurse Practitioners and Physician Assistants, 2001 to 2013*, 177 JAMA INTERN. MED. 1213 (2017); Grant R. Martsolf et al., *Employment of Advanced Practice Clinicians in Physician Practices*, 178 JAMA INTERN. MED. 988 (2018); Roderick S. Hooker &

practitioners have reduced the cost of care and expanded access to care, and their use and deployment appears correlated with changes in legal scope of care regulation (how much care APPs can provide without physician supervision or participation) as well as whether their care is reimbursed at the full Medicare rate for the visit or procedure or at some lower level.¹⁰¹ These trends reflect the overall pressure throughout the healthcare system to reduce costs associated with physician wages compare to the lower levels of wages APPs command, and are true across the board in both primary and specialty care.¹⁰² Nonetheless, the pattern is more pronounced in private-equity as opposed to non-private equity acquired practices. When private equity acquires a physicians' private practice, they expand the use of APPs to a greater than matched practices that were not acquired do.¹⁰³ As with the impact of private equity on outcomes, this may result in increased access for patients, which may improve care if lack of access is a primary constraint on access to care and early intervention, or it may reduce the quality of care, if the expanded use of paraprofessionals results in reduced access to physicians. In all events, it certainly enables the private-equity owned practices to increase profitability per visit as the wages for APPs are considerably lower than those of physicians, and this effect is intensified where reimbursement rates for APP-provided visits and interventions are reimbursed at 100% of physician-provided visits, and where scope of practice regulations enable the owners of the practice to provide the services without physician participation.

3. *Pharmacists and pharmacy technicians*

To be added. Longest-term change. Drug interaction software and professional training of DPharma and technicians as core example of upskilling of the clerical to paraprofessional, and paraprofessional to professional class roles.

4. *Investment advisors*

The clearest example of a professional sector pointing toward further concentration of returns to capitalist and professional class workers relative to clerical and paraprofessional workers is the business of investment advisors. Since the Global Financial Crisis and the regulatory reaction, there has been a consistent shift of investment advisors from the large wirehouses and broker-dealers to registered investment advisors (RIAs), a shift projected by industry observers to

Linda F. McCaig, *Use Of Physician Assistants And Nurse Practitioners In Primary Care, 1995–1999*, 20 HEALTH AFF. (MILLWOOD) 231 (2001).

¹⁰¹ Hilary Barnes et al., *Effects of Regulation and Payment Policies on Nurse Practitioners' Clinical Practices*, 74 MED. CARE RES. REV. MCRR 431 (2017); Esita Y. Patel, Victoria Petermann & Barbara A. Mark, *Does State-Level Nurse Practitioner Scope-of-Practice Policy Affect Access to Care?*, 41 WEST. J. NURS. RES. 488 (2019).

¹⁰² Martsof et al., *supra* note 100; Sadiq Y. Patel et al., *Provision of Evaluation and Management Visits by Nurse Practitioners and Physician Assistants in the USA from 2013 to 2019: Cross-Sectional Time Series Study*, 382 BMJ e073933 (2023); Alicia Quella, Douglas M. Brock & Roderick S. Hooker, *Physician Assistant Wages and Employment, 2000-2025*, 28 J. AM. ACAD. PHYSICIAN ASSIST. 56 (2015); Ray et al., *supra* note 100.

¹⁰³ Joseph Dov Bruch et al., *Workforce Composition In Private Equity–Acquired Versus Non–Private Equity–Acquired Physician Practices: Study Examines Physician Workforce Composition Comparing Private Equity–Acquired with Non-Private Equity–Acquired Practices.*, 42 HEALTH AFF. (MILLWOOD) 121 (2023).

continue.¹⁰⁴ Over 70% of these new RIAs are LLCs or Limited Partnerships.¹⁰⁵ Many operate with a thin staff, on average employing, in 2023, a staff of three professional employees and one clerical and administrative support staff, and even for firms with over \$500 million assets under management employing 11 advisors and staff overall.¹⁰⁶ About as many investment advisors are too small to register with the SEC as operate in firms large enough to require SEC registration, and these operate in average firm size of two non-clerical employees (who may be an advisor and a different professional specialist) advising on average 52 clients.¹⁰⁷ While the research on the employment structure in healthcare and law is more extensive, the investment advisor space seems to reflect a model in which professionals increasingly minimize the use of clerical workers and provide professional advice directly to clients, in both large-scale professional firms and small scale partnership and LLC models, as professional-proprietors. I use these here not as evidence but as an intuitive marker for the model in which AI and automation of clerical work can fully replace clerical workers, leaving only professional class workers to operate in individual or small-scale partnership models.

B. Techno-institutional pathways for AI in Cognitive Labor

One core lesson from the history of technological change in capitalism is that the technologies that develop and dominate a regime are those that are most profitable, not those that are technically superior, much less those that achieve the maximum technological possibility frontier. The Concord proved both that supersonic commercial air travel was technically possible and that technologies that are at the technological frontier, but not the profitability frontier, will not become the dominant technologies in a capitalist economy unless funded by government fiscal sources, most prominently in the cases of military technology and drugs with large human benefits but no ready profitable market. Profitability, in turn, is a function of the combined effect of productivity and power asymmetries produced by the new techno-institutional configuration, which is itself a function of the opposing forces struggling over the potential forms of adoption and adaptation of the new ensemble. Several core constraints on the profitability of AGI as full replacement of labor suggests that a laissez faire attitude toward AI development is unlikely to result in a socially stable outcome, and provide insights into the potential directions that targeted programmatic interventions could offer for the trajectory of development.

¹⁰⁴ Kenneth Corbin, *Independent Broker-Dealers Are Losing Advisors to the RIA Channel, Cerulli Says*, BARRONS, <https://www.barrons.com/advisor/articles/financial-advisors-leaving-independent-broker-dealers-rias-cerulli-750e43ca> (last visited Oct. 27, 2025); 2025 SIFMA CAPITAL MARKETS FACT BOOK 100–101 (2025), <https://www.sifma.org/wp-content/uploads/2024/07/2025-SIFMA-Capital-Markets-Factbook.pdf>; Cerulli: *Independent RIAs to Grow 4% Through 2028*, WEALTH MANAGEMENT, <https://www.wealthmanagement.com/ria-news/cerulli-independent-rias-to-outpace-all-other-channels-by-2028> (last visited Oct. 27, 2025).

¹⁰⁵ THE DIVISION OF INVESTMENT MANAGEMENT’S ANALYTICS OFFICE OF THE U.S. SECURITIES AND EXCHANGE COMMISSION, INVESTMENT ADVISER STATISTICS: FORM ADV DATA, PERIOD ENDING DECEMBER 2023 (2024), <https://www.sec.gov/files/im-investment-adviser-statistics-20240515.pdf>.

¹⁰⁶ CERULLI ASSOCIATES, THE CERULLI REPORT: U.S. RIA MARKETPLACE 2023 RESILIENCY IN THE PURSUIT OF SCALE (2023), https://s3.us-east-1.amazonaws.com/cerulli-website-assets/documents/Info-Packs/2023/Cerulli-US-RIA-Marketplace-2023-Information-Packet.pdf?utm_source=chatgpt.com.

¹⁰⁷ Meredith Wise, *2024 Investment Adviser Industry Snapshot Shows Increase in Demand for Adviser Services*, INVESTMENT ADVISER ASSOCIATION (June 20, 2024), <https://www.investmentadviser.org/iaatoday/press-release/2024-investment-adviser-industry-snapshot-shows-increase-in-demand-for-adviser-services/>.

New AI systems aiming to achieve AGI with superior capabilities to humans in all domains so as to fully displace labor never actually compete against humans as they were in 2020 or 2000, but always against humans as they are trained and equipped to use the most recent generation of AI. The question is never whether AGI could in theory X, but whether it could X at lower cost than a different AI system could do in collaboration with a suitably trained human being, or at sufficiently higher quality to command prices able to overcome the higher cost. Let's take healthcare applications. Imagine AGI reaches a point that it designs a set of standard diagnostic tests any patient entering the emergency room would have to take, as well as which tests would need to be run in response to the new patient's description of their symptoms. Given initial results and further questions for the patient, the model could determine what further tests to order, instruct a robot to take the patient's blood and send it to the lab, which is also fully automated, receive the results, interpret them correctly, communicate these to the patient in a way the patient will understand, and prescribe a regimen the patient would accept. The question is not simply whether this would be technically feasible, but whether the service provided (emergency room intake, processing, and recommendation) could be more profitably provided by designing a system that would leverage suitably trained models in collaboration with newly-licensed and trained professionals or paraprofessionals empowered by the model to provide services that once required an emergency room physician.

More profitable includes the cost of developing the "last mile" of intelligence, vision, sensing, and communication to overcome barriers to machine interaction of an aging generation who may be resistant to machine-only communication, but open to machine-assisted human service. If two for-profit medical healthcare delivery systems are competing with each other to offer ambulatory care services, the system they would adopt is the one that would draw more patients and provide them with sufficient care to retain them as patients at the lowest cost. This may mean a fully automated process, but it would depend on whether the last mile could be overcome to the satisfaction of patients, and on winning the inevitable battles with the healthcare professionals over regulation and with the insurance industry over coverage, liability, and the cost of errors. If AGI advances at the rate the optimists predict, it will have to do this replacement long before a generation of patients trained from a young age to work without human beings in the loop is the largest consumer of healthcare services. In the meantime, those companies that managed to design human-computer collaborative systems would likely get the lion's share of the market, feeding back into the market for healthcare-oriented AI systems development. The same is true for law firms seeking to fully replace human research and document analysis competing against law firms leveraging existing AI systems in collaboration with increasing the ratio of suitably trained paralegals capable of providing higher quality service than the artificial system alone could at a competitive price.

We saw earlier that the healthcare system in general has expanded the use of APPs to drive down total cost of healthcare delivery in the teeth of expanded coverage and shrinking reimbursement rates since the 1990s. We saw too that these changes are sensitive to the institutional environment—the legal definitions of scope of service and the insurance industry's rules regarding reimbursement rates, and changed paraprofessional training trajectories. Specifically, we saw that private-equity owned services have increased the use of APPs relative to

physicians in specialty and primary care practices, particularly focused on performing procedures commanding high-reimbursement rates, but also relative to semi-skilled care workers providing crucial care that cannot be billed separately—to the great detriment of nursing home patients where semi-skilled daily physical ameliorative care is necessary and not separately billable.

In law, we saw displacement of legal secretaries by paralegals in law practices tracing a related but different arc to that of the APPs in healthcare, with paraprofessional displacing clerical class workers rather than taking over more roles previously reserved for the licensed professionals. We have less direct evidence of the relationship between the increasing portion of paralegals and the change in apprenticeship model for young lawyers. Rather, what we saw was that the past two decades have seen a shift from a craft-like professional partnership model to a capitalist model employing a class of permanent journeymen lawyers as one outlet for young lawyers who serve their apprenticeship at the large, increasingly capitalist firms, and spinning off to small firms as professional partnerships as the other primary outlet for lawyers after their first few years of practice. Finally, we briefly saw that investment advisers appear to have mostly dispensed with clerical class workers, and increased their use of both registered advisors and non-clerical professionals and paraprofessionals who are not registered advisors.

Across these three sectors of cognitive work the pattern has been one of expanding paraprofessional work, at the expense of shrinking clerical class work across the board, but differentially according to institutional structure in relation to the professional class workers in the enterprises. It seems safe to assume that, even if increased improvement of LLMs slows down in the near future, leaving us with only incremental improvements over the dramatic leap in capabilities AI models have had since 2022, these emerging systems will make suitably-trained paraprofessionals capable of delivering services much closer to the quality of services professionals could give without these new models than was true in the past.

For non-profit or public healthcare providers, this could mean containing costs while increasing access to services for a larger population. With suitable training, it would also enable more meaningful, autonomous, and socially valuable work for the newly empowered paraprofessionals. In private-equity owned practices, this is one possible outcome, but equally possible is that paraprofessionals will replace the number of physicians, creating a conflict between the capitalist owners of the practices and the professional employees by reducing the need for, and wages of, the professionals, increasing profitability, but without meaningfully expanding access to care or its quality for the population of patients as a whole. In a public national health system, these effects would all be part of the same internal calculus, subject to political contestation, and the expansion of paraprofessional class workers and access to care at lower total cost is the more likely outcome. The same is far from assured in a system that relies primarily on profit-driven care and insurance systems that compete against each other and can profit as much or more from distributive gains—with adverse selection among high- and low-cost patients, and between the investors and the workers—both professional class and paraprofessionals.

Figures 6 to 9 sketch out possible potential pathways that the development and integration of AI into cognitive labor, abstracted from the early pathways we see in, on the one hand, robotics, and on the other hand the less-than-AI computer-assisted pathways in the professions in the past

decade. These represent the following four ideal types: AGI Capitalist Utopia, PMC-led, Capitalist-led trifurcated PMC model, and a social-democratic model.

1. AGI Capitalist Utopia

The capitalist utopia is the model driving the pursuit of AGI and the insane stock valuations any company remotely claiming to be AI linked seems to command. I take Dario Amodei's late 2024 "reasonable" description of what "powerful AI" would be as my framework. In this model, an infinite number of instances of AGI will be running, with access to all real world operating systems and the ability to design, implement, and oversee human labor, if any, to fulfill tasks human beings once did better—from medical research to managing large organizations. In this capitalist utopia, it is possible that some elite PMC incumbents, like senior partners at law firms or established surgeons, will escape to become capitalists, but mostly the PMC and paraprofessionals will be replaced by artificial cognitive systems that can perform better than humans do, leaving mostly unskilled and semi-skilled or clerical workers, whether as part of the working classes or status-subordinated underclass, to provide the ghost work necessary as long as reinforcement learning still benefits from human feedback (RLHF). The extent to which the same can happen to skilled manual workers depends on how advanced robotics will become to instantiate AGI in a body that can manipulate the physical world. There is no particular reason to think that there will not be instantiations of AGI available for sale to smallholders to improve their own operations, as long as they focus on local, geographically-present services, while freeing them from having to rely on the services of the more expensive cognitive workers.

Dario Amodei, CEO of Anthropic, provides a nice encapsulation of the anticipated effects as seen from Silicon Valley.

I think that the human economy may continue to make sense even a little past the point where we reach "a country of geniuses in a datacenter." However, I do think in the long run AI will become so broadly effective and so cheap that this will no longer apply. At that point our current economic setup will no longer make sense, and there will be a need for a broader societal conversation about how the economy should be organized.

While that might sound crazy, the fact is that civilization has successfully navigated major economic shifts in the past: from hunter-gathering to farming, farming to feudalism, and feudalism to industrialism. I suspect that some new and stranger thing will be needed, and that it's something no one today has done a good job of envisioning. *It could be as simple as a large universal basic income for everyone*, although I suspect that will only be a small part of a solution. *It could be a capitalist economy of AI systems, which then give out resources* (huge amounts of them, since the overall economic pie will be gigantic) *to humans based on some secondary economy of what the AI systems think makes sense to reward in humans* (based on some judgment ultimately derived from human values). Perhaps the economy runs on Whuffie points. *Or perhaps humans will continue to be economically valuable after all, in some way not anticipated by the usual economic models*. All of these solutions have tons of possible problems, and it's not possible to know whether they will make sense without lots of iteration and experimentation. And as with some of the other challenges, *we will likely have to fight to*

get a good outcome here: exploitative or dystopian directions are clearly also possible and have to be prevented.¹⁰⁸

Amodei's musings reveal two remarkable things. First, the basic assumption of the pursuit of AGI is that it will render human labor superfluous. Second, his solutions suppose at least a social democratic system of sorts (universal basic income), possibly a socialist economy where the AGI has solved the socialist calculation system, or possibly a post-scarcity world where reputation alone is the universal currency.

The simple macroeconomic question of who will buy all the stuff that AGI will produce, with what money, simply disappears in this analysis. Where will profit come from if everyone is wallowing in superfluities and rewarded in reputational tokens? Where will the government revenues to pay the universal basic income come from? Why would capitalists, now the soul sources of production, agree to be taxed at levels sufficient to pay the rest of society to hang around contemplating life, the universe, and everything (as though AGI couldn't do it better than they could anyway)? If not basic income, where will household income to support demand come from? Why would other countries buy our stuff if AGI can produce equally well anywhere? Who would have the money to buy in other countries, if their capitalists too pursued this same strategy? Never mind, just buy more stock and join the rentiers who will float above the waves.

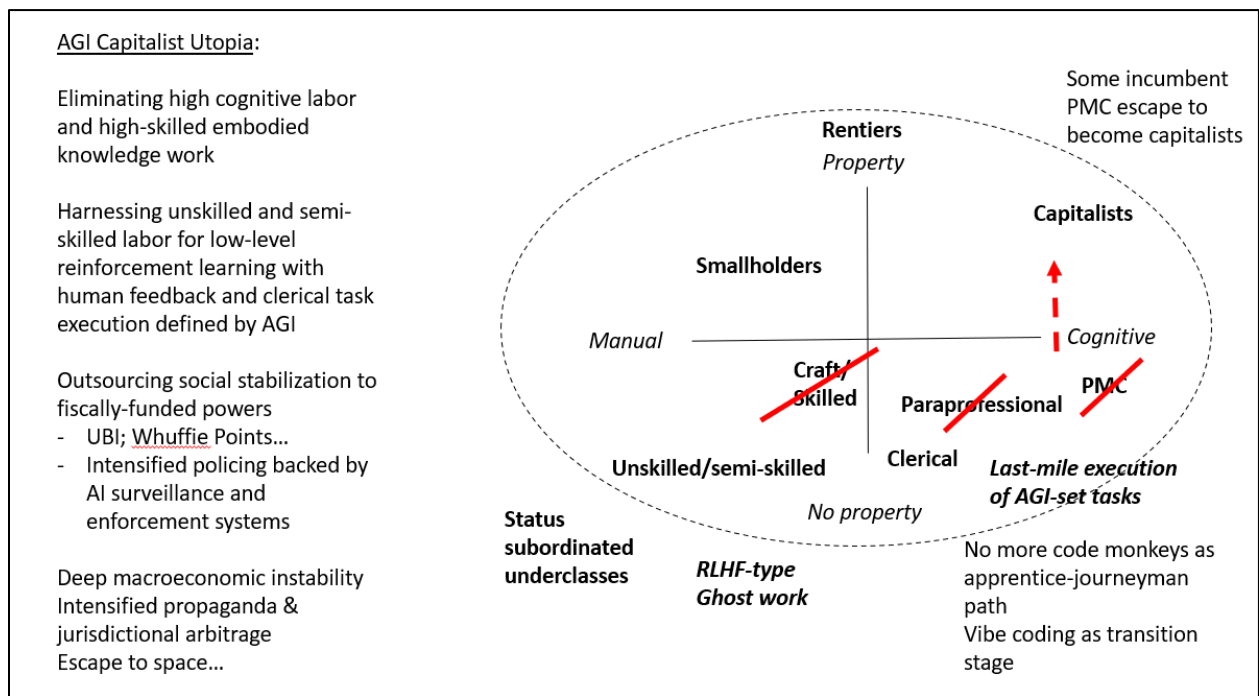


Figure 6: AGI Capitalist Utopia

Any path that leads in this direction will encounter deep resistance long before it reaches this point. The deep social dislocation of job loss, loss of a horizon for a life plan, and deep uncertainty about the future is already triggering a wave of revanchist populism throughout the richer

¹⁰⁸ Dario Amodei — Machines of Loving Grace, *supra* note 1.

countries. Building an alliance with politicians who catch the wave of anger to distract the voters, as Silicon Valley leaders are currently doing in the United States, complemented by misdirection and opinion manipulation to distract attention from the cause of the dislocation and support for more intensive violent suppression of dissent, supported by improved surveillance techniques, are the most obvious solutions. Locating operations in other jurisdictions to avoid legal restrictions, moving the centers of operation for the same reasons, and ultimately imagining an escape to Mars, leaving the unwashed masses behind, all seem to be within contemplation. What seems clear is that there is no smooth path from here to Capitalist Utopia. The only question is whether the frustration, fear, and rejection will take the form of political and economic resistance that is mobilized effectively and directed at institutional changes that could, in fact, shift the trajectory.

As we saw with the difference between the United States and several European countries, how the model actually unfolds will depend on the distribution of political and economic power through the predictable conflicts, given the inherited institutional political economy of the society in question. In the case of cognitive work, the most obvious potential counterpower is the PMC itself, with lawyers more directly in control of regulating their own profession than doctors, and possessing more control over the institutional structure of the service and terms of pay. The next two figures abstract from the current trends in law and medicine in the United States, respectively, based on the relative institutional powerbase each profession has and on the foothold non-professional capitalists have in each sector. The last figure sketches what a social democratic model might look like, one that all other classes should find would be worth fighting for.

2. An incumbent-PMC led model and breakdown of the apprenticeship model

Machines can be property in the way people cannot, and so the shift from high embodied knowledge to materially-embedded-knowledge underwrites a shift in power in favor of capitalists who own the tools, as well as doing cognitive work. Current patterns in the legal profession and as well as higher education suggests a potential trifurcation of the PMC in reorienting the relations of those who do cognitive labor and those who own capital that produces outputs equivalent or complementary to cognitive work. In this model, some of the more senior incumbent professionals shift to become capitalists—primarily engaging in ownership and management of the professional service firm while extracting high income from it in the form of income or pass-through profits. These now take the place of the capitalists, particularly if there are strong institutional reasons for the persistence of the institutional relations in which they previously practiced despite a shift in the locus of cognitive labor. In the case of lawyers and law firms, it is the hold the bar has on the provision of legal services, the regulation of that provision by anyone not licensed to practice law, and the prohibitions, where they remain, on ownership of a legal services firm by non-lawyers. In the case of universities, it is often the non-profit structure, the accreditation, and the reputational roots of the system. Here, the “capitalists” are mostly the rising managerial class of university administrators who become the residual owners of non-profit universities, or the capitalists who open for-profit investor-owned universities.

In this model, the newly capitalist professionals restructure cognitive work in to (a) increase the prevalence and contribution of paraprofessionals, who are cognitive workers with substantive skills in a dependent relation to the professionals and ineligible to full professional position in the

cognitive production system; (b) cut off the apprenticeship model that historically characterized the professions, leaving the professionals who do not become capitalists in a position of either (i) permanent journeymen—doing professional-level cognitive work in a dependent relation to the capitalist professionals or (ii) setting up as smallholder professionals, alone or in small partnerships, and able to expand the services they provide by themselves deploying machine-generated cognitive work equivalents and a smaller number of paraprofessionals, at lower levels of capital expenditure and scope of service. This is the model we already see in law, as well as in large segments of the investment advisor sector.

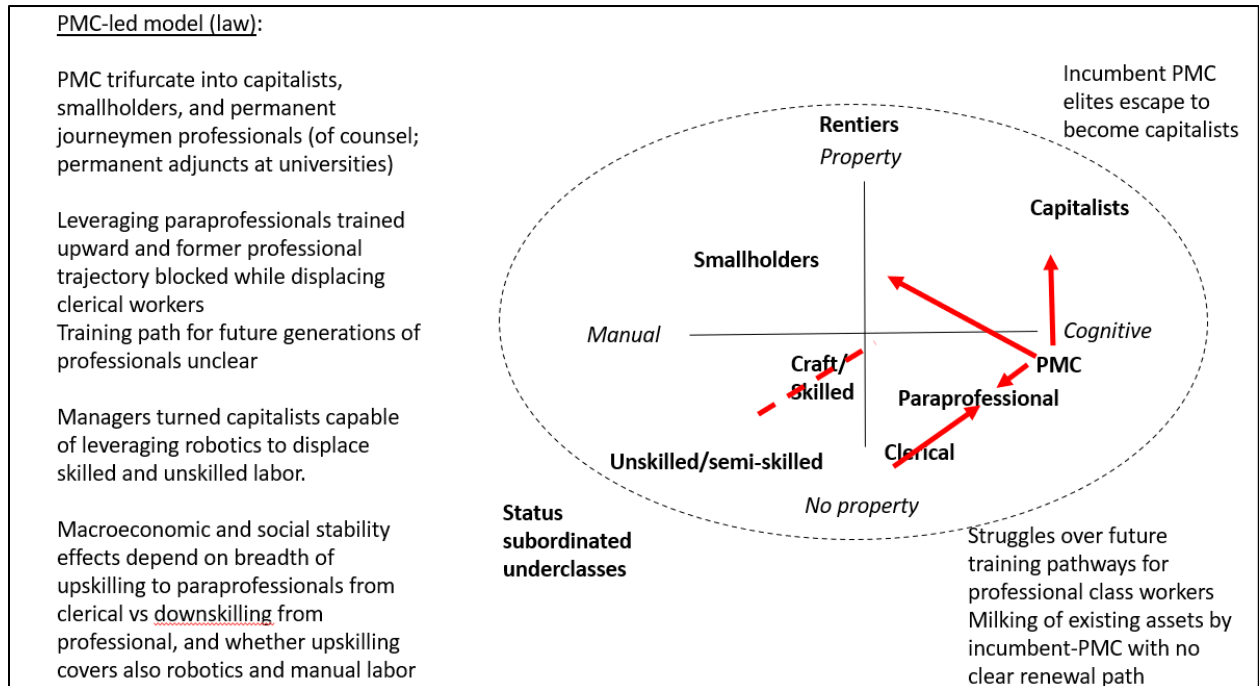


Figure 7: Incumbent PMC-led trifurcation of the professional and managerial class

The implications of this model for macroeconomic and social stability depend on the breadth of expansion of the roles of paraprofessionals relative to the loss of clerical jobs, and in particular whether the pattern of increasing the skill quotient of the workers working with the machines also develops for robotics and displacement of unskilled or semi-skilled manual labor with skilled labor at higher productivity and higher wages.

A major concern triggered within the professional class itself by early examples of this pattern is the disruption to the historical apprenticeship path for professional training. This is true whether the apprenticeship is in the form known from the traditional professions—as with young lawyers and even more so medical interns, residents, and fellows—but also prevails in newer, less formally structured professions. The most immediately impacted sector among these is software development, which has seen the availability of entry-level coder positions shrink dramatically. With comparatively weak institutionalized structures for the profession, little by way of institutionalized worker power in terms of unions, and a mostly capitalist structure for firms that employ software developers, this sector seems particularly vulnerable to displacement. Current

industry newsletters suggest a decline of about one third in new job listings for software developers, and even the BLS, which projects growth in software development as a whole, projects decline in entry level positions for computer programmers with only a bachelor's degree from 2024 to 2034, although to a less extreme extent than the industry publications worry.¹⁰⁹

By contrast to the programmers, university teachers have been experiencing a process of shifting from primarily tenure and tenure-track model to non-tenure track full, part time, and contingent contracts since the late 1980s.¹¹⁰ These positions now make up over two-thirds of faculty at American colleges and universities. Unionization rates among non-tenure track teachers are much higher than unionization among tenure-track college and university teachers, and proportions are higher in public than in private universities.¹¹¹ Overall, unionization rates among college and university teachers are higher than those of nurses, which are in turn higher than those of advanced practitioner paraprofessionals.¹¹² Because of the large weight that public and non-profit institutions play in higher education, capitalists are comparatively constrained in leading the integration of AI into this already-bifurcated model. Instead, the administrative leadership of universities plays the role of incumbent PMC-led transformation.¹¹³ How AI gets deployed in higher education is already emerging as a major point of contention between faculty and administrators, with the potential path of milking of existing educational reputation and accreditation functions by the incumbent administrators and tenured professoriate without providing an established apprenticeship path presenting a real possibility, subject to organized opposition by the increasingly-unionized contingent teachers. The point, for our narrow purposes here, is to underscore the centrality of the inherited institutionalized class power in the sector of cognitive labor under consideration to predict the likely future development of the techno-institutional structure and to identify the relevant possible class formations that could shape that development path.

3. Capitalist-led trifurcation of professional cognitive labor

¹⁰⁹ *Computer Programmers*, BUREAU OF LABOR STATISTICS, <https://www.bls.gov/ooh/computer-and-information-technology/computer-programmers.htm> (last visited Oct. 29, 2025); *Software Engineering Job Openings Hit 5-Year Low amid Industry Shift*, TECHSPOT (Feb. 22, 2025), <https://www.techspot.com/news/106878-software-engineering-job-openings-plummet-35-five-year.html>; *Charted: The Decline of U.S. Software Developer Jobs*, https://www.visualcapitalist.com/charted-the-decline-of-u-s-software-developer-jobs/?utm_source=chatgpt.com (last visited Oct. 29, 2025).

¹¹⁰ *Data Snapshot: Tenure and Contingency in US Higher Education | AAUP*, <https://www.aaup.org/academe/issues/spring-2023/data-snapshot-tenure-and-contingency-us-higher-education> (last visited Oct. 29, 2025).

¹¹¹ WILLIAM A. HERBERT, JACOB APKARIAN & JOSEPH VAN DER NAAL, 2024 DIRECTORY OF BARGAINING AGENTS AND CONTRACTS IN INSTITUTIONS OF HIGHER EDUCATION, https://research-data.hunter.cuny.edu/ncscbhep/2024DirectoryofBargainingAgentsandContractsinInstitutionsofHigherEducation.pdf?utm_source=chatgpt.com.

¹¹² Ahmed M. Ahmed et al., *Trends in Labor Unionization Among US Health Care Workers, 2009-2021*, 328 JAMA 2404 (2022).

¹¹³ Benjamin S. Kay & Andrea Honeycutt, *Trends in Administrative Expenses in American Higher Education* (Jan. 30, 2025), <https://papers.ssrn.com/abstract=5279495>. (finding significant rise in broad measures of expenditure proportionally between the 1980s and 2020s, and more moderate but still significant growth in “core-administrative” or upper and middle management expenditures in universities over the course of the period).

A variant of the PMC-led trifurcation is suggested by the private equity trend in physician practices and hospitals—a buyout of the incumbent PMC enterprises by capitalists and transformation of the cognitive work similar to that led in the incumbent-PMC model. The major difference here is that profit imperatives are likely to drive the harnessing of paraprofessionals to displace professionals for their cost-reduction characteristics—because of lower wage expectations. The concerns regarding quality of outcomes in the sector, by comparison to professional-led models, replicate those we saw in the studies of PE-owned physicians’ practices and hospitals—the possibility of expanded access to care for those services that are highly-reimbursed under the extant institutionalized payments system, strategic investment in emphasizing the most profitable services rather than those with the highest public benefit—most clearly present in healthcare.

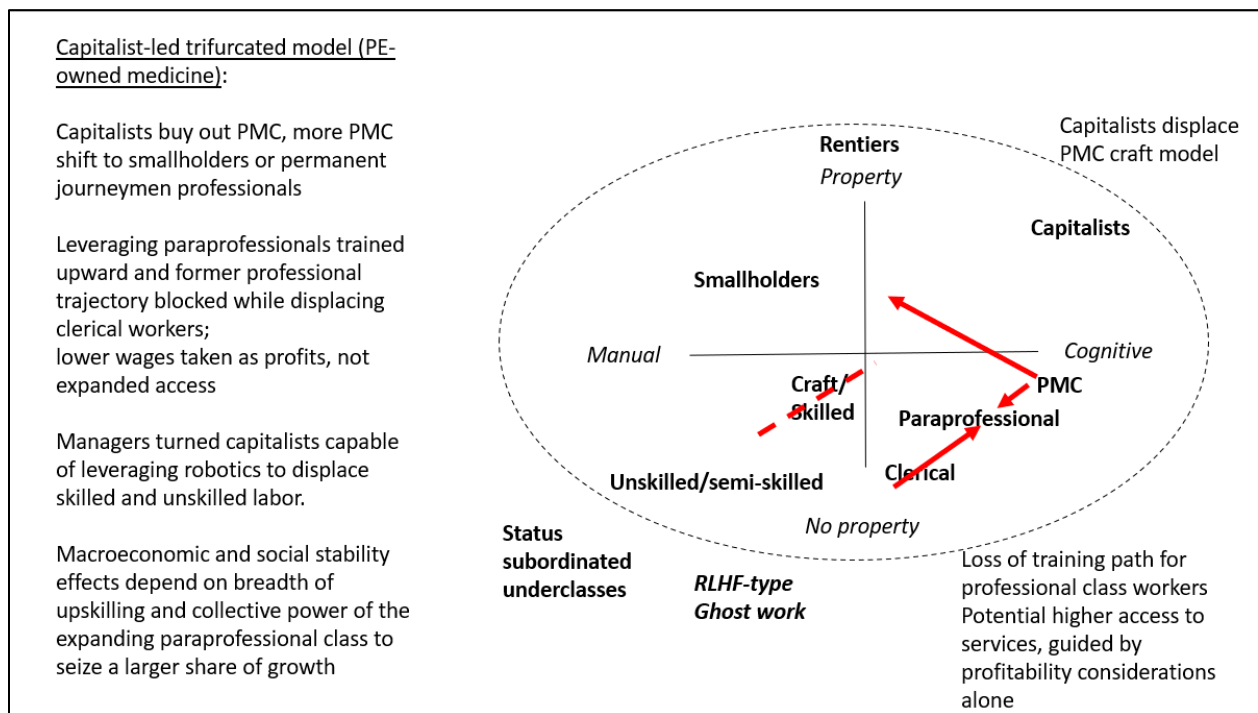


Figure 8: Capitalist-led model

4. A social-democratic model

The major alternative to any of the models that rely on one or another of the profit-reaping classes to take the lead in AI development, instantiation, and deployment is to follow a social-democratic model. This is of course easier and more natural for the handful of countries that are already social democratic, but certain elements can be implemented by corporatist oligarchic and even liberal oligarchic societies like the United States as well. The crucial dependency is that the present transition will intensify the political instability and crisis we are already observing, and the crisis, for its part, particularly the inevitable collapse of the present AI bubble, may open up an opportunity for meaningful institutional transformation. If that comes to pass, having a roadmap to what is needed is a precondition to not dissipating the political opportunity the crisis will provide on half measures and structurally weak changes.

The first measure would be the adoption of socialized provisioning of datacenters, compute, and model development and application access, as described in Part IV. As I explained there, directly provisioning the core infrastructure of the new techno-institutional wave provides far greater leverage to implement a social agenda than any efforts to impose a behavioral regulatory framework on capitalist enterprises operating across the globe, running an extremely opaque resource system, and able to bring to bear vast amounts of money and lobbying power, as well as engage in jurisdictional arbitrage, to evade, undermine, or outright block or reverse externally-imposed regulations. At a bare minimum, some system of public utility regulation, with cost-based open access to computational resources for competing enterprises (or small enterprises, if that is the primary concern of the relevant political system) and price and access regulation for households insofar as uses determined to be basic needs are concerned. More expansively, a publicly-provisioned infrastructure, as I suggested in Part IV, can be used as a form of development bank to change the cost of different models of deploying AI in relation to cognitive workers, emphasizing high road strategies that implement continuous educational improvement for a broad population of workers earning dignified wages and doing meaningful work over low-road strategies focused on displacing labor.

In particular, preferential access to compute and models can provide a means to raise prices and decrease profitability for enterprises that pursue low-road strategies based on minimizing labor input and harnessing and squeezing relatively weaker workers, and increase the profitability of enterprises that meet some prespecified set of conditions reflecting continuous skills improvement and integration of new artificial cognitive outputs with improved human cognitive labor. Another mechanism of containing the profitability of capitalist enterprises that depend on upward redistribution and identifying and exploiting labor weakness is to preferentially support worker cooperatives in cognitive labor. The fact that at the professional level we are seeing an increase in the number of smallholder enterprises suggests that one way in which the new technology can help expand the autonomy and self-direction of cognitive workers is by supporting cognitive-worker-controlled enterprises—small partnerships or, for larger groups, cooperatives.

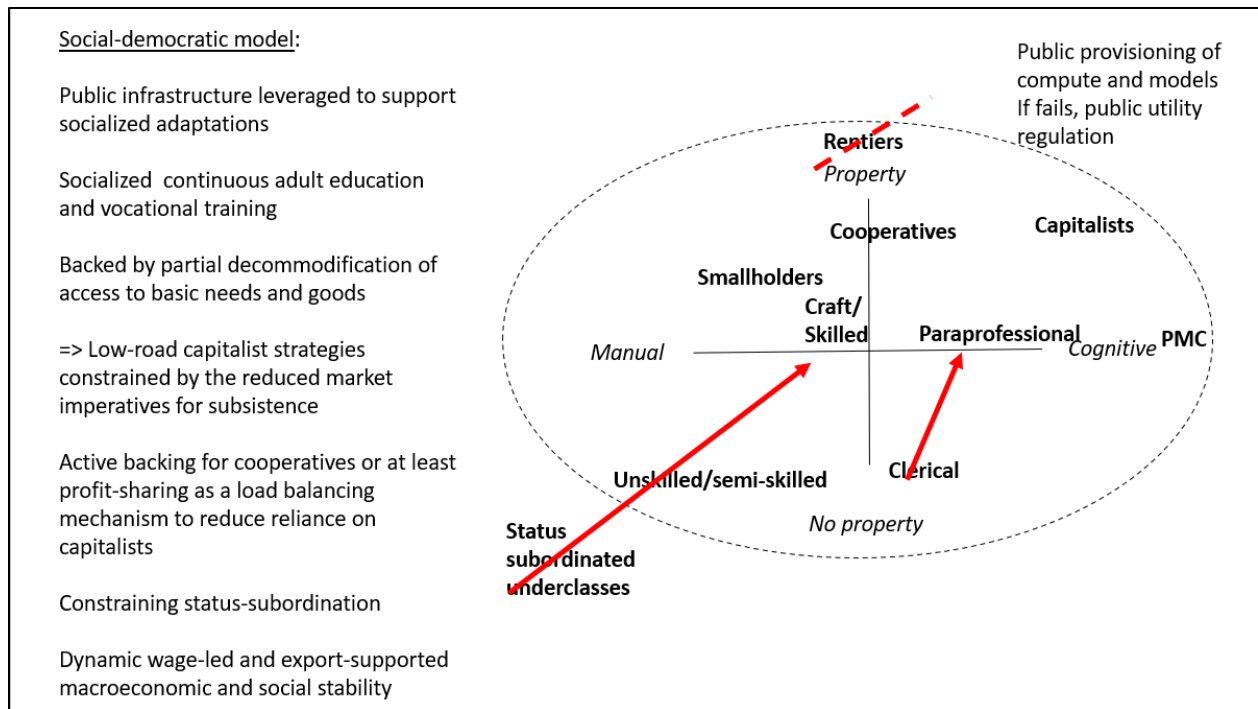


Figure 9: A Social Democratic Model

Continuous training and adult education at socialized cost, and broad-based socialization of the risk of having one's skills rendered obsolete by rapidly changing techno-institutional waves and configurations is a precondition for managing the social dislocation attendant to all such transitions and maintaining dynamic productivity growth and macroeconomic stability in the teeth of rapid change. In the specific case of the new AI-based systems, what professional class workers now see as a threat is, from a broader population perspective, an opportunity—both in terms of potential jobs and in terms of access to basic services. Shifting more of the clerical class to being able to perform paraprofessional level work by being trained on AI-enabled cognitive support systems, training unskilled and semi-skilled workers to become technically capable to work with, supervise, and operate as skilled workers with robots instead of being displaced by them, all suggest a potential path for eliminating or shrinking the classes operating under the greatest asymmetric power relations.

Nudging the instantiation of machine learning systems in services sectors to leverage an increasingly skilled clerical and paraprofessional class of workers reaps social gains both in social relations of production and in provisioning of universal access to basic needs and goods. The more APPs there are, and the more of the universe of healthcare needs these paraprofessional can perform without depending on physician oversight or approval, the broader access to healthcare can be made. The school nurse, the community healthcare worker, or the around the corner clinic can offer better, cheaper access to basic preventive and routine care throughout a society. Again, this is all much easier to achieve in a society that already has universal healthcare and an infrastructure of local clinics than it is in a society whose healthcare is optimized for capitalist profits, as the U.S. healthcare system uniquely is. Adding continuous skills training to AI will not

of its own transform a healthcare system otherwise optimized for profit after decades of resisting the universal model adopted in all other wealthy countries and many middle- and low-income countries. But the proportion of public and non-profit organizations in healthcare does provide a pathway. A similar pathway can be imagined for access to basic legal services by enabling paraprofessionals without a full legal degree to provide a broad range of services that could improve the interaction between most of the population and the regulatory and legal system.¹¹⁴ Just as the range of approaches to Active Labor Market Policies (ALMP) in Europe socialized the risk of technological change and the cost of continuous retraining and adult education that the perennial gale produces,¹¹⁵ so too in this new wave those countries, most prominently Denmark and Sweden that already have a high degree of ALMP expenditures as a percent of GDP, will have an easier time incorporating AI into changing labor markets, just as they did with robotics.

The constraint is not fiscal, but political will. However expensive, annual Danish spend on active measures as a proportion of GDP since 2019 was well below what the United States spend on active measures in the last year of the first Trump Administration, and similar to what the U.S. spent in the first year of the Biden Administration, in response to the Covid-19 pandemic.¹¹⁶ It is not, then, a matter of capacity, but of political will and guiding theory about how the economy works and how to make sure workers in all classes are the most productive they can be in a rapidly changing techno-institutional structure. And if political will is the question, then one must recognize that ethnocentric anti-immigrant propaganda and racialized ideology have been an enormously powerful political force in both the United States and Western Europe, including the Nordic social democracies, with right wing political parties funneling the frustrations of a broad range of working classes and smallholders at their declining prospects in an increasingly congested world where much of the growth that buoyed the life prospects of their parents' and grandparents' generation has shifted to Asia. Racialized politics will continue to exert a major drag on any meaningful possibility of social democratic transformation across these societies.

V. Conclusion

Technology is not an exogenous force operating on society from outside. Technological change is a product of the endogenous dynamics of capitalism, which are characterized by continuous change and destruction in a Schumpeterian dynamic where “destruction” is socially and politically destructive as well as the destruction of one technologically-instantiated business model to the next. Technological change has always involved class struggle, and always involves not only

¹¹⁴ Well before the present wave of AI enthusiasm, this overall model of broad upskilling of paraprofessionals integrated with computerized expert system to expand access to crucial professional services was proposed by Richard Susskind in several books over the 2000s and 2010s, the broadest of which was co-authored with his son. RICHARD E. SUSSKIND & DANIEL SUSSKIND, *THE FUTURE OF THE PROFESSIONS: HOW TECHNOLOGY WILL TRANSFORM THE WORK OF HUMAN EXPERTS* (First edition ed. 2015).

¹¹⁵ JONAS PONTUSSON, *INEQUALITY AND PROSPERITY: SOCIAL EUROPE VS. LIBERAL AMERICA* (2006), https://muse.jhu.edu/pub/255/edited_volume/book/52768; THELEN, *supra* note 43.

¹¹⁶ *OECD Data Explorer • Labour Market Programmes*, [https://data-explorer.oecd.org/vis?tm=labour%20market%20policy&pg=0&hc\[Unit%20of%20measure\]=&snb=30&df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_LMP%40DF_LMP&df\[ag\]=OECD.ELS.JAI&df\[vs\]=1.0&dq=DNK%2BFRA%2BDEU%2BNLD%2BSWE%2BCHE%2BUS%2BAUS.EXP..PT_B1GQ&lom=LASTNPERIODS&lo=5&to\[TIME_PERIOD\]=false&vw=tb](https://data-explorer.oecd.org/vis?tm=labour%20market%20policy&pg=0&hc[Unit%20of%20measure]=&snb=30&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_LMP%40DF_LMP&df[ag]=OECD.ELS.JAI&df[vs]=1.0&dq=DNK%2BFRA%2BDEU%2BNLD%2BSWE%2BCHE%2BUS%2BAUS.EXP..PT_B1GQ&lom=LASTNPERIODS&lo=5&to[TIME_PERIOD]=false&vw=tb) (last visited Oct. 31, 2025).

productivity-increasing changes, but changes aimed to redistribute power in social relations of production, almost always in favor of the capitalists who adopt the new technology and the financial rentiers who finance them. Endogenous as it is to capitalism, technology is always already embedded in inherited institutional and ideological structures, and each wave of change is techno-institutional, not purely material. As a result, the same technical components can be instantiated with different emphases and configurations based on the historically inherited distribution of institutionalized power in social relations of production in different societies. Within this framework, the observation that several societies in Europe see an increase in labor productivity and increasing wages, as well as at least no decrease and in some cases increase in employment with the introduction of robots, while in the United States, American firms deploying robots (mostly made by Japanese or European firms) lead to declining jobs and wages, is a feature of the different trajectories of different societies' institutional political economy over time. We can expect to see similar differentiation between the ways in which machine learning based systems are integrated into cognitive work with the emergence of generative AI, and that these are amenable to policy, but primarily to structural institutional change not to surface adaptation that can only be applied from the outside in, to a fully capitalist model of model development and application.

In the last part of the paper, I outlined a social democratic agenda for AI, whose detailed contours need to be developed before the next crash. These include publicly provisioned AI, which is expensive but well within normal levels of infrastructure investment even in the United States, whose compute and models can be deployed strategically to incentivize high-road instantiations of generative AI in cognitive work and reduce the profitability of low-road strategies. Such a core structural program needs to be complemented by significant increases in socializing the cost and risk of continuous change by implementing expanded lifelong educational access to continuous learning backed by significant support for job changes—significant unemployment and retraining funds to maintain a dignified level of life between jobs. In combination, leveraging machine-embedded cognitive outputs with continuously learning human beings can be deployed to shrink that portion of the population stuck within an unskilled and semi-skilled category, and expand broad social access to affordable services that depend on cognitive work—most importantly healthcare, financial or legal advice, and, again, lifelong learning. All these interventions will be easier to implement and have a more robust effect in societies that already have meaningful access to a dignified level of access to basic needs and goods, because that reduces the imperatives that market dependence for subsistence imposes on all households in capitalist societies. In those societies, employers following low-road strategies necessarily find it harder to hire workers into bad jobs that are worse than staying unemployed longer, retraining, until a better job is available. Nevertheless, even in societies that do not have such a baseline provisioning system, the combination of interventions proposed can shift the dynamic in the direction of broader participation in the labor force at higher levels of skill and worker autonomy, in turn underwriting a broader, more dynamically stable growth model.